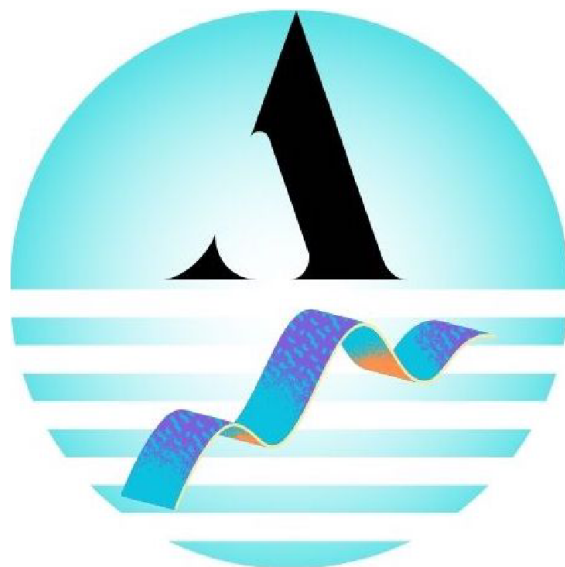




Advanced Graph Theory (MMTHDGT325)

||||||| Written by: **DR. AIJAZ** |||||

ASSISTANT PROFESSOR OF MATHEMATICS
JK HIGHER EDUCATION DEPARTMENT

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ADVANCED GRAPH THEORY

Course No: **MMTHDGT325**

Examination:

(a). Assessment

(b). Theory

Time Duration: 2.5 hrs

Total Credits: **04**

Total Marks: 100

Max. Marks: 28

Max. Marks: 72

Min. Pass Marks: 40

Objectives: To expose the student to the various concepts of graph theory to model many types of relations and processes in physical, biological, social and information systems.

UNIT -I

Colorings

Vertex coloring, chromatic number $\chi(G)$, bounds for $\chi(G)$, Brook's theorem, edge coloring, Vizing's theorem, map coloring, six color theorem, five color theorem, every graph is four colourable iff every cubic bridgeless plane map is 4-colorable, every planar graph is 4-colorable iff $\chi'(G) = 3$, Heawood map-coloring theorem, uniquely colorable graphs

UNIT -II

Matchings

Matchings and 1-factors, Berge's theorem, Hall's theorem, 1-factor theorem of Tutte, antifactor sets, f-factor theorem, f-factor theorem implies 1-factor theorem, Erdos-Gallai theorem follows from f-factor theorem, degree factors, k -factor theorem, factorization of K_n .

UNIT -III

Edge graphs and eccentricity sequences

Edge graphs, Whitney's theorem on edge graphs, Beineke's theorem, edge graphs of trees, edge graphs and traversability, total graphs, eccentricity sequences and sets, Lesniak's theorem for trees, construction of trees, neighbourhoods, Lesniak theorem graphs.

UNIT -IV

Groups in graphs and graph spectra

Automorphism groups of graphs, graphs with a given group, Frucht's theorem, Cayley digraph, spectrum of a graph, spectrum of some graphs-regular graph, complement of a graph, edge graph, complete graph, complete bipartite, cycle and path, Laplacian spectrum, energy of a graph, Laplacian energy.

Recommended Books:

1. R. Balakrishnan, K. Ranganathan, A Text Book of Graph Theory, Springer-Verlag, New York
2. B. Bollobas, Extremal Graph Theory, Academic Press.
3. F. Harary, Graph Theory, Addison-Wesley.
4. Narsingh Deo, Graph Theory with Applications to Engineering and Computer Science, Prentice Hall
5. S. Pirzada, An Introduction to Graph Theory, Universities Press, Orient Blackswan, 2012
6. W. T. Tutte, Graph Theory, Cambridge University Press.
7. D. B. West, Introduction to Graph Theory, Prentice Hall

Chapter 1

Coloring of Graphs

INTRODUCTION

Graph coloring originated with the celebrated **Four Color Problem**. In 1852, Francis Guthrie observed that four colors appeared sufficient to color the regions of a map so that adjacent regions received different colors. The problem was later popularized by Augustus De Morgan and Arthur Cayley and became one of the central questions in graph theory.

Alfred Kempe proposed an influential proof in 1879. Although the proof was later shown to be incorrect, several of its ideas survived. In 1890, Percy Heawood used related techniques to prove the **Five Color Theorem** and to extend map-coloring ideas to surfaces of higher genus. Later developments shifted attention from planar maps to vertex and edge colorings of general graphs. Brooks' theorem (1941) related the chromatic number $\chi(G)$ to the maximum degree $\Delta(G)$, while Vizing's theorem (1964) showed that the edge chromatic number of a simple graph is either $\Delta(G)$ or $\Delta(G) + 1$.

The Four Color Theorem was finally proved by Appel and Haken in 1976 using computer-assisted methods. The ideas developed around the problem—proper colorings, critical graphs, degree bounds, and structural characterizations—remain fundamental in modern graph theory.

1.1 Vertex Coloring

1.1.1 Basic Definitions

Definition 1.1.1 (Vertex Coloring). Let $G = (V, E)$ be an undirected graph. A **proper vertex coloring** with k colors is a function $c : V \rightarrow \{1, 2, \dots, k\}$ such that $uv \in E$ implies $c(u) \neq c(v)$. A proper coloring using at most k colors is called a **k -coloring**.

The color labels themselves are irrelevant; positive integers are used only for convenience.

Definition 1.1.2 (Chromatic Number). The **chromatic number** of G , denoted by $\chi(G)$, is the minimum number of colors required for a proper coloring; equivalently,

$$\chi(G) = \min\{k : G \text{ admits a proper } k\text{-coloring}\}.$$

Elementary Examples

- If G has no edges and at least one vertex, then $\chi(G) = 1$.
- If G contains at least one edge, then $\chi(G) \geq 2$.
- For the complete graph K_n , $\chi(K_n) = n$.

- For the cycle C_n ($n \geq 3$),

$$\chi(C_n) = \begin{cases} 2, & n \text{ even,} \\ 3, & n \text{ odd.} \end{cases}$$

Proof for cycles. Label the vertices $v_1, \dots, v_n$ consecutively. If n is even, alternating two colors around the cycle gives a proper coloring, and since C_n contains an edge, $\chi(C_n) = 2$. If n is odd, alternating two colors forces v_n and v_1 to receive the same color although they are adjacent, so two colors are impossible; three colors suffice. Hence $\chi(C_n) = 3$. ■

1.1.2 Trees and Bipartite Graphs

Theorem 1.1.3. *Every tree with at least two vertices is 2-chromatic.*

Proof. Let T be a tree with at least two vertices. Since T contains an edge, $\chi(T) \geq 2$. Choose a root $r \in V(T)$ and color each vertex u according to the parity of its distance from r : use color 1 if $d(r, u)$ is even and color 2 if it is odd.

If $xy \in E(T)$, then the unique paths from r to x and r to y differ in length by exactly one. Thus $d(r, x)$ and $d(r, y)$ have opposite parity, so adjacent vertices receive different colors. Hence T is 2-colorable and therefore $\chi(T) = 2$. ■

Theorem 1.1.4. *A graph G is 2-colorable if and only if it contains no odd cycle.*

Proof. If G is 2-colorable, the colors must alternate around every cycle; therefore every cycle has even length.

Conversely, suppose G has no odd cycle. It suffices to consider each connected component separately. Choose a root r in a connected component and define

$$V_1 = \{u : d(r, u) \text{ is even}\}, \quad V_2 = \{u : d(r, u) \text{ is odd}\}.$$

No two vertices in the same set can be adjacent, for otherwise the corresponding shortest paths from r , together with that edge, would create an odd cycle. Hence every edge joins V_1 to V_2 , so coloring V_1 and V_2 with two different colors gives a proper 2-coloring. ■

Theorem 1.1.5. *A nonempty graph G is 2-chromatic if and only if G is bipartite.*

Proof. If G is bipartite with bipartition $V(G) = V_1 \cup V_2$, color V_1 with color 1 and V_2 with color 2. Since G is nonempty, it contains an edge, so $\chi(G) = 2$.

Conversely, if G has a proper 2-coloring, let V_1 and V_2 be the sets of vertices receiving the two colors. No two vertices within the same set are adjacent, so every edge joins V_1 to V_2 . Hence G is bipartite. ■

Remark 1.1.6. Thus, for a nonempty graph,

$$G \text{ is bipartite} \iff G \text{ is 2-colorable} \iff G \text{ has no odd cycle.}$$

Theorem 1.1.7 (A General Upper Bound). *For every graph G , $\chi(G) \leq \Delta(G) + 1$.*

Proof. We induct on $n = |V(G)|$. The case $n = 1$ is immediate. Assume the result holds for graphs with $n - 1$ vertices, and choose $v \in V(G)$. By induction,

$$\chi(G - v) \leq \Delta(G - v) + 1 \leq \Delta(G) + 1.$$

Thus $G - v$ can be colored with at most $\Delta(G) + 1$ colors. Since v has at most $\Delta(G)$ neighbors, at most $\Delta(G)$ colors are forbidden at v , leaving at least one available color. Hence the coloring extends to G , proving $\chi(G) \leq \Delta(G) + 1$. ■

1.1.3 Color-Critical Graphs

Definition 1.1.8 (Color-Critical Graph). A graph G is k -critical if $\chi(G) = k$ and $\chi(G - v) = k - 1$ for every $v \in V(G)$.

Theorem 1.1.9. If G is k -critical, then:

- (a) G is connected;
- (b) $\delta(G) \geq k - 1$;
- (c) there do not exist proper subgraphs G_1, G_2 such that $G = G_1 \cup G_2$ and $G_1 \cap G_2$ is complete;
- (d) if $k > 1$, then $G - v$ is connected for every $v \in V(G)$.

Proof. a. Assume G is not connected. Since $\chi(G) = k$, there is a component G_1 of G such that $\chi(G_1) = k$. If v is any vertex of G which is not in G_1 , then G_1 is a component of the subgraph $G - v$. Therefore, $\chi(G - v) = \chi(G_1) = k$. This contradicts the fact that G is k -critical.

b. Assume to the contrary that v be a vertex of G so that $d(v) < k - 1$. Since G is k -critical, the subgraph $G - v$ has a $(k - 1)$ -colouring. As v has at most $k - 2$ neighbours, these neighbours use at most $k - 2$ colours in this $(k - 1)$ -colouring of $G - v$. Now, colour v with the unused colour and this gives a $(k - 1)$ -colouring of G . This contradicts the given assumption that $\chi(G) = k$. Hence every vertex v has degree at least $k - 1$.

c. Let $G = G_1 \cup G_2$, where G_1 and G_2 are subgraphs with $G_1 \cap G_2 = K_r$. Since G is k -critical, therefore G_1 and G_2 both have chromatic number at most $k - 1$. Consider a $(k - 1)$ -colouring of G_1 and a $(k - 1)$ -colouring of G_2 . As $G_1 \cap G_2$ is complete, in the overlap, every vertex in $G_1 \cap G_2$ has a different colour (in each of the $(k - 1)$ -colourings). This implies that colours in the $(k - 1)$ -colouring of G_2 can be rearranged such that it assigns the same colour to each vertex in $G_1 \cap G_2$, as is given by the colouring of G_1 . Combining the two colourings then produces a $(k - 1)$ -colouring of all of G . This is impossible, since $\chi(G) = k$. Thus no subgraphs of the type G_1 and G_2 exist.

d. Assume $G - v$ is disconnected, for some vertex v of G . Then $G - v$ has a subgraph H_1 and H_2 with $H_1 \cup H_2 = G - v$ and $H_1 \cap H_2 = \emptyset$. Let G_1 and G_2 be the subgraphs of G , where G_1 is induced by H_1 and v , while G_2 is induced by H_2 and v . Then $G = G_1 \cup G_2$ and $G_1 \cap G_2 = K_1$ (with K_1 as a single vertex). This contradicts (c) and thus $G - v$ is connected. ■

Corollary 1.1.10. Every k -chromatic graph has at least k vertices of degree at least $k - 1$.

Proof. Starting from a k -chromatic graph G , repeatedly delete vertices whenever the chromatic number remains k . This process ends with a k -critical subgraph H . By Theorem 1.1.9, $\delta(H) \geq k - 1$, and since every k -chromatic graph has at least k vertices, H contains at least k vertices of degree at least $k - 1$. Their degrees in G are at least as large. ■

Theorem 1.1.11. Let

$$k = \max\{\delta(G') : G' \text{ is a nonempty subgraph of } G\}.$$

Then $\chi(G) \leq k + 1$.

Proof. Let $\chi(G) = r$ and let H be an r -critical subgraph of G . By Theorem 1.1.9, $\delta(H) \geq r - 1$. Since H is a subgraph of G , the definition of k gives $\delta(H) \leq k$. Hence $r - 1 \leq k$, so $\chi(G) = r \leq k + 1$. ■

Theorem 1.1.12. Let G have degree sequence $d_1 \geq d_2 \geq \dots \geq d_n$. Then

$$\chi(G) \leq \max_{1 \leq i \leq n} \min\{i, d_i + 1\}.$$

Proof. Let $\chi(G) = k$. By Corollary 1.1.10, G has at least k vertices of degree at least $k - 1$, so $d_k \geq k - 1$. Hence $\min\{k, d_k + 1\} = k$, and therefore

$$\max_{1 \leq i \leq n} \min\{i, d_i + 1\} \geq k = \chi(G).$$

■

Algorithm 1.1.13 (The Greedy Coloring Algorithm). The **greedy coloring algorithm** is a simple method for graph vertex coloring that processes vertices in a specific order and assigns each vertex the smallest possible color not used by its already-colored neighbors.

1. Order the vertices $v_1, v_2, \dots, v_n$
2. For $i = 1$ to n :

Assign v_i the smallest color $c \in \mathbb{N}$ such that no adjacent vertex already has color c .

The algorithm produces a proper coloring but may not use the minimum number of colors (chromatic number $\chi(G)$). The number of colors used depends heavily on the vertex ordering. (See Figures 1.1 and 1.2).

Algorithm 1.1.14. A **Kempe chain** is a maximal connected component of a graph induced by vertices colored with two given colors.

The **Kempe chain argument** is a recoloring technique used in graph coloring where colors are swapped along an alternating path between two colors. By swapping these colors along the entire component, one obtains another proper coloring.

This method was famously used in:

- The proof of the **Five Color Theorem** for planar graphs
- Alfred Kempe's flawed but influential attempt at the **Four Color Theorem**
- Various graph coloring algorithms and proofs

How to use Kempe Chain Argument: Let G be a graph that uses at least two colors for its proper coloring. Choose two colors, say i & j and construct a subgraph H of G induced by the colors i & j . Let K be a connected component in H . Interchange the colors i and j in K . This produces a new coloring of G . In this process, the subgraph K of G is known as Kempe chain and this technique to obtain a new coloring of G is called the Kempe chain argument. (See Figures 1.1.3, 1.1.3).

Theorem 1.1.15 (Brooks' Theorem). *Let G be a connected graph with maximum degree Δ . If G is neither a complete graph nor an odd cycle, then*

$$\chi(G) \leq \Delta.$$

Proof. If $\Delta \leq 2$, then G is a path or a cycle. Paths and even cycles are 2-colorable, while odd cycles are excluded. Hence assume

$$\Delta \geq 3.$$

Case 1: G is not Δ -regular.

Choose a vertex v with $\deg(v) < \Delta$. Order the vertices by decreasing distance from v , so that v is colored last.

Color greedily using the colors $1, \dots, \Delta$. Every vertex $x \neq v$ has a neighbor closer to v , and this neighbor occurs later in the ordering. Thus, when x is colored, at most $\Delta - 1$ of its neighbors have already been colored. Hence a color is available for x . Finally, since $\deg(v) < \Delta$, a color is also available for v . Therefore

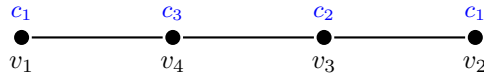
$$\chi(G) \leq \Delta.$$

We may now assume that G is Δ -regular.

Case 2: G has a cut-vertex.

1. Greedy Vertex Coloring

Processing Order: $v_1 \rightarrow v_2 \rightarrow v_3 \rightarrow v_4$



Result:

Uses 3 colors (c_1, c_2, c_3).

Optimal was 2.

2. Kempe Chain (Edge Coloring)

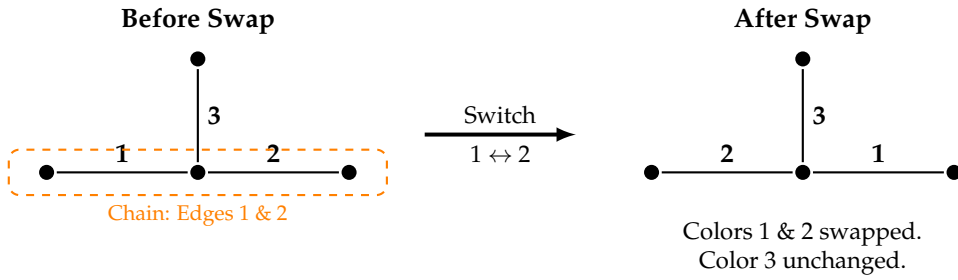


Figure 1.1: Top: Greedy Vertex Coloring. Bottom: Kempe Chain Argument on Edge Coloring.

Let v be a cut-vertex, and let

$$C_1, \dots, C_k$$

be the components of $G - v$. For each i , consider

$$G_i = G[C_i \cup \{v\}].$$

Since v has neighbors in at least two components,

$$\deg_{G_i}(v) < \Delta.$$

Applying the greedy argument of Case 1 to each G_i , we obtain a Δ -coloring of every G_i . By permuting the colors, we may assume that v receives the same color in every G_i . These colorings then combine to give a proper Δ -coloring of G .

Thus it remains to consider the case in which G is Δ -regular and 2-connected.

We first identify three useful vertices. We claim that there exist vertices u, v, w such that

$$uv, vw \in E(G), \quad uw \notin E(G),$$

and

$$G - \{u, w\}$$

is connected.

If G is 3-connected, choose any induced path $u - v - w$; such a path exists because G is connected and not complete. Since G is 3-connected, deleting u and w leaves a connected graph.

Suppose instead that G is 2-connected but not 3-connected. Then there are vertices v and x such that $G - \{v, x\}$ is disconnected. Hence $G - v$ is connected but has a cut-vertex.

Choose two distinct endblocks B_1 and B_2 of $G - v$, and let b_i be the cut-vertex through which B_i is attached to the rest of $G - v$. Since G itself has no cut-vertex, v must have a neighbor in each set

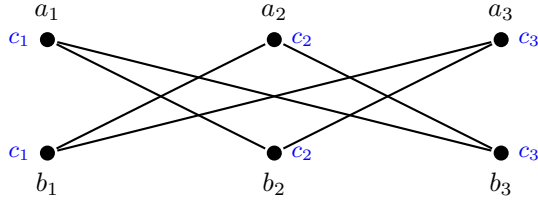
$$B_i - \{b_i\}.$$

Choose

$$u \in B_1 - \{b_1\}, \quad w \in B_2 - \{b_2\}$$

1. Greedy Coloring (Crown Graph)

Order: $a_1, b_1, a_2, b_2, a_3, b_3$



Result:

Used 3 Colors (c_1, c_2, c_3)
(Optimal is 2)

2. Kempe Chain (Edge Coloring)

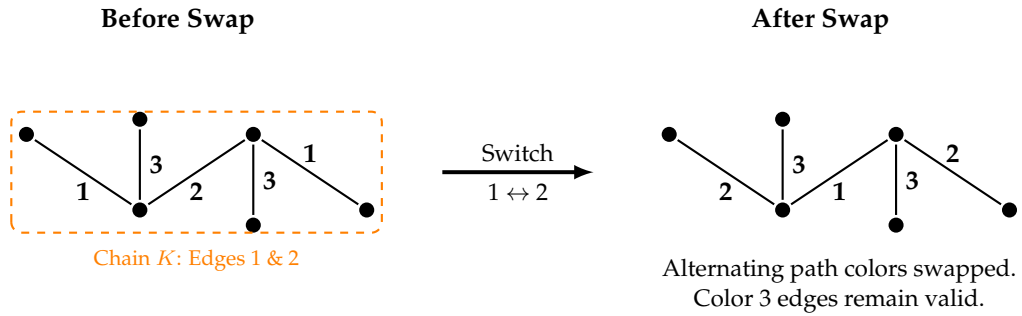


Figure 1.2: Top: Greedy coloring on a Crown Graph (6 vertices) forcing a sub-optimal 3rd color. Bottom: Kempe Chain argument on a larger path, swapping edge colors 1 and 2.

such that

$$uv, vw \in E(G).$$

Because u and w belong to distinct endblocks, they are not adjacent.

Moreover, u and w are non-cut vertices lying in distinct endblocks. Deleting them does not disconnect $G - v$. Thus

$$G - \{u, v, w\}$$

is connected. Since $\deg(v) = \Delta \geq 3$, the vertex v has a neighbor other than u and w , so adding v back shows that

$$G - \{u, w\}$$

is connected. This proves the claim.

Now give the nonadjacent vertices u and w the same color, say color 1.

Since

$$H = G - \{u, w\}$$

is connected, order the vertices of H by decreasing distance from v , with v last. Color them greedily with the colors $1, \dots, \Delta$.

Every vertex $x \neq v$ has a neighbor closer to v , hence a neighbor that is still uncolored when x is considered. Therefore at most $\Delta - 1$ of its neighbors have already been colored, and a color is available.

Finally, consider v . It has exactly Δ neighbors, but two of them, namely u and w , have the same color. Hence the neighbors of v use at most

$$\Delta - 1$$

distinct colors. Thus one of the Δ colors is available for v .

Consequently, G has a proper Δ -coloring, and therefore

$$\chi(G) \leq \Delta.$$

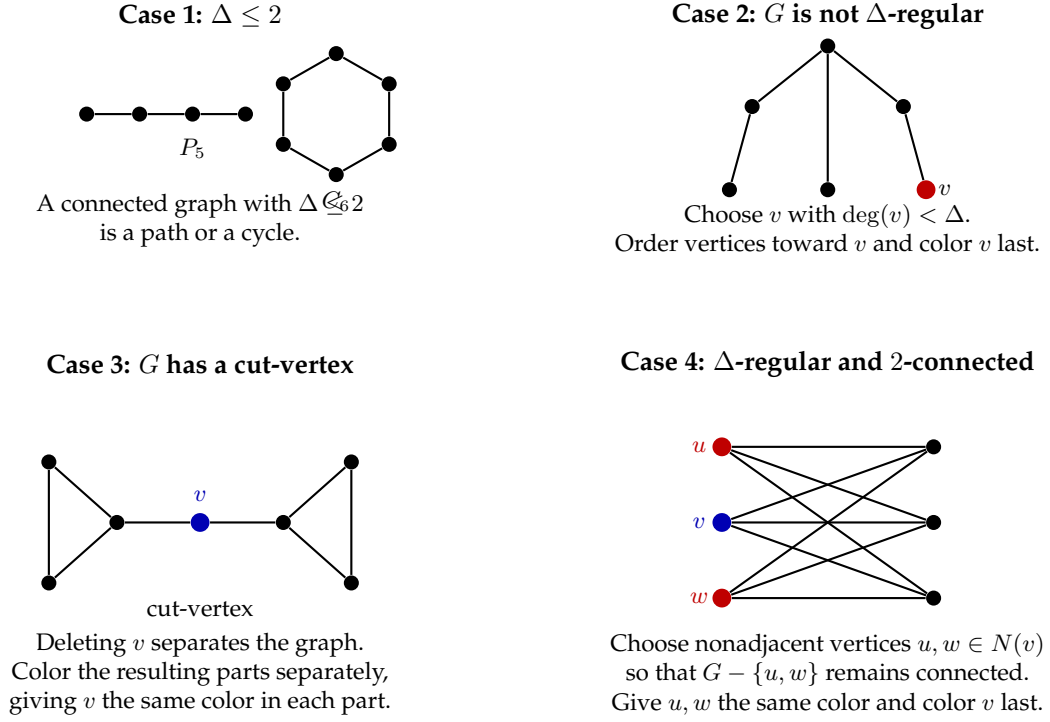


Figure 1.3: Graphical illustration of the main cases in the proof of Brooks' theorem.

Theorem 1.1.16. For any graph G of order n ,

$$\frac{n}{\chi(G)} \leq \alpha_0(G) \leq n - \chi(G) + 1.$$

Proof. Let $\chi(G) = k$. Then the vertices of G can be partitioned into k colour classes

$$V_1, V_2, \dots, V_k,$$

where each V_i is an independent set.

Let $|V_i| = n_i$. Since $\alpha_0(G)$ is the maximum size of an independent set in G , we have

$$n_i \leq \alpha_0(G) \quad \text{for each } i.$$

Therefore,

$$n = \sum_{i=1}^k n_i \leq \sum_{i=1}^k \alpha_0(G) = k\alpha_0(G).$$

Since $k = \chi(G)$, it follows that

$$\frac{n}{\chi(G)} \leq \alpha_0(G).$$

For the upper bound, let S be a maximum independent set of G , so that

$$|S| = \alpha_0(G).$$

Removing S can reduce the chromatic number by at most one, because all vertices of S can be assigned a single colour. Hence

$$\chi(G - S) \geq \chi(G) - 1.$$

On the other hand, $G - S$ has $n - \alpha_0(G)$ vertices, and therefore

$$\chi(G - S) \leq n - \alpha_0(G).$$

Combining these inequalities gives

$$\chi(G) - 1 \leq \chi(G - S) \leq n - \alpha_0(G).$$

Thus,

$$\chi(G) - 1 \leq n - \alpha_0(G),$$

and hence

$$\boxed{\alpha_0(G) \leq n - \chi(G) + 1}.$$

■

Theorem 1.1.17. For any graph of order n ,

$$2\sqrt{n} \leq \chi + \bar{\chi} \leq n + 1, \text{ and } n \leq \chi\bar{\chi} \leq \left(\frac{n+1}{2}\right)^2, \quad (1.1)$$

where $\chi = \chi(G)$ and $\bar{\chi} = \chi(\bar{G})$.

Proof. Let G be k -chromatic and let $v_1, v_2, \dots, v_k$ be the colour classes of G , where $|V_i| = n_i$. Then $\sum n_i = n$ and $\max n_i \geq n/k$. Since each V_i induces a complete subgraph of $\bar{G}$, $\bar{\chi} \geq \max n_i/n_i$ so that $\bar{\chi} \geq n/n_i$. Thus for each i , $n_i \geq n/\bar{\chi}$. Since the geometric mean of two positive numbers is always less or equal to their arithmetic mean, it follows that $\chi + \bar{\chi} \geq 2\sqrt{n}$.

To prove $\chi + \bar{\chi} \leq n + 1$, we induct on n . Clearly, the equality holds for $n = 1$. We assume that $\chi(G) + \bar{\chi}(G) \leq n$ for all graphs G having fewer than n vertices. Let H and $\bar{H}$ be complementary graphs with n vertices and let v be a vertex of H . Then $G = H - v$ and $\bar{G} = \bar{H} - v$ are complementary graphs with $n - 1$ vertices. Let the degree of v in H be d , so that degree of v in $\bar{H}$ is $n - d - 1$. Clearly, $\chi(H) \leq \chi(G) + 1$ and $\bar{\chi}(H) \leq \bar{\chi}(G) + 1$. If either $\chi(H) < \chi(G) + 1$ or $\bar{\chi}(H) < \bar{\chi}(G) + 1$, then $\chi(H) + \bar{\chi}(H) \leq n + 1$. If $\chi(H) = \chi(G) + 1$ and $\bar{\chi}(H) = \bar{\chi}(G) + 1$, then the removal of v from H , producing G , decreases the chromatic number, so that $d \geq \chi(G)$. Similarly, $n - d - 1 \geq \bar{\chi}(G)$. Thus, $\chi(G) + \bar{\chi}(G) \leq n - 1$. Therefore we always have $\chi(H) + \bar{\chi}(H) \leq n + 1$. Applying now the inequality $4\chi\bar{\chi} \leq (\chi + \bar{\chi})^2$, we get $\chi\bar{\chi} \leq \left(\frac{n+1}{2}\right)^2$. ■

2nd Proof. Let G be a graph of order n . Let $\chi = \chi(G)$ and $\bar{\chi} = \chi(\bar{G})$.

(i) $n \leq \chi\bar{\chi}$: Consider a χ -coloring of G with color classes $V_1, \dots, V_\chi$. Each V_i is an independent set in G . Consider a $\bar{\chi}$ -coloring of $\bar{G}$ with color classes $\bar{V}_1, \dots, \bar{V}_{\bar{\chi}}$. Each $\bar{V}_j$ is an independent set in $\bar{G}$, which means it is a clique in G . Every vertex $v \in V(G)$ belongs to some V_i and some $\bar{V}_j$. If u, v are two distinct vertices such that they are in the same color class V_i in G AND in the same color class $\bar{V}_j$ in $\bar{G}$, then u and v are not adjacent in G (because V_i is an independent set) and u and v are not adjacent in $\bar{G}$ (because $\bar{V}_j$ is an independent set). This is impossible, as any two distinct vertices must be adjacent in either G or $\bar{G}$. Therefore, each vertex must be assigned a unique ordered pair of colors (i, j) where $1 \leq i \leq \chi$ and $1 \leq j \leq \bar{\chi}$. The total number of such unique pairs is $\chi\bar{\chi}$. Since there are n vertices, $n \leq \chi\bar{\chi}$.

(ii) $2\sqrt{n} \leq \chi + \bar{\chi}$: Using the AM-GM inequality: $\frac{\chi + \bar{\chi}}{2} \geq \sqrt{\chi\bar{\chi}}$. From the previous part, $n \leq \chi\bar{\chi}$, so $\sqrt{n} \leq \sqrt{\chi\bar{\chi}}$. Combining these, $\frac{\chi + \bar{\chi}}{2} \geq \sqrt{n}$, which implies $2\sqrt{n} \leq \chi + \bar{\chi}$.

(iii) $\chi + \bar{\chi} \leq n + 1$: This is a standard result, often attributed to Nordhaus and Stewart (1963). The maximum value of $\chi(G) + \chi(\bar{G})$ is $n + 1$, achieved by complete graphs (K_n) and empty graphs ($\bar{K}_n$). For K_n , $\chi(K_n) = n$ and $\bar{\chi}(K_n) = 1$, so $\chi + \bar{\chi} = n + 1$. For $\bar{K}_n$, $\chi(\bar{K}_n) = 1$ and $\bar{\chi}(\bar{K}_n) = n$, so $\chi + \bar{\chi} = n + 1$.

(iv) $\chi\bar{\chi} \leq \left(\frac{n+1}{2}\right)^2$: From $\chi + \bar{\chi} \leq n + 1$, and using the property that for any real numbers a, b , $4ab \leq (a + b)^2$: $4\chi\bar{\chi} \leq (\chi + \bar{\chi})^2 \leq (n + 1)^2$. Dividing by 4, we get $\chi\bar{\chi} \leq \frac{(n+1)^2}{4} = \left(\frac{n+1}{2}\right)^2$. ■

1.2 Edge Colouring

An edge colouring of a non-empty graph G is a labelling $f : E(G) \rightarrow \{1, 2, \dots\}$ such that adjacent edges are assigned different colours. The labels are called *colours*. A k -edge colouring of G is an edge colouring of G that uses exactly k different colours, and in this case, G is said to be k -edge colourable.

This definition implies that a k -edge colouring of a graph $G(V, E)$ partitions the edge set E into k independent sets $E_1, E_2, \dots, E_k$ such that $E = E_1 \cup E_2 \cup \dots \cup E_k$. These independent sets E_i , for $1 \leq i \leq k$, are called the *colour classes*. The function $f : E(G) \rightarrow \{1, 2, \dots, k\}$ such that $f(e) = i$ for each $e \in E_i$ (where $1 \leq i \leq k$) is called the *colour function*. The minimum number k for which a k -colouring of G exists is called the *edge chromatic number* (or *edge chromatic index*) and is denoted by $\chi'(G)$.

We have the following observations:

1. If H is a subgraph of a graph G , then $\chi'(H) \leq \chi'(G)$.
2. For any graph G , $\chi'(G) \geq \Delta(G)$, where $\Delta(G)$ is the maximum degree of G .

If v is any vertex of G with degree $d(v) = \Delta(G)$, then the $\Delta(G)$ edges incident with v must all have different colours in any edge colouring of G .

3. For a cycle graph C_n of order n :

$$\chi'(C_n) = \begin{cases} 2, & \text{if } n \text{ is even,} \\ 3, & \text{if } n \text{ is odd.} \end{cases}$$

Definition: Let i be a colour used in an edge colouring of a graph G . If there is an edge coloured i incident to a vertex v of G , we say i is *present* at v . If there is no edge coloured i incident to v , we say i is *absent* from v .

Theorem 1.2.1 (König, 1916). For a non-empty bipartite graph G , $\chi'(G) = \Delta(G)$.

Proof. The proof is by induction on the number of edges of G . If G has only one edge, the result is trivial since $\chi'(G) = 1$ and $\Delta(G) = 1$.

Assume that G has more than one edge, and that the theorem holds for all non-empty bipartite graphs having fewer edges than G . Since we know $\Delta(G) \leq \chi'(G)$ (Observation 2), it suffices to prove that G has a $\Delta(G)$ -edge colouring. Let $\Delta(G) = k$.

Consider an arbitrary edge $e = uv$ in G . The graph $G - e$ is bipartite and has fewer edges than G . By the induction hypothesis, $G - e$ has a $\Delta(G - e)$ -edge colouring. Since $\Delta(G - e) \leq \Delta(G) = k$, $G - e$ has a k -edge colouring. We will show how to extend this k -edge colouring of $G - e$ to a k -edge colouring of G by assigning a colour to e .

Since $d_G(u) \leq k$ and $d_G(v) \leq k$, in the k -edge colouring of $G - e$, there is at least one colour absent at vertex u and at least one colour absent at vertex v .

1. **Case 1: There is a colour c that is absent at both u and v .** In this case, we can assign colour c to edge $e = uv$. This results in a k -edge colouring of G .
2. **Case 2: There is no colour absent at both u and v .** This implies that for every colour c , if c is absent at u , it must be present at v , and vice versa. Let i be a colour absent at v (and thus present at u) and j be a colour absent at u (and thus present at v).

Consider the subgraph $H(i, j)$ formed by all edges in $G - e$ coloured either i or j . Let K be the connected component of $H(i, j)$ containing u . We claim that v does not belong to K .

Proof of claim: Assume, for contradiction, that v belongs to K . Then there exists a path P in K between u and v . Since u and v are adjacent in G (via edge e), they belong to different partitions in the bipartite graph G . Therefore, any path between u and v in G must have odd length. The

path P is composed of alternately coloured edges i and j . Since colour i is present at u , the first edge of P incident to u must be coloured i . As P has an odd number of edges and alternates colours, its last edge incident to v must also be coloured i . This implies that colour i is present at v , which contradicts our initial assumption that i is absent from v . Thus, v does not belong to the Kempe chain K .

Now, we apply the Kempe chain argument to the component K . We interchange the colours i and j on all edges in K .

- At vertex u : Since $u \in K$ and the colour i was present at u , after the recolouring, colour j becomes present at u and colour i becomes absent at u .
- At vertex v : Since $v \notin K$, the colours of edges incident to v are unaffected by this interchange. Therefore, i remains absent from v .

After this recolouring, colour i is now absent at both u and v . We are now back to Case 1, where we can assign colour i to edge $e = uv$. This yields a k -edge colouring of G .

Therefore, in all cases, G has a $\Delta(G)$ -edge colouring. ■

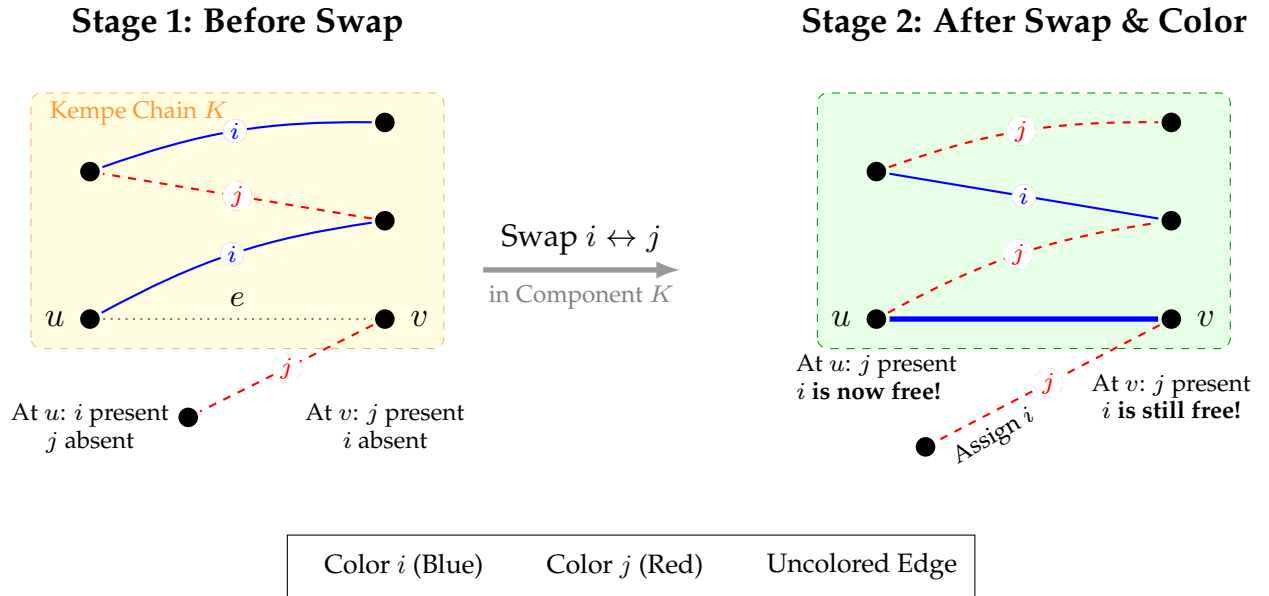


Figure 1.4: Visualization of the Kempe Chain swap. By drawing the highlight background first and using standard arrow syntax, this code is robust and compatible with standard LaTeX installations.

Theorem 1.2.2. If $G = K_n$ is a complete graph with n vertices, $n \geq 2$, then

$$\chi'(G) = \begin{cases} n-1, & \text{if } n \text{ is even,} \\ n, & \text{if } n \text{ is odd.} \end{cases}$$

Proof. (See Figure 1.5).

Since every vertex of K_n has degree $n-1$, we have $\Delta(K_n) = n-1$. Hence, $\chi'(K_n) \geq n-1$.

Case 1: n is odd. Suppose that K_n can be edge-coloured with $n-1$ colours. Each colour class is a matching, and since n is odd, a matching contains at most $(n-1)/2$ edges. Therefore, $n-1$ colours can colour at most

$$(n-1) \frac{n-1}{2} = \frac{(n-1)^2}{2}$$

edges.

However,

$$|E(K_n)| = \frac{n(n-1)}{2} > \frac{(n-1)^2}{2}.$$

This is a contradiction. Hence $\chi'(K_n) \geq n$.

To show that n colours are sufficient, label the vertices by $0, 1, \dots, n-1$, with arithmetic modulo n . Colour the edge joining i and j with the colour

$$i + j \pmod{n}.$$

If two edges ij and ik incident with the same vertex i received the same colour, then $i + j \equiv i + k \pmod{n}$, and hence $j = k$. Thus adjacent edges receive different colours. Therefore K_n has a proper n -edge colouring.

Hence

$$\chi'(K_n) = n$$

when n is odd.

Case 2: n is even. We already know that $\chi'(K_n) \geq n-1$. It remains to show that $n-1$ colours are sufficient.

Place $n-1$ vertices at the vertices of a regular $(n-1)$ -gon and place the remaining vertex at the centre. Join the central vertex to one outer vertex, and pair the remaining $n-2$ outer vertices symmetrically. These edges form a perfect matching, so they may all be assigned the same colour.

Now rotate this arrangement successively through the $n-1$ positions of the outer polygon. Each rotation gives a new perfect matching, and the resulting $n-1$ perfect matchings together contain every edge of K_n exactly once. Assigning a different colour to each matching gives a proper edge colouring of K_n with $n-1$ colours.

Therefore

$$\chi'(K_n) = n-1$$

when n is even.

Thus,

$$\chi'(K_n) = \begin{cases} n-1, & \text{if } n \text{ is even,} \\ n, & \text{if } n \text{ is odd.} \end{cases}$$

■

Theorem 1.2.3 (Vizing). For any graph G ,

$$\Delta(G) \leq \chi'(G) \leq \Delta(G) + 1.$$

Proof. Let G be a simple graph. We always have $\Delta(G) \leq \chi'(G)$. It remains to prove that $\chi'(G) \leq \Delta(G) + 1$.

We use induction on the number of edges of G . Let $\Delta(G) = k$. If G has only one edge, then $k = 1 = \chi'(G)$. Therefore, assume that G has more than one edge and that the result is true for all graphs having fewer edges than G .

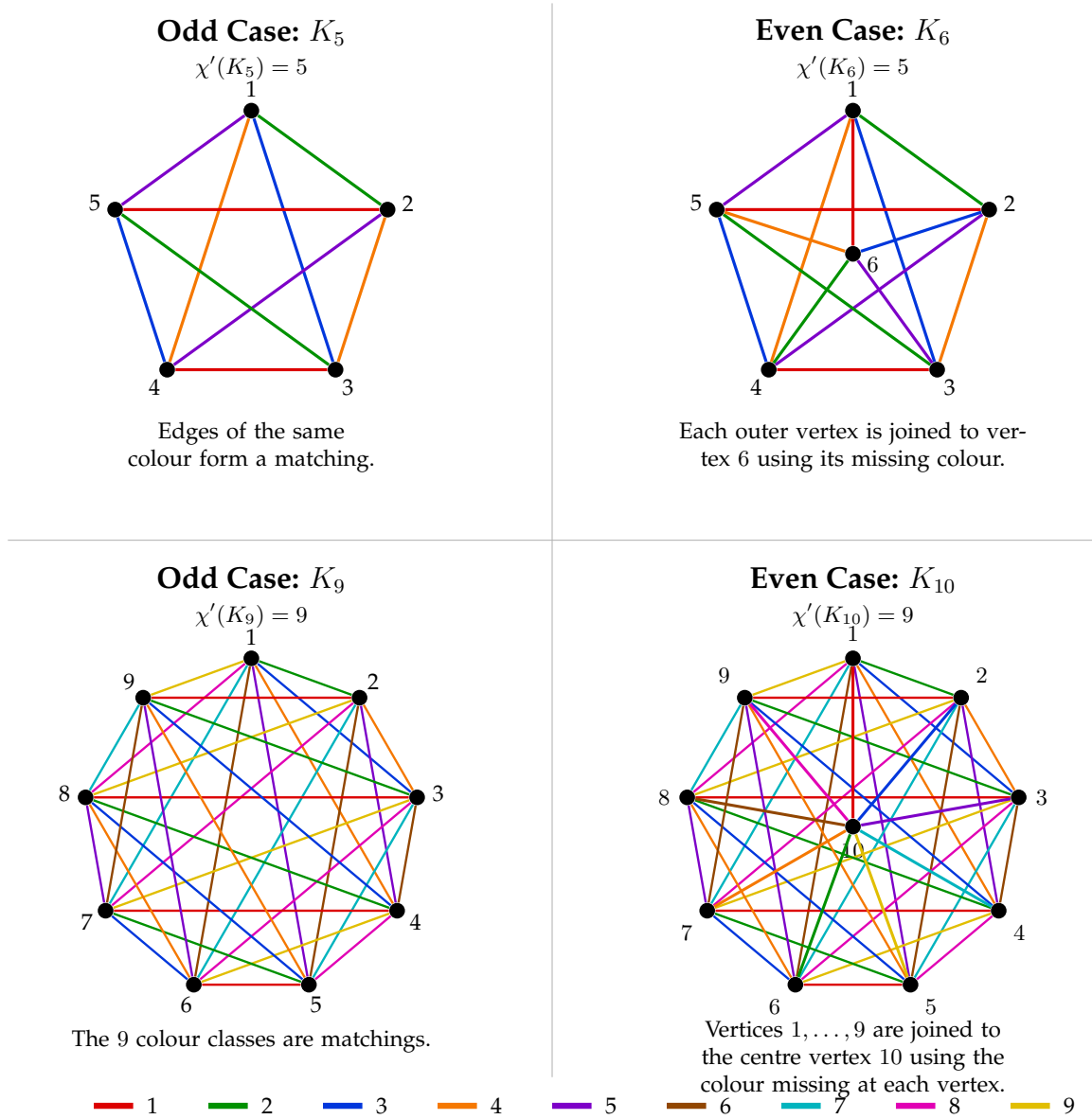
Let $e = v_1v_2$ be an edge of G . By the induction hypothesis, the subgraph $G - e$ has a $(k+1)$ -edge colouring. Let the colours used be $1, 2, \dots, k+1$.

Since $d(v_1) \leq k$ and $d(v_2) \leq k$, out of these $k+1$ colours, at least one colour is absent from v_1 and at least one colour is absent from v_2 . If there is a common colour absent from both v_1 and v_2 , then we use this colour to colour e and obtain a $(k+1)$ -edge colouring of G . Therefore, in this case, $\chi'(G) \leq k+1$.

We now assume that there is a colour, say 1, absent from v_1 but present at v_2 , and a colour, say 2, absent from v_2 but present at v_1 . We start from v_1 and v_2 and construct a sequence of distinct vertices $v_1, v_2, \dots, v_j$, where each v_i , for $i \geq 2$, is adjacent to v_1 .

Let v_1v_3 be coloured 2. This vertex v_3 exists because colour 2 is present at v_1 . We observe that not all the $k+1$ colours are present at v_3 , and assume that colour 3 is absent from v_3 . If colour 3 is present at v_1 , choose the vertex v_4 so that v_1v_4 is coloured 3. Continuing in this way, we choose a new colour i absent from v_i but present at v_1 , so that v_1v_{i+1} is the edge coloured i .

In this way, we obtain a sequence of vertices $v_1, v_2, \dots, v_{j-1}, v_j$ such that

Figure 1.5: Proper edge colourings of K_5 , K_6 , K_9 , and K_{10} .

- (a) v_i is adjacent to v_1 for each $i > 1$;
- (b) colour i is absent from v_i for $i = 1, 2, \dots, j-1$;
- (c) the edge v_1v_{i+1} is coloured i for $i = 1, 2, \dots, j-1$.

Since $d(v_1) \leq k$, condition (a) implies that such a sequence has at most $k+1$ terms; that is, $j \leq k+1$. Assume that $v_1, v_2, \dots, v_j$ is a longest such sequence; that is, it is not possible to find a new colour j , absent from v_j , together with a new neighbour v_{j+1} of v_1 such that v_1v_{j+1} is coloured j .

We first assume that, for some colour j absent from v_j , there is no edge of colour j incident with v_1 . We colour the edge $e = v_1v_2$ by 2, and then recolour the edges v_1v_i by i , for $i = 3, \dots, j-1$. Since colour i is absent from v_i for each $i = 2, \dots, j-1$, this gives a $(k+1)$ -edge colouring of the subgraph $G - v_1v_j$. Since colour j is absent from both v_1 and v_j , we colour v_1v_j by j . This gives a $(k+1)$ -edge colouring of G .

Now assume that whenever colour j is absent from v_j , it is present at v_1 . If v_1v_{j+1} is an edge joining v_1 to a new neighbour v_{j+1} and is coloured j , then we can extend our sequence to $v_1, v_2, \dots, v_j, v_{j+1}$, contradicting the assumption that $v_1, v_2, \dots, v_j$ is a longest sequence. Thus one of the edges $v_1v_3, \dots, v_1v_{j-1}$ must be coloured j ; say v_1v_ℓ , where $3 \leq \ell \leq j-1$.

Now colour $e = v_1v_2$ by 2, and for $i = 3, \dots, \ell-1$, recolour each edge v_1v_i by i , while leaving the

colours of the edges v_1v_i , for $i = \ell + 1, \dots, j$, unchanged. Removing colour j from v_1v_ℓ , we obtain a $(k + 1)$ -edge colouring of the edge-deleted subgraph $G - v_1v_\ell$.

Let $H(1, j)$ represent the subgraph of G induced by the edges coloured 1 or j in this partial colouring of G . Since the degree of every vertex in $H(1, j)$ is at most 2, each component of $H(1, j)$ is either a path or a cycle. Since colour 1 is absent from v_1 and colour j is absent from both v_j and v_ℓ , it follows that these three vertices do not all belong to the same connected component of $H(1, j)$. Therefore, if K and L represent the corresponding Kempe chains containing v_j and v_ℓ , respectively, then either $v_1 \notin K$ or $v_1 \notin L$.

Suppose that $v_1 \notin L$. Interchanging colours 1 and j on L , the Kempe chain argument gives a $(k + 1)$ -edge colouring of $G - v_1v_\ell$ in which colour 1 is absent from both v_1 and v_ℓ . Colouring v_1v_ℓ by 1 gives a $(k + 1)$ -edge colouring of G .

Now suppose that $v_1 \notin K$. Colour the edge v_1v_ℓ by ℓ , recolour the edges v_1v_i by i , for $i = \ell, \dots, j - 1$, and remove colour $j - 1$ from v_1v_j . Then, from the definition of the sequence $v_1, v_2, \dots, v_j$, we obtain a $(k + 1)$ -edge colouring of $G - v_1v_j$ without affecting the two-coloured subgraph $H(1, j)$. Using the Kempe chain argument to interchange the colours on K , we obtain a $(k + 1)$ -edge colouring of $G - v_1v_j$ in which colour 1 is absent from both v_1 and v_j . Therefore, colouring v_1v_j by 1 gives a $(k + 1)$ -edge colouring of G .

Hence $\chi'(G) \leq k + 1 = \Delta(G) + 1$. Together with $\Delta(G) \leq \chi'(G)$, we obtain

$$\Delta(G) \leq \chi'(G) \leq \Delta(G) + 1.$$

■

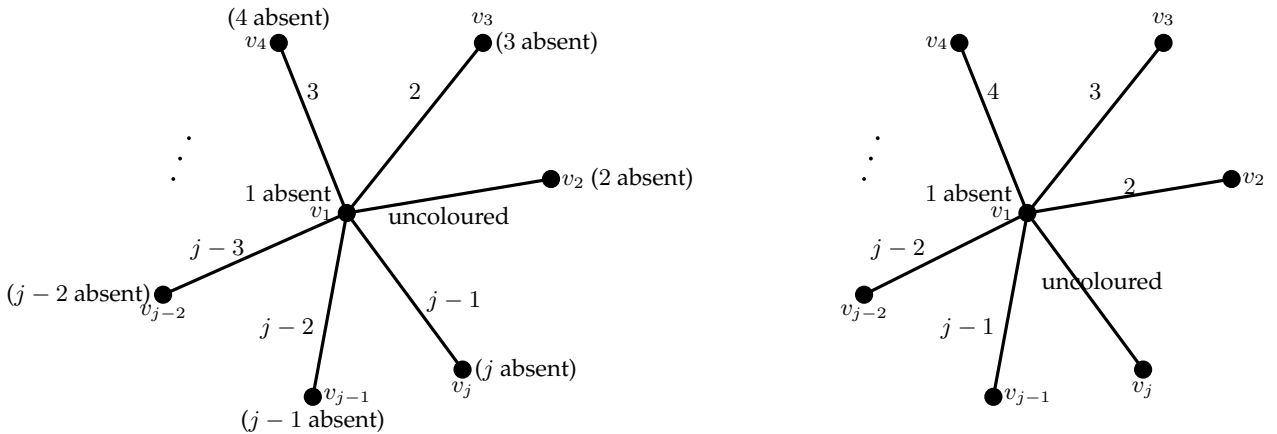


Figure 1.6: Illustration of Vizing's theorem.

Second Proof. The lower bound $\Delta(G) \leq \chi'(G)$ is straightforward. At any vertex v with degree $\Delta(G)$, there are $\Delta(G)$ edges incident with it. Since all these edges must receive distinct colours in a proper edge colouring, at least $\Delta(G)$ colours are necessary. Thus $\chi'(G) \geq \Delta(G)$.

The more challenging part is to prove the upper bound $\chi'(G) \leq \Delta(G) + 1$. This part of the proof typically involves a constructive algorithm or an inductive argument. Here, we provide a high-level constructive proof sketch.

Let $\Delta = \Delta(G)$. We want to show that G can be $(\Delta + 1)$ -edge coloured. We proceed by induction on the number m of edges of G . The base case is trivial for $m = 0$ or $m = 1$.

Assume that any graph with fewer than m edges can be $(\Delta + 1)$ -edge coloured. Consider a graph G with m edges. Let $e = uv$ be an uncoloured edge of G . We try to colour e with a colour $c \in \{1, \dots, \Delta + 1\}$. If we can colour all the edges one by one, we obtain a $(\Delta + 1)$ -edge colouring.

Consider a partial edge colouring c' of G , where some edges may be uncoloured. For any vertex $x \in V(G)$, let $C(x)$ be the set of colours used on edges incident with x , and let

$$M(x) = \{1, \dots, \Delta + 1\} \setminus C(x)$$

be the set of colours missing at x . Since the degree of every vertex is at most Δ and we have $\Delta + 1$ available colours, every vertex x has at least one missing colour; that is, $M(x) \neq \emptyset$.

Now let $e_0 = uv$ be an uncoloured edge. We want to assign a colour to e_0 . If there is a colour $c \in M(u) \cap M(v)$, we can assign colour c to e_0 . If no such colour exists, then $M(u)$ and $M(v)$ are disjoint. Since $|M(u)| \geq 1$ and $|M(v)| \geq 1$, we have $|C(u)| \leq \Delta$ and $|C(v)| \leq \Delta$, while $|M(u)| \geq (\Delta + 1) - \Delta = 1$ and $|M(v)| \geq (\Delta + 1) - \Delta = 1$.

If $M(u) \cap M(v) = \emptyset$, we try to rearrange the colours to make a colour available for uv . Let $a \in M(u)$ and $b \in M(v)$. Since $a \notin M(v)$, there is an edge vw coloured a . Since $b \notin M(u)$, there is an edge ux coloured b . We consider an (a, b) -alternating path or cycle starting from v . Such a path has edges whose colours alternate between a and b .

The proof involves a technique called colour shifting or fan colouring. Let $a \in M(u)$ and $b \in M(v)$. Since $a \notin M(v)$, there exists a vertex w such that the edge vw is coloured a . Consider the path P starting at w whose edges alternate between colours a and b . If this path does not reach u , we can interchange colours a and b along P . This frees colour a at v and allows us to colour uv with a .

The essential point is that interchanging the two colours along an alternating path does not create a conflict. The argument considers the possible (a, b) -alternating paths and cycles arising from the vertices involved. Since $\Delta + 1$ colours are available while the maximum degree is only Δ , there is sufficient flexibility to rearrange the colours and make a colour available for the uncoloured edge.

By systematically applying this procedure, colouring one uncoloured edge at a time and using colour shifting whenever necessary, we obtain a proper $(\Delta + 1)$ -edge colouring of G . Hence

$$\chi'(G) \leq \Delta(G) + 1.$$

Together with $\Delta(G) \leq \chi'(G)$, this proves the result. ■

Theorem 1.2.4. *Let G be a graph of order n and let $\overline{G}$ be the complement of G . Then*

(a) *if n is even, then*

$$n - 1 \leq \chi'(G) + \chi'(\overline{G}) \leq 2(n - 1),$$

and

$$0 \leq \chi'(G)\chi'(\overline{G}) \leq (n - 1)^2;$$

(b) *if n is odd, then*

$$n \leq \chi'(G) + \chi'(\overline{G}) \leq 2n - 3,$$

and

$$0 \leq \chi'(G)\chi'(\overline{G}) \leq (n - 1)(n - 2).$$

Further, the bounds are best possible for every positive integer $n \neq 2$.

Proof. Let G be a graph of order n and let $\overline{G}$ be its complement. Clearly,

$$\Delta(G) \geq n - 1 - \Delta(\overline{G}),$$

and hence

$$\Delta(G) + \Delta(\overline{G}) \geq n - 1.$$

Since $\chi'(G) \geq \Delta(G)$ and $\chi'(\overline{G}) \geq \Delta(\overline{G})$, it follows that

$$\chi'(G) + \chi'(\overline{G}) \geq n - 1.$$

Part 1: Even n . Let n be even. Then

$$\chi'(G) + \chi'(\overline{G}) \geq n - 1$$

and

$$\chi'(G)\chi'(\overline{G}) \geq 0.$$

The lower bounds are attained when $G = K_n$ or $G = \overline{K_n}$.

For the upper bounds, both G and $\overline{G}$ are subgraphs of K_n . Since $\chi'(K_n) = n - 1$ for even n , we have

$$\chi'(G) \leq n - 1 \quad \text{and} \quad \chi'(\overline{G}) \leq n - 1.$$

Therefore,

$$\chi'(G) + \chi'(\overline{G}) \leq 2(n - 1)$$

and

$$\chi'(G)\chi'(\overline{G}) \leq (n - 1)^2.$$

These upper bounds are attained for suitable graphs, including the complete bipartite graph $K_{1,n-1}$ when $n \neq 2$.

Part 2: Odd n . Let n be odd. We claim that

$$\chi'(G) + \chi'(\overline{G}) \geq n.$$

For example, if $G = K_n$, then $\chi'(K_n) = n$, while $\overline{G}$ is the empty graph and has edge chromatic number 0. Thus

$$\chi'(G) + \chi'(\overline{G}) = n.$$

Similarly, the same value is obtained when G is the empty graph.

Clearly,

$$\chi'(G)\chi'(\overline{G}) \geq 0,$$

and this lower bound is also attained for complete or empty graphs.

We now prove the upper bounds

$$\chi'(G) + \chi'(\overline{G}) \leq 2n - 3$$

and

$$\chi'(G)\chi'(\overline{G}) \leq (n - 1)(n - 2).$$

For $n = 1$ and $n = 3$, the inequalities are immediate. Thus assume that $n \geq 5$. If

$$\Delta(G) + \Delta(\overline{G}) \leq 2n - 5,$$

then by Vizing's theorem,

$$\chi'(G) + \chi'(\overline{G}) \leq \Delta(G) + \Delta(\overline{G}) + 2 \leq 2n - 3.$$

Otherwise, we consider the following cases.

- (i) Suppose $\Delta(G) = n - 1$ and $\Delta(\overline{G}) = n - 2$. Then G has a pendant vertex v . Also, $\Delta(G - v) \geq n - 2$ and hence $\chi'(G - v) \geq n - 2$. Since $\overline{G} - v$ is a subgraph of K_{n-1} ,

$$\chi'(\overline{G} - v) \leq \chi'(K_{n-1}) = n - 2.$$

As $\overline{G}$ is the disjoint union of an isolated vertex and a subgraph of K_{n-1} , we obtain the required bound.

- (ii) Suppose $\Delta(G) = n - 2$ and $\Delta(\overline{G}) = n - 2$. Again, G has a pendant vertex v , and as before,

$$\chi'(G - v) \leq n - 2 \quad \text{and} \quad \chi'(\overline{G}) \leq n - 2.$$

Similarly, $\chi'(\overline{G}) = n - 2$. Thus

$$\chi'(G) + \chi'(\overline{G}) = 2n - 4 < 2n - 3.$$

(iii) Suppose $\Delta(G) = n - 1$ and $\Delta(\overline{G}) = n - 3$. In this case, G has a vertex v of degree 2, and so $\chi'(G - v) = n - 2$.

Let vu and vw be the edges incident with v in G , where $d(u) = n - 1$. In an $(n - 2)$ -edge colouring of $G - v$, change the colour i of an edge uu' to a new colour $n - 1$. Now colour vu by i and vw by $n - 1$. This gives an $(n - 1)$ -edge colouring of G . By Vizing's theorem,

$$\chi'(\overline{G}) \leq n - 2.$$

Hence

$$\chi'(G) + \chi'(\overline{G}) \leq 2n - 3.$$

Since

$$\chi'(G) + \chi'(\overline{G}) \leq 2n - 3,$$

the possible largest product under the stated bounds is

$$\chi'(G)\chi'(\overline{G}) \leq (n - 1)(n - 2).$$

The upper bounds are attained, for example, by suitable complete bipartite graphs such as $K_{1,n-1}$.

Thus all the stated inequalities follow. ■

A more complete argument for the upper bound when n is odd involves considering the possible maximum degrees of G and $\overline{G}$. By Vizing's theorem,

$$\chi'(G) \leq \Delta(G) + 1 \quad \text{and} \quad \chi'(\overline{G}) \leq \Delta(\overline{G}) + 1.$$

Hence

$$\chi'(G) + \chi'(\overline{G}) \leq \Delta(G) + \Delta(\overline{G}) + 2.$$

For every graph G of order n ,

$$\Delta(G) + \Delta(\overline{G}) \geq n - 1.$$

Also,

$$\Delta(G) + \Delta(\overline{G}) \leq 2n - 2.$$

The sharper bound $2n - 3$ for odd n requires a more detailed examination of the possible degree configurations in G and $\overline{G}$.

Similarly, the maximum value

$$\chi'(G)\chi'(\overline{G}) = (n - 1)(n - 2)$$

is obtained when the two edge chromatic numbers take the values $n - 1$ and $n - 2$, respectively. The determination of such sharp bounds belongs to the general study of graph invariants of complementary graphs.

1.3 Region Colouring (Map Colouring)

A region colouring of a planar graph is a labelling of its regions $f : R(G) \rightarrow \{1, 2, \dots\}$; the labels called *colours*, such that no two adjacent regions get the same colour. A k -region colouring of a planar graph G consists of k different colours and G is then called k -region colourable. From the definition, it follows that the k -region colouring of a planar graph G partitions the region set R into k independent sets $R_1, R_2, \dots, R_k$, so that $R = R_1 \cup R_2 \cup \dots \cup R_k$. The independent sets are called the *colour classes*, and the function $f : R(G) \rightarrow \{1, 2, \dots, k\}$ such that $f(r) = i$, for each $r \in R_i, 1 \leq i \leq k$, is called the *colour function*. The minimum number k for which there is a k -region colouring of the planar graph G is called the *region-chromatic number* of G , and is denoted by $\chi''(G)$. The colouring of regions is also called *map colouring*, because of the fact that in an atlas different countries are coloured such that countries with common boundaries are shown in different colours.

Theorem 1.3.1. 1. A planar graph G is k -region colourable if and only if its dual G^* is k -vertex colourable.
 2. If G is a plane connected graph without loops, then G has a k -vertex colouring if and only if its dual G^* has a k -region colouring.

Proof. a. Let the regions and edges of G be respectively denoted by $r_1, \dots, r_t$ and $e_1, \dots, e_m$. Let the vertices of G^* be $r_1^*, \dots, r_t^*$ and edges be $e_1^*, \dots, e_m^*$. Then the vertices and edges of G^* are in one-to-one correspondence with the regions and edges of G , and two vertices r^* and s^* in G^* are joined by an edge e^* if and only if the corresponding regions r and s in G have the corresponding edge e as a common edge on their boundary.

Let G be k -region colourable. We colour the vertices in G^* such that each vertex r^* in G^* gets the same colour as assigned to the region r in G . Since the vertices r^* and s^* are only adjacent in G^* if the corresponding regions r and s are adjacent in G , G^* is k -vertex colourable.

Conversely, let G^* be k -vertex colourable. Now colour the regions of G such that the region r in G gets the same colour as the vertex r^* in G^* . This gives a k -region colouring of G , since the regions r and s are adjacent in G only if the corresponding vertices r^* and s^* are adjacent in G^* .

b. Since G has no loops its dual G^* has no bridges, and therefore G^* is planar. Thus by (a), G^* is k -region colourable if and only if the double dual G^{**} is k -vertex colourable. Since G is connected, G is isomorphic to G^{**} and hence the result follows. ■

Theorem 1.3.2. A planar graph is bicolourable if and only if it is an Euler graph.

Proof. Suppose G is bicolourable. Around each vertex, the colours of the incident faces alternate. Hence every vertex has even degree, so G is an Euler graph.

Conversely, suppose G is an Euler graph. Then every vertex of G has even degree, so every face of the dual graph G^* has even length. Hence G^* is bipartite. Therefore its vertices can be coloured with two colours, which gives a two-colouring of the faces of G .

Thus G is bicolourable if and only if it is an Euler graph. ■

Second Proof. Suppose G is bicolourable, that is, its dual G^* is 2-vertex-colourable. So G^* is bipartite, so every face of G^* has even length. Thus, every vertex of the dual of G^* has even degree. Since the bidual G^{**} is isomorphic to G , every vertex of G has even degree. Hence G is an Euler graph.

Conversely, suppose G is an Euler graph. Since $G^{**} \cong G$, every vertex of G^{**} has even degree. Thus every face of G^* has even length, and hence G^* is bipartite. Therefore the faces of G can be coloured with two colours, so G is bicolourable. ■

Theorem 1.3.3. Every planar graph is 6-colourable.

Proof. We use induction on the number of vertices. Every planar graph has a vertex v of degree at most 5. By induction, $G - v$ is 6-colourable. Since v has at most 5 neighbours, at most 5 colours appear on them, so one of the 6 colours is available for v . Hence G is 6-colourable. ■

Theorem 1.3.4 (Heawood). Every planar graph is 5-colourable.

Proof. (See Figure 1.7). Let G be a planar graph with n vertices. We use induction on n , the order of G . The result is obvious for $n \leq 5$. So, let $n \geq 6$. Assume the result is true for all planar graphs with fewer than n vertices.

Let G' be the graph obtained from G by deleting the vertex v and removing all the edges incident with v . The graph G' with order $n - 1$ is clearly planar and by induction hypothesis is 5-colourable. Let the colours used to colour G' be c_1, c_2, c_3, c_4, c_5 . We know for a planar graph with $n \geq 6$ vertices, there exists a vertex, say v , such that $d(v) \leq 5$. Thus v has utmost five neighbours in G and all of these neighbours are the already coloured vertices in G' . If in G' less than five colours are used to colour these neighbours, then the 5-colouring of G is obtained by using the colouring for G' on all vertices, and by colouring v with the colour not used to colour the neighbours of v .

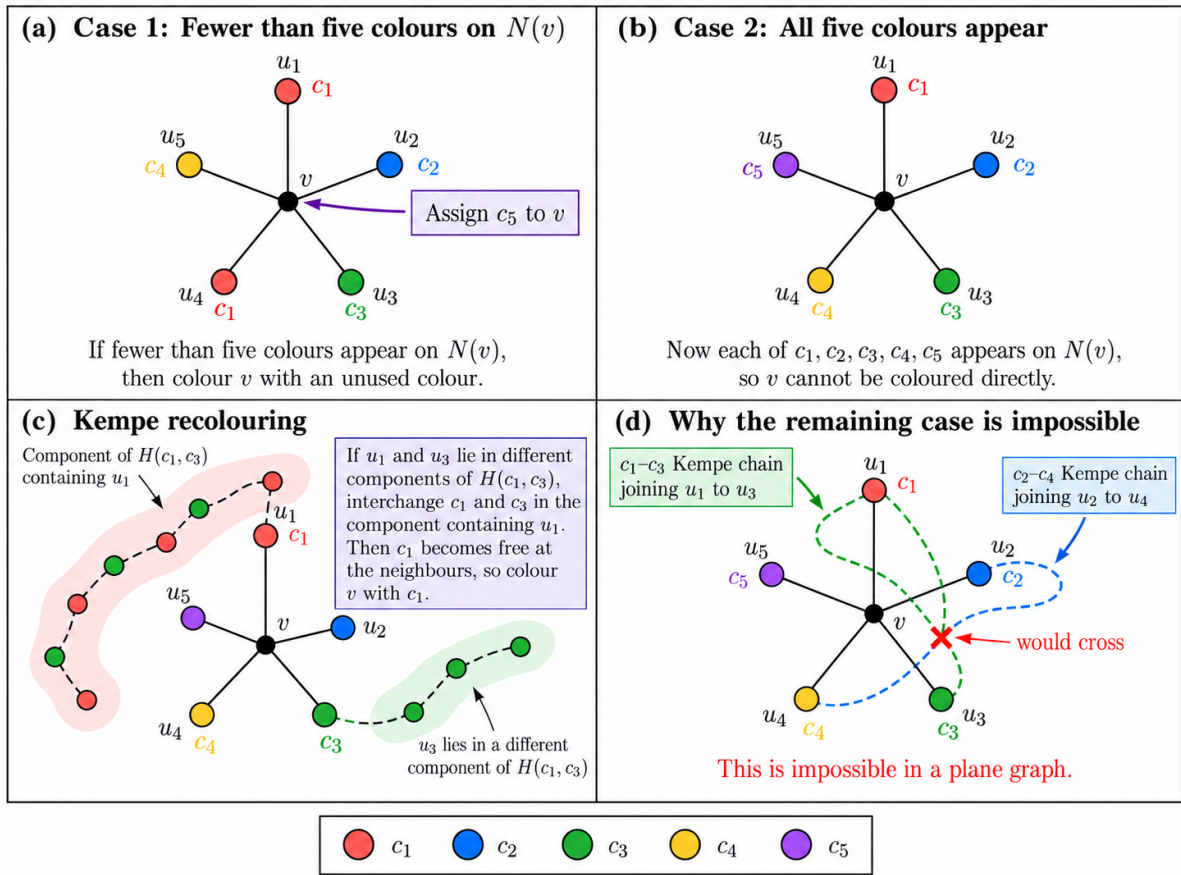


Figure 1.7: Heawood 5-color Theorem

Let all the five colours be used in G' , to colour the neighbours of v . This implies that there are exactly five neighbours of v , say u_1, u_2, u_3, u_4 and u_5 . Assume without loss of generality that u_i is coloured with c_i , for each i , $1 \leq i \leq 5$.

Consider all the vertices of G' that are coloured with colour c_1 and c_3 . If u_1 and u_3 are in different components of the Kempe subgraph $H(c_1, c_3)$ induced by those vertices coloured c_1 and c_3 , then using the Kempe chain argument to interchange the colours c_1 and c_3 in the component containing u_1 leaves c_1 unused on the set $\{u_1, u_2, u_3, u_4, u_5\}$. We colour v by c_1 to get a 5-colouring of G . Finally, if u_1 and u_3 are in the same component of $H(c_1, c_3)$, and u_2 and u_4 are in the same component of $H(c_2, c_4)$, then any component of $H(c_2, c_4)$ can not contain both u_2 and u_4 since otherwise the $c_1 - c_3$ Kempe chain from u_1 to u_3 crosses the $c_2 - c_4$ Kempe chain from u_2 to u_4 . This is impossible as the graph is a plane graph. Thus, we always have a pair of colors, say c_i and c_j , such that the induced graph $H(c_i, c_j)$ has a component K which does not contain both u_i and u_j . Interchanging the colors on any such component releases a color for the last uncolored vertex. ■

Theorem 1.3.5. *Every planar graph is four colourable if and only if every cubic bridgeless planar graph is 4-colourable.*

Proof. We observe that every planar graph is four colourable if and only if every bridgeless planar graph (without cut edges) is four colourable. This is because if G has a bridge e and G' is obtained from G by contracting e , then $\chi''(G') = \chi''(G)$ (as the elementary contraction of identifying the end vertices of a bridge affects neither the number of regions in the planar graph nor the adjacency of any of the regions). We now prove that every bridgeless planar graph is 4-colourable if and only if every cubic bridgeless planar graph is 4-colourable. If every bridgeless planar graph is 4-colourable, then evidently every cubic bridgeless planar graph is 4-colourable. Conversely, let all cubic bridgeless planar graphs be 4-colourable. Let G be a bridgeless planar graph. We obtain G' from G by performing the following operations. In case G has a vertex v of degree two, let xv and

yv be the edges incident with v . Subdivide xv at u and yv at w . Take two new vertices a and b and add the edges au, aw, bu, bw and ab . Now remove v (and uv and uw). In doing so, we have replaced v by a $K_4 - e$ and we see that each new vertex has degree three.

If G has a vertex v so that $d(v) \geq 4$, let $vx_1, vx_2, \dots, vx_d$ be the edges incident with v . Subdivide each vx_i producing a new vertex v_i , for $1 \leq i \leq d$. We then remove v and add the new edges $v_1v_2, v_2v_3, \dots, v_{d-1}v_d, v_dv_1$. Here we have replaced v by C_d and again the degree of each new vertex is three.

In both cases the graph G' formed is a bridgeless cubic graph. If these $K_4 - e$ and C_d introduced are contracted, we get the original graph G . Hence G is 4-colourable, since we have assumed that G' is 4-colourable. ■

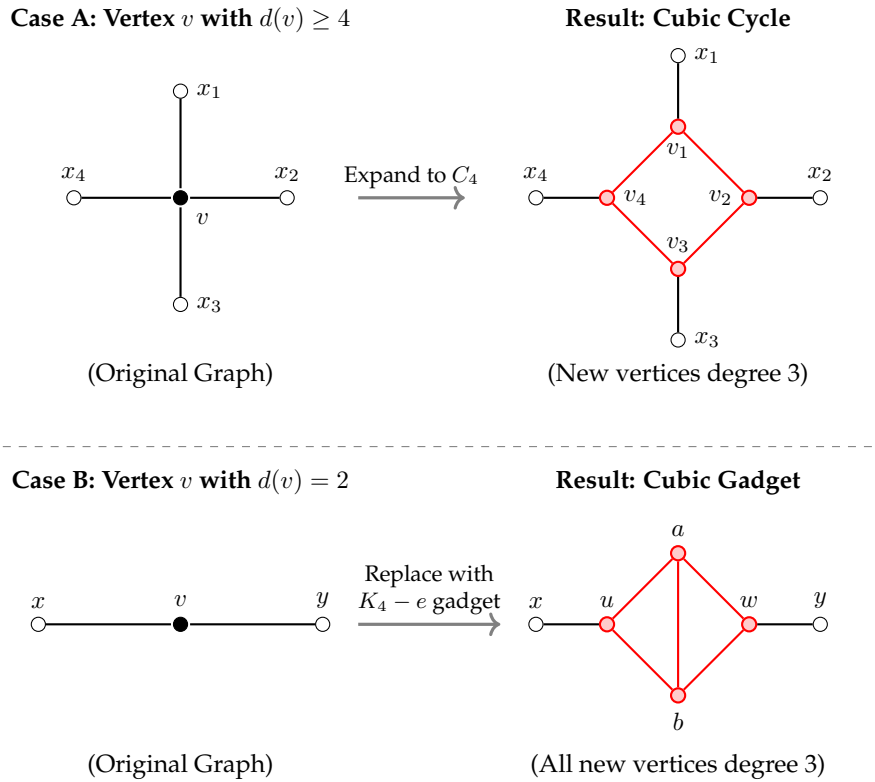


Figure 1.8: Visualisation of the operations described in Theorem 1.3.5. Top: Expanding a vertex of degree 4 into a cycle. Bottom: Replacing a vertex of degree 2 with a specialized gadget.

Theorem 1.3.6 (Tait's Theorem). *A cubic bridgeless, planar graph G is 4 region colorable if and only if, G is 3-edge colorable.*

Proof. Suppose G is 4-region colorable. Let the 4-region coloring of G be given, where the colors are the four elements of $\mathbb{Z}_2 \times \mathbb{Z}_2$. We now define an edge coloring of G . We define the color of an edge as the sum of colors of regions adjacent to it, noting that each edge lies on the boundary of at most two regions (because the given graph is cubic and bridgeless). Thus, we are done.

Conversely: [Refer to Figure above]. Suppose that G is 3-edge colorable. Let a 3-edge coloring of G be given using the colors $c_1 = (0, 1), c_2 = (1, 0), c_3 = (1, 1)$. This produces a partition of $E(G)$ into three perfect matchings E_1, E_2, E_3 where E_i is the set of edges colored c_i . Let G_1 be a spanning subgraph of G with edge set $E(G_1) = E_1 \cup E_2$ and G_2 be the spanning subgraph of G with edge set $E(G_2) = E_2 \cup E_3$. Then, both G_1 and G_2 are 2-regular spanning subgraphs of G so each of G_1 and G_2 is the disjoint union of even cycles. For $i = 1, 2$ each region of G_i is the union of regions of G . For each cycle C in the graph G_i , every region of G either lies interior or exterior to C . Furthermore, each edge of G belongs to a cycle C' of G_i and is on the boundary of two distinct regions of G_i , one of which lies interior to C' and the other exterior to C' .

We now define a 4-region coloring of G . We assign to region R of G the color (a_1, a_2) with $a_i \in \{0, 1\}$ for $i = 1, 2$, where $a_i = 0$ if R lies interior to even number of cycles in G_i and $a_i = 1$ if R lies interior to odd number of cycles in G_i .

Next, we shall show that this 4-region coloring of G is proper. Let R_1 and R_2 be two adjacent regions with common edge e . Then clearly either e is in G_1 or in G_2 or in both. Thus, e lies on a cycle in G_1 or a cycle in G_2 or both. Let C be a cycle in G_1 containing the edge e . Exactly, one of R_1 and R_2 lies interior to C , while for every other cycle C' , either both lie interior to C' or both exterior to C . Hence, the first coordinate of the colours of R_1 and R_2 are different. Therefore, the colors of every two adjacent regions of G differ in the first coordinate or the second coordinate or both. Hence, the 4-region coloring of G is proper. ■

Theorem 1.3.7 (Tait's Theorem). *A cubic bridgeless planar graph G is 4-region colorable if and only if G is 3-edge colorable.*

Proof. We prove the result in both directions.

Suppose first that G is 4-region colorable.

Let the four region colors be identified with the four elements of $\mathbb{Z}_2 \times \mathbb{Z}_2$, namely

$$(0, 0), \quad (0, 1), \quad (1, 0), \quad (1, 1).$$

We define a color for each edge of G as follows. If an edge e separates two regions having colors a and b , assign to e the color

$$a + b,$$

where the addition is taken in $\mathbb{Z}_2 \times \mathbb{Z}_2$.

Since the region coloring is proper, the two regions adjacent to an edge have different colors. Therefore, $a \neq b$, and hence

$$a + b \neq (0, 0).$$

Thus every edge receives one of the three nonzero elements

$$(0, 1), \quad (1, 0), \quad (1, 1).$$

We now show that the three edges incident with any vertex receive three different colors.

Let v be a vertex of G . Since G is cubic, exactly three edges are incident with v . Let the three regions occurring around v have colors a, b, c . Because the region coloring is proper, these three colors are pairwise distinct.

The three incident edges receive the colors

$$a + b, \quad b + c, \quad c + a.$$

These three colors are nonzero and pairwise distinct. For example, if

$$a + b = b + c,$$

then, since every element of $\mathbb{Z}_2 \times \mathbb{Z}_2$ is its own additive inverse, cancellation gives $a = c$, which is impossible. The same argument applies to the other pairs.

Hence the three edges incident with every vertex receive three distinct colors. Therefore, G is 3-edge colorable.

Conversely, suppose that G is 3-edge colorable.

Let the three edge colors be

$$c_1 = (0, 1), \quad c_2 = (1, 0), \quad c_3 = (1, 1).$$

Let

$$E_i = \{e \in E(G) : e \text{ has color } c_i\}, \quad i = 1, 2, 3.$$

Since G is cubic and the edge coloring is proper, exactly one edge of each color is incident with every vertex. Hence each set E_i is a perfect matching.

Now define two spanning subgraphs G_1 and G_2 of G by

$$E(G_1) = E_1 \cup E_2$$

and

$$E(G_2) = E_2 \cup E_3.$$

At each vertex, exactly two edges belong to G_1 , one from E_1 and one from E_2 . Therefore, every vertex has degree 2 in G_1 . Similarly, every vertex has degree 2 in G_2 .

Thus both G_1 and G_2 are 2-regular spanning subgraphs. Consequently, each of them is a disjoint union of cycles.

Moreover, the cycles are even. Indeed, in a cycle of G_1 , the edge colors must alternate between c_1 and c_2 , so the cycle has even length. Similarly, the colors on every cycle of G_2 alternate between c_2 and c_3 , and hence those cycles are also even.

Because G is planar, every cycle divides the plane into an interior region and an exterior region.

We now use the cycles of G_1 and G_2 to construct a 4-region coloring of G .

Let R be any region of G . Assign to R the ordered pair

$$(a_1, a_2),$$

where $a_1, a_2 \in \{0, 1\}$ are defined as follows:

$$a_1 = \begin{cases} 0, & \text{if } R \text{ lies inside an even number of cycles of } G_1, \\ 1, & \text{if } R \text{ lies inside an odd number of cycles of } G_1, \end{cases}$$

and

$$a_2 = \begin{cases} 0, & \text{if } R \text{ lies inside an even number of cycles of } G_2, \\ 1, & \text{if } R \text{ lies inside an odd number of cycles of } G_2. \end{cases}$$

Thus every region receives one of the four possible colors

$$(0, 0), \quad (0, 1), \quad (1, 0), \quad (1, 1).$$

It remains to prove that this region coloring is proper.

Let R_1 and R_2 be two adjacent regions of G , and let e be their common boundary edge.

Every edge of G belongs to at least one of the subgraphs G_1 and G_2 . More precisely,

$$e \in E_1 \Rightarrow e \in G_1,$$

$$e \in E_2 \Rightarrow e \in G_1 \cap G_2,$$

and

$$e \in E_3 \Rightarrow e \in G_2.$$

Suppose first that $e \in G_1$. Since G_1 is a disjoint union of cycles, the edge e lies on exactly one cycle C of G_1 .

The two regions R_1 and R_2 lie on opposite sides of the edge e . Hence one of them lies inside the cycle C and the other lies outside C .

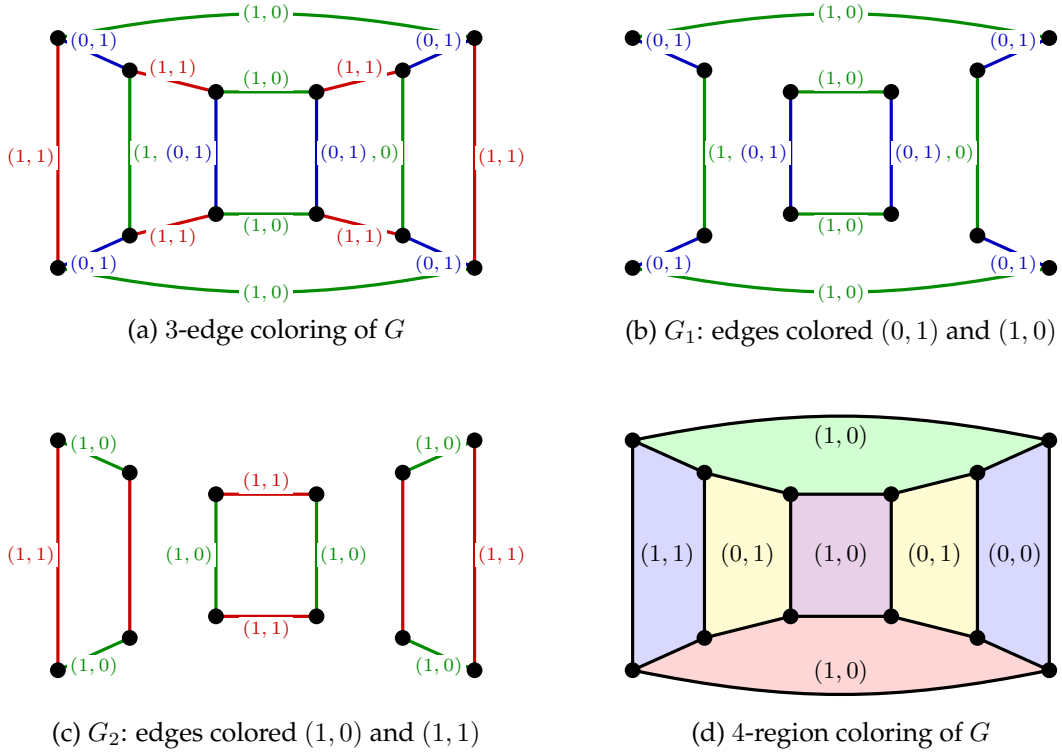
For every other cycle C' of G_1 , the regions R_1 and R_2 lie on the same side of C' . Therefore, when we pass from R_1 to R_2 across the edge e , the number of cycles of G_1 containing the region changes by exactly one.

Hence the parity changes. Therefore, the first coordinates of the colors of R_1 and R_2 are different:

$$a_1(R_1) \neq a_1(R_2).$$

Similarly, if $e \in G_2$, then e lies on a cycle of G_2 , and the same argument shows that

$$a_2(R_1) \neq a_2(R_2).$$

Figure 1.9: Tait's Theorem: $\chi'(G) = 3 \implies G$ is 4-region colorable.

Thus we have three cases.

If $e \in E_1$, then $e \in G_1$, so the first coordinates of the colors of R_1 and R_2 are different.

If $e \in E_3$, then $e \in G_2$, so the second coordinates are different.

If $e \in E_2$, then e belongs to both G_1 and G_2 , so both coordinates are different.

Therefore, in every case, two adjacent regions receive different colors. Hence the constructed 4-region coloring is proper.

Thus G is 4-region colorable.

Therefore, a cubic bridgeless planar graph G is 4-region colorable if and only if it is 3-edge colorable. ■

Figure 1.10: Tait's Theorem

1.4 Heawood Map-Colouring Theorem

Let S_p denote the orientable surface of genus p . Topologically, S_p may be regarded as a sphere with p handles attached. The chromatic number of S_p , denoted by $\chi(S_p)$, is defined to be the maximum chromatic number among all graphs that can be embedded on S_p . In particular, S_0 is the sphere, and the Four Colour Theorem asserts that $\chi(S_0) = 4$.

Definition 1.4.1 (Genus of a Graph). The *genus* of a graph G , denoted by $p(G)$, is the minimum non-negative integer p such that G can be embedded on the orientable surface S_p .

Thus $p(G)$ is the minimum number of handles that must be added to a sphere in order to obtain a surface on which G can be embedded. In particular, $p(G) = 0$ if and only if G is planar.

If a graph is embedded on S_p with n vertices, m edges, and f faces, then Euler's formula takes the form

$$n - m + f = 2 - 2p.$$

Theorem 1.4.2 (Genus of the Complete Graph). *For every integer $n \geq 3$, the orientable genus of the complete graph K_n is*

$$p(K_n) = \left\lceil \frac{(n-3)(n-4)}{12} \right\rceil.$$

Proof. Let K_n be embedded on an orientable surface S_p of genus p . Let m and f denote the numbers of edges and faces, respectively. Since K_n has n vertices, we have $m = n(n-1)/2$.

Every face boundary contains at least three edges, and each edge is incident with two faces. Hence $3f \leq 2m$, so that $f \leq 2m/3$.

By Euler's formula, $n - m + f = 2 - 2p$. Therefore

$$2 - 2p = n - m + f \leq n - m + \frac{2m}{3} = n - \frac{m}{3}.$$

Using $m = n(n-1)/2$, we get

$$2 - 2p \leq n - \frac{n(n-1)}{6}.$$

On simplification, this gives $12p \geq (n-3)(n-4)$. Hence $p \geq (n-3)(n-4)/12$. Since p is an integer,

$$p(K_n) \geq \left\lceil \frac{(n-3)(n-4)}{12} \right\rceil.$$

For the reverse inequality, Ringel and Youngs proved that K_n can be embedded on an orientable surface of genus $\lceil (n-3)(n-4)/12 \rceil$. Therefore

$$p(K_n) \leq \left\lceil \frac{(n-3)(n-4)}{12} \right\rceil.$$

Combining the two inequalities, we obtain

$$p(K_n) = \left\lceil \frac{(n-3)(n-4)}{12} \right\rceil.$$

■

Detailed proof of the Ringel-Youngs Theorem is as follows.

Theorem 1.4.3 (Ringel-Youngs). *For every integer $n \geq 3$, the orientable genus of the complete graph K_n is*

$$p(K_n) = \left\lceil \frac{(n-3)(n-4)}{12} \right\rceil.$$

Proof. Let K_n be embedded on an orientable surface S_p of genus p . Since K_n has n vertices and $m = n(n-1)/2$ edges, let f denote the number of faces of the embedding.

Every face boundary contains at least three edges, and every edge is incident with two faces. Hence $3f \leq 2m$. By Euler's formula, $n - m + f = 2 - 2p$, and therefore

$$2 - 2p \leq n - m + \frac{2m}{3} = n - \frac{m}{3}.$$

Substituting $m = n(n-1)/2$ and simplifying, we obtain $12p \geq (n-3)(n-4)$. Since p is an integer,

$$p(K_n) \geq \left\lceil \frac{(n-3)(n-4)}{12} \right\rceil.$$

It remains to prove that this lower bound is attainable. This is the main part of the Ringel-Youngs theorem.

For an embedding of a graph, define its *face excess* by

$$\varepsilon = \sum_F (|F| - 3),$$

where $|F|$ denotes the length of the boundary of the face F . Since the sum of the lengths of all face boundaries is $2m$, we have $3f + \varepsilon = 2m$. Consequently, $f = (2m - \varepsilon)/3$. Substitution in Euler's formula gives

$$p = \frac{(n-3)(n-4) + 2\varepsilon}{12}.$$

Thus, to obtain an embedding of minimum genus, it is sufficient to construct an embedding of K_n for which

$$\varepsilon = \frac{1}{2} \left(12 \left\lceil \frac{(n-3)(n-4)}{12} \right\rceil - (n-3)(n-4) \right).$$

The value of ε depends only on the residue class of n modulo 12. The required values are

$n \pmod{12}$	0	1	2	3	4	5	6	7	8	9	10	11
ε	0	3	5	0	0	5	3	0	2	3	3	2.

Ringel and Youngs constructed embeddings having exactly these face excesses by means of *current graphs*. We describe the construction.

A current graph is an oriented graph whose directed edges are labelled by elements of a cyclic group $\mathbb{Z}_q$. If an edge receives the current a in one direction, it receives the current $-a$ in the opposite direction. At each ordinary vertex the currents are chosen to satisfy Kirchhoff's law: the sum of the currents entering the vertex is 0 in $\mathbb{Z}_q$.

The boundary of a face of the current graph is read successively, and the sequence of currents so obtained is called its *log*. The current graphs used by Ringel and Youngs are chosen so that every nonzero element of the current group occurs in the log in the required way. From such a log a rotation system is obtained as follows.

If

$$(a_1, a_2, \dots, a_r)$$

is the log at the vertex 0, then the rotation at a vertex $i \in \mathbb{Z}_q$ is obtained by adding i to every numerical entry:

$$i : (i + a_1, i + a_2, \dots, i + a_r),$$

where the additions are taken modulo q . Thus the single current graph generates simultaneously the rotations at all the numbered vertices.

Because every nonzero group element occurs in the log, every difference $j - i$, with $i \neq j$, occurs. Hence the derived graph contains every edge joining two numbered vertices. In other words, the numbered vertices induce a complete graph.

At an ordinary vertex of degree three, Kirchhoff's law implies that the corresponding faces in the derived embedding are triangles. Thus, when there are no exceptional vertices, the derived rotation system gives a triangular embedding of a complete graph.

This immediately settles the residue classes $n \equiv 0, 3, 4, 7 \pmod{12}$. In these cases $(n-3)(n-4)$ is divisible by 12, so that $\varepsilon = 0$, and Ringel-Youngs current graphs give triangular embeddings of K_n . For such an embedding, $3f = 2m$, and hence

$$p = \frac{(n-3)(n-4)}{12}.$$

For the remaining residue classes a triangular embedding of K_n is arithmetically impossible, because $(n-3)(n-4)$ is not divisible by 12. Ringel and Youngs therefore introduced certain exceptional vertices, called *vortices*, into the current graph. A vortex does not satisfy the ordinary Kirchhoff condition and produces one or more large faces in the derived embedding. New vertices are inserted into these faces and joined to the vertices occurring on their boundaries.

The resulting derived graph is either K_n itself or a graph of the form $K_n - L$, where L is a small graph consisting of the few edges between the newly introduced vertices that have not yet been obtained. The missing edges of L are then inserted by local modifications of the embedding. These modifications consist of diagonal changes in quadrilateral faces and, when necessary, the addition of handles. Each added handle is used to insert the required missing adjacencies without creating crossings.

The constructions are arranged so that the resulting embeddings have precisely the following face excesses:

$n \pmod{12}$	ε
0, 3, 4, 7	0
8, 11	2
1, 6, 9, 10	3
2, 5	5.

Consequently, in every residue class the constructed embedding satisfies

$$p = \frac{(n-3)(n-4) + 2\varepsilon}{12} = \left\lceil \frac{(n-3)(n-4)}{12} \right\rceil.$$

Thus, for every $n \geq 3$, Ringel and Youngs obtained an embedding of K_n on an orientable surface of genus $\lceil (n-3)(n-4)/12 \rceil$. Therefore

$$p(K_n) \leq \left\lceil \frac{(n-3)(n-4)}{12} \right\rceil.$$

Combining this with the previously established lower bound gives

$$p(K_n) = \left\lceil \frac{(n-3)(n-4)}{12} \right\rceil.$$

■

The following inequality is due to Heawood [113].

Theorem 1.4.4. *The chromatic number of the orientable surface of positive genus p has the upper bound*

$$\chi(S_p) \leq \frac{7 + \sqrt{1 + 48p}}{2}, \quad p > 0.$$

Proof. Let G be an (n, m) graph embedded on S_p . Since any graph can be augmented to a triangulation of the same genus by adding edges without reducing χ , we assume G to be a triangulation. Let d' be the average degree of the vertices of G . Then n, m and f (the number of regions) are related by the equations

$$d'n = 2m = 3f.$$

Solving for m and f in terms of n and using Euler's equation $n - m + f = 2 - 2p$, we get

$$d' = \frac{12(p-1)}{n} + 6. \tag{1.2}$$

As $d' \leq n-1$, this gives $n-1 \geq \frac{12(p-1)}{n} + 6$. Solving for n and taking the positive square root, we obtain $n \geq \frac{7 + \sqrt{1 + 48p}}{2}$.

Let $H(p) = \frac{7 + \sqrt{1 + 48p}}{2}$. Then we show that $H(p)$ colours are sufficient to colour the vertices of G . If $n = H(p)$, obviously we have sufficient colours. In case $n > H(p)$, we have $\frac{1}{n} < \frac{1}{H(p)}$ then equation (1.2) gives obtain

$$d' < \frac{12(p-1)}{H(p)} + 6 \tag{1.3}$$

Now,

$$\begin{aligned}
 2H(p) &= 7 + \sqrt{1 + 48p} \\
 \text{or } (2H(p) - 7)^2 &= 1 + 48p \\
 \text{or } 4H^2(p) - 28H(p) + 49 &= 1 + 48p \\
 \text{or } H(p) [H(p) - 7] &= 12(p - 1) \\
 \text{or } H(p) - 7 &= \frac{12(p - 1)}{H(p)} \\
 \text{i.e. } H(p) - 1 &= \frac{12(p - 1)}{H(p)} + 6
 \end{aligned}$$

Hence, equation 1.3 gives, $d' < H(p) - 1$. Therefore, when $n > H(p)$, there is a vertex v of degree at most $H(p) - 2$. Identify v and any adjacent vertex by an elementary contraction to obtain a new graph G' . If $n' = n - 1 = H(p)$, then G' can be coloured in $H(p)$ colours. If $n' > H(p)$, repeat the argument and evidently we get an $H(p)$ colourable graph. It is then easy to see that the colouring of this graph induces a colouring of the preceding one in $H(p)$ colours, and so forth, so that G itself is $H(p)$ -colourable. Hence the result follows. ■

Theorem 1.4.5 (Heawood Map-Colouring Theorem). *For every positive integer p , the chromatic number of the orientable surface S_p of genus p is*

$$\chi(S_p) = \left\lfloor \frac{7 + \sqrt{1 + 48p}}{2} \right\rfloor.$$

Proof. By the upper-bound theorem, every graph G embeddable on S_p satisfies $\chi(G) \leq \left\lfloor \frac{7 + \sqrt{1 + 48p}}{2} \right\rfloor$. Hence $\chi(S_p) \leq \left\lfloor \frac{7 + \sqrt{1 + 48p}}{2} \right\rfloor$.

It remains to establish the reverse inequality. Recall that the orientable genus of the complete graph K_n is

$$p(K_n) = \left\lceil \frac{(n - 3)(n - 4)}{12} \right\rceil.$$

Let n be the largest integer such that K_n is embeddable on S_p . Then $p(K_n) \leq p$, whereas $p(K_{n+1}) > p$. Since p is an integer, the genus formula gives

$$\frac{(n - 3)(n - 4)}{12} \leq p < \frac{(n - 2)(n - 3)}{12}.$$

Thus $n^2 - 7n + 12 \leq 12p < n^2 - 5n + 6$. The first inequality yields $n \leq \frac{7 + \sqrt{1 + 48p}}{2}$, while the second yields $n > \frac{5 + \sqrt{1 + 48p}}{2}$. Therefore

$$\frac{5 + \sqrt{1 + 48p}}{2} < n \leq \frac{7 + \sqrt{1 + 48p}}{2}.$$

Since the two bounds differ by 1 and n is an integer,

$$n = \left\lfloor \frac{7 + \sqrt{1 + 48p}}{2} \right\rfloor.$$

Now K_n is embeddable on S_p and $\chi(K_n) = n$. Hence $\chi(S_p) \geq n = \left\lfloor \frac{7 + \sqrt{1 + 48p}}{2} \right\rfloor$.

Combining the two inequalities, we obtain

$$\chi(S_p) = \left\lfloor \frac{7 + \sqrt{1 + 48p}}{2} \right\rfloor.$$

■

1.5 Uniquely Colorable Graphs

Definition 1.5.1 (Uniquely Colorable Graph). A graph $G = (V, E)$ is said to be *uniquely k -colorable* if there is a unique partition of V into k independent sets. Equivalently, every proper k -coloring of G determines the same partition of V into color classes, apart from a permutation of the colors.

Theorem 1.5.2. *If G is uniquely k -colorable, then every vertex of G has degree at least $k - 1$.*

Proof. Let c be the unique k -coloring of G , and suppose that some vertex v has degree less than $k - 1$. Then at most $k - 2$ colors occur among the neighbours of v . Hence, among the k available colors, there is a color different from $c(v)$ which is not used on any neighbour of v . Recoloring v with this color gives another proper k -coloring of G , and the resulting partition into color classes is different from the original one. This contradicts the uniqueness of the k -coloring. Therefore, $d(v) \geq k - 1$ for every $v \in V(G)$. ■

Theorem 1.5.3. *In a uniquely k -colorable graph G , the subgraph induced by the union of any two color classes is connected.*

Proof. Let C_i and C_j be two distinct color classes in the unique k -coloring of G , and let $H = G[C_i \cup C_j]$. Suppose that H is disconnected, and let H_1 be one of its components. Interchange the colors i and j on all vertices of H_1 , leaving the colors of all other vertices unchanged. Since H_1 is a component of H , no edge of H joins a vertex of H_1 to a vertex of another component of H . Hence the interchange preserves proper coloring and produces another proper k -coloring of G inducing a different partition of the vertices into color classes. This contradicts the assumption that G is uniquely k -colorable. Therefore, $G[C_i \cup C_j]$ is connected. ■

Theorem 1.5.4. *A uniquely k -colorable graph is $(k - 1)$ -connected.*

Proof. Let G be uniquely k -colorable. If G is complete, the result is immediate. Suppose that G is not complete. Assume, to the contrary, that G is not $(k - 1)$ -connected. Then there exists a vertex cut X with $|X| \leq k - 2$ such that $G - X$ is disconnected. Since at most $k - 2$ vertices occur in X , at least two colors, say c_1 and c_2 , are not used on any vertex of X . Let C_1 and C_2 denote the corresponding color classes. By the preceding theorem, the subgraph $G[C_1 \cup C_2]$ is connected. As no vertex of X has color c_1 or c_2 , the entire subgraph $G[C_1 \cup C_2]$ lies in $G - X$ and therefore in a single component, say G_1 . Let G_2 be another component of $G - X$, and choose $v \in V(G_2)$. Its color is different from c_1 and c_2 . Consider the subgraph induced by the color of v together with c_1 , and interchange these two colors on the component containing v . This produces another proper k -coloring of G with a different color-class partition, contradicting uniqueness. Hence G is $(k - 1)$ -connected. ■

Theorem 1.5.5. *Every planar graph is four-colorable if and only if every Hamiltonian planar graph is four-colorable.*

Proof. If every planar graph is four-colorable, then every Hamiltonian planar graph is four-colorable, since every Hamiltonian planar graph is planar. Conversely, suppose that every Hamiltonian planar graph is four-colorable. Using the standard fact that every planar graph can be embedded as a subgraph of a Hamiltonian planar graph, let G be an arbitrary planar graph and let G' be a Hamiltonian planar graph containing G as a subgraph. By hypothesis, G' is four-colorable. Restricting such a coloring to G gives a proper four-coloring of G . Hence every planar graph is four-colorable. ■

Chapter 2

Matchings and Factors

Historical Note and Motivation

The theory of matchings began with **Julius Petersen's** 1891 study of regular graphs and evolved through **Philip Hall's** 1935 matching criteria for bipartite graphs. The field was revolutionized by **W.T. Tutte** in 1947, who characterized 1-factors in general graphs and later extended this to f -factors. These structural foundations were complemented by **Claude Berge's** algorithmic approach using augmenting paths and the **Erdős-Gallai** characterization of degree sequences. The study culminates in the factorization of K_n , a problem rooted in 19th-century tournament designs.

This chapter explores the fundamental tension between **local constraints** (vertex degrees) and **global structure** (factors). By mastering these topics, we learn how to determine when local demands can be universally satisfied, a concept essential for solving real-world assignment and scheduling problems. Theoretically, this section highlights the power of reduction, demonstrating how complex f -factor problems can be solved using the simpler mechanics of 1-factors.

2.1 Matchings

Definition 2.1.1 (Matching, Maximal, and Maximum). A **matching** in a graph is a set of independent edges. That is, a subset M of the edge set E of a graph $G(V, E)$ is a matching if no two edges of M have a common vertex.

- A matching M is said to be **maximal** if there is no matching M' strictly containing M . That is, M is maximal if it cannot be enlarged (you cannot add another edge without breaking the matching property).
- A matching M is said to be **maximum** if it has the largest possible cardinality. That is, M is maximum if there is no matching M' such that $|M'| > |M|$.

Definition 2.1.2 (Saturated and Unsaturated Vertices). Let M be a matching in a graph G .

- A vertex v in G is said to be **M -saturated** (or saturated by M) if there is an edge $e \in M$ incident with v .
- A vertex which is not incident with any edge of M is said to be **M -unsaturated**.

In other words, given a matching M in a graph G , the vertices belonging to the edges of M are M -saturated and the vertices not belonging to the edges of M are M -unsaturated.

Consider the graph shown in Figure 2.1 below. Clearly, $M = \{v_1v_2, v_3v_7, v_4v_5\}$ is a matching. The vertices $v_1, v_2, v_3, v_4, v_5, v_7$ are M -saturated, but v_6 is M -unsaturated.

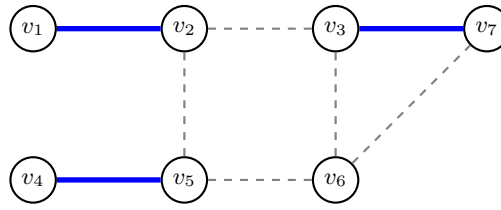


Figure 2.1: Visual representation of the matching M . The blue edges form the matching.

Definition 2.1.3 (Perfect Matching). A matching M in a graph G is said to be a **perfect matching** if M saturates every vertex of G .

Definition 2.1.4 (Complete Matching in Bipartite Graphs). If G is a bipartite graph with bipartition $V = V_1 \cup V_2$ and $|V_1| = n_1$, $|V_2| = n_2$, then a matching M of G saturating all the vertices in V_1 is called a **complete matching** (or V_1 matched into V_2).

Definition 2.1.5 (Alternating Path). Let H be a subgraph of a graph G . An H -**alternating path** (cycle) in G is a path (cycle) whose edges are alternately in $E(G) \setminus E(H)$ and $E(H)$.

Specifically, if M is a matching in a graph G , then an M -**alternating path** (cycle) in G is a path (cycle) whose edges are alternately in $E(G) \setminus M$ and M . That is, in an M -alternating path, the edges alternate between M -edges and non- M -edges.

Definition 2.1.6 (Augmenting Path). An M -alternating path whose end vertices are M -unsaturated is said to be an M -**augmenting path**.

Example 2.1.7. Examples to explain definitions given above.

- **Matching:** A set of edges without common vertices.
- **Maximal Matching:** A matching to which no more edges can be added.
- **Maximum Matching:** A matching with the largest possible number of edges.



Figure 2.2: **Maximal Matching.** Size = 1. We cannot add (A, B) or (C, D) because B and C are already busy.

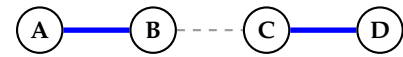


Figure 2.3: **Maximum Matching.** Size = 2. This is the largest possible matching for this graph.

Saturated vs. Unsaturated Vertices

A vertex is **Saturated** if it touches a matched edge (blue). Otherwise, it is **Unsaturated**.

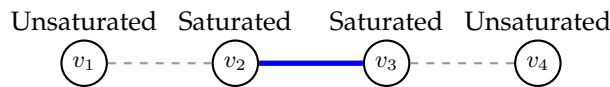


Figure 2.4: Vertices v_2, v_3 are saturated by the blue edge. v_1, v_4 are unsaturated.

Perfect Matching

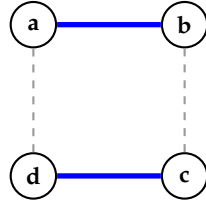
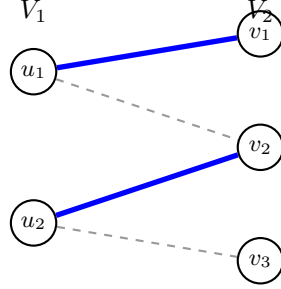
A matching that saturates **every** vertex in the graph.

Complete Matching (Bipartite)

In a bipartite graph $G = (V_1 \cup V_2, E)$, a matching is **complete from V_1 to V_2** if every vertex in V_1 is saturated.

Alternating and Augmenting Paths

- **Alternating Path:** Edges alternate between Non-Matching and Matching.
- **Augmenting Path:** An alternating path that starts and ends at unsaturated vertices.

Figure 2.5: A Perfect Matching on C_4 . All 4 vertices are covered.Figure 2.6: Complete Matching of V_1 into V_2 . All vertices in V_1 are blue-connected. Vertex v_3 is left over, which is fine.Figure 2.7: **Alternating Path** (1 - 2 - 3).
Starts Unsat(1) $\rightarrow$ Sat(3).Figure 2.8: **Augmenting Path** (1 - 2 - 3 - 4).
Starts Unsat(1) $\rightarrow$ Ends Unsat(4).

Theorem 2.1.8 (Berge). *A matching M of a graph G is maximum if and only if G contains no M -augmenting path.*

Proof. Suppose first that M is a maximum matching and that $P = v_1v_2 \cdots v_k$ is an M -augmenting path. Then the edges of P alternate between edges outside M and edges in M , with the first and last edges outside M . Thus $v_2v_3, v_4v_5, \dots, v_{k-2}v_{k-1} \in M$, while $v_1v_2, v_3v_4, \dots, v_{k-1}v_k \notin M$.

Replacing the edges of M on P by the remaining edges of P produces a matching

$$M_1 = [M - \{v_2v_3, \dots, v_{k-2}v_{k-1}\}] \cup \{v_1v_2, \dots, v_{k-1}v_k\},$$

with $|M_1| = |M| + 1$, contradicting the maximality of M . Hence no M -augmenting path exists.

Conversely, suppose that G contains no M -augmenting path. If M were not maximum, there would exist a matching M' with $|M'| > |M|$. Let H be the spanning subgraph of G whose edge set is $M \triangle M'$.

Since each vertex is incident with at most one edge of M and at most one edge of M' , every vertex of H has degree at most 2. Hence each component of H is an isolated vertex, a path, or a cycle, and its edges alternate between M and M' .

Every cycle contains the same number of edges from M and M' . Since $|M'| > |M|$, some path component must contain one more edge from M' than from M . Such a path begins and ends with edges of M' , and therefore its end vertices are unsaturated by M . Hence it is an M -augmenting path, a contradiction.

Therefore, M is maximum. ■

Theorem 2.1.9 (Hall's Marriage Theorem). *A bipartite graph $G = (X \cup Y, E)$ has a matching that covers all vertices in X if and only if for every $S \subseteq X$, $|N(S)| \geq |S|$.*

Proof. Assume a matching M covers all vertices in X . For any $S \subseteq X$, each vertex in S is matched by M to a distinct vertex in Y . Let $M(S)$ be this set of matched vertices in Y . Then $|M(S)| = |S|$. Since every vertex in S is connected to its matched partner, $M(S) \subseteq N(S)$. Thus, $|N(S)| \geq |M(S)| = |S|$.

Assume $|N(S)| \geq |S|$ for all $S \subseteq X$. We prove by contradiction that a matching covering X exists. Suppose no such matching exists. Let M be a maximum matching in G . Since M does not cover

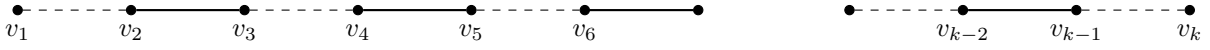


Figure 2.9:

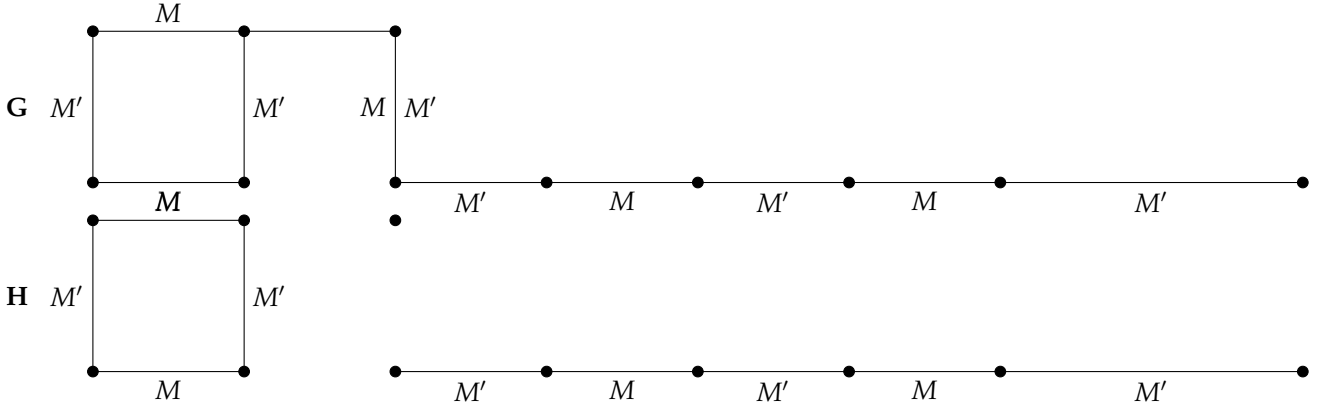


Figure 2.10:

X , there must be at least one unmatched vertex $u \in X$. By Berge's Lemma, since M is a maximum matching, there are no M -augmenting paths in G . Let A be the set of all vertices reachable from u by an M -alternating path. Let $X' = A \cap X$ and $Y' = A \cap Y$. Since u is unmatched and starts an M -alternating path, $u \in X'$. For any $y \in Y'$, it must be connected to some $x \in X'$ by an edge not in M . M must match this x to some $y' \in Y'$, otherwise we could extend the path to x to form an M -augmenting path from u to x , a contradiction (since x would be unmatched). Specifically, any vertex $x \in X'$ (other than u) must be matched by M to some $y \in Y'$. If x were unmatched, then an M -alternating path from u to x would be an M -augmenting path, a contradiction. Similarly, any vertex $y \in Y'$ must be matched by M to some $x \in X'$. If y were matched to $x \notin X'$, then combining the path from u to y (ending with an M -edge) with the edge (y, x) (which is in M) would form an M -alternating path from u to x ending with an M -edge. This would imply $x \in X'$, a contradiction. Therefore, all edges of M that are incident to A must connect a vertex in $X' \setminus \{u\}$ to a vertex in Y' . This implies that M provides a perfect matching between $X' \setminus \{u\}$ and Y' . Thus, $|X'| - 1 = |Y'|$.

Now consider the set $S = X'$. By construction of A , all neighbors of X' must be in Y' . Thus $N(X') \subseteq Y'$. From our previous deduction, $|Y'| = |X'| - 1$. Therefore, $|N(X')| \leq |Y'| = |X'| - 1$. This means $|N(X')| < |X'|$, which contradicts the given condition that $|N(S)| \geq |S|$ for all $S \subseteq X$. Our initial assumption that no matching covering X exists must be false. Hence, such a matching exists. ■

2nd Proof using Augmenting Paths. Assume a matching M covers all vertices in X . For any $S \subseteq X$, each $x \in S$ is matched to a distinct $y_x \in Y$. These y_x form the set $M(S)$. Clearly, $|M(S)| = |S|$ and $M(S) \subseteq N(S)$. Thus, $|N(S)| \geq |S|$.

Assume $|N(S)| \geq |S|$ for all $S \subseteq X$. We prove by contradiction. Suppose no matching covers X . Let M be a maximum matching. Since it doesn't cover X , there's an unmatched $u \in X$. By the given theorem (Berge's Lemma), since M is maximum, there are no M -augmenting paths.

Let A be the set of vertices reachable from u by M -alternating paths. Let $X_A = A \cap X$ and $Y_A = A \cap Y$. (See Figure 2.11). Since there are no M -augmenting paths, u is the only vertex in A that is M -unsaturated. Furthermore, if $y \in Y_A$ is matched by M , its matched partner x' must be in X_A . (If $x' \notin X_A$, then extending the M -alternating path from u to y with the edge $(y, x') \in M$ would make x' reachable, implying $x' \in X_A$, a contradiction). Thus, M perfectly matches Y_A with $X_A \setminus \{u\}$. Therefore, $|Y_A| = |X_A| - 1$.

Now, let $S = X_A$. We claim $N(S) \subseteq Y_A$. Take any $x \in S$ and its neighbor $y \in N(x)$. If $y \notin Y_A$, then the edge (x, y) cannot be in M (because if it were, then x would be matched to y , but $x \in X_A$ and $y \notin Y_A$ implies x is not matched within A , a contradiction to M matching $X_A \setminus \{u\}$ to Y_A). If

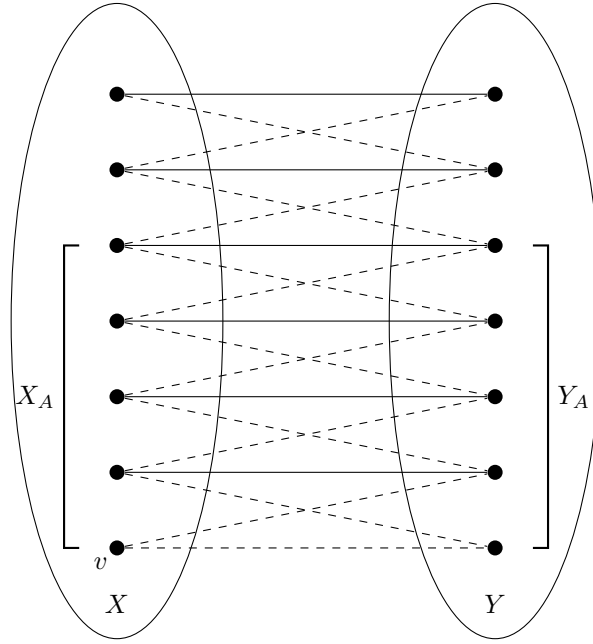


Figure 2.11: Bipartite graph illustrating sets $X_A \subseteq X$ and $Y_A \subseteq Y$ relative to a matching.

$(x, y) \notin M$, then combining the M -alternating path from u to x with (x, y) would make y reachable, implying $y \in Y_A$, a contradiction. Thus, $N(S) \subseteq Y_A$.

Combining these, $|N(S)| \leq |Y_A| = |X_A| - 1 = |S| - 1$. So, $|N(S)| < |S|$ for $S = X_A$. This contradicts our initial assumption. Therefore, a matching covering X must exist. ■

Theorem 2.1.10 (Hall's SDR Theorem). *Let $\mathcal{A} = \{A_i : 1 \leq i \leq r\}$ be a finite family of nonempty finite sets. Then $\mathcal{A}$ has a system of distinct representatives if and only if, for every subset $I \subseteq \{1, 2, \dots, r\}$,*

$$\left| \bigcup_{i \in I} A_i \right| \geq |I|.$$

Equivalently, the union of any k members of $\mathcal{A}$ contains at least k elements, for $1 \leq k \leq r$.

Proof. Construct a bipartite graph $G = (X \cup Y, E)$ by taking $X = \{1, 2, \dots, r\}$ and $Y = \bigcup_{i=1}^r A_i$, and joining $i \in X$ to $y \in Y$ precisely when $y \in A_i$.

A system of distinct representatives for $\mathcal{A}$ is exactly a matching in G that saturates every vertex of X . Indeed, choosing a representative $x_i \in A_i$ corresponds to choosing the edge ix_i , and the representatives are distinct precisely when the chosen edges form a matching.

By Hall's Marriage Theorem, such a matching exists if and only if $|N(S)| \geq |S|$ for every subset $S \subseteq X$. But, by the construction of G ,

$$N(S) = \bigcup_{i \in S} A_i.$$

Hence Hall's condition is equivalent to

$$\left| \bigcup_{i \in S} A_i \right| \geq |S|$$

for every $S \subseteq X$.

Therefore, $\mathcal{A}$ has an SDR if and only if the union of any k members of $\mathcal{A}$ contains at least k elements. ■

Theorem 2.1.11. *Let G be a bipartite graph with bipartite sets V_1 and V_2 of orders n_1 and n_2 , respectively, where $n_1 \leq n_2$. Suppose that G satisfies Hall's condition*

$$|S| \leq |N(S)|$$

for every subset $S \subseteq V_1$. Then G has at least

$$r(\delta, n_1) = \prod_{i=1}^{\min\{\delta, n_1\}} (\delta - i + 1)$$

complete matchings, where

$$\delta = \min\{d(v) : v \in V_1\}.$$

Proof. We proceed by induction on n_1 .

If $n_1 = 1$, then $V_1 = \{v_1\}$ and $d(v_1) = \delta$. Each edge incident with v_1 gives a complete matching. Hence there are δ complete matchings. Since $r(\delta, 1) = \delta$, the result holds.

Assume that the result is true for bipartite graphs whose first partite set has $n_1 - 1$ vertices. Let $v_1 \in V_1$. Since $d(v_1) \geq \delta$, there are at least δ choices for a vertex $u_1 \in N(v_1)$.

For each such choice of u_1 , consider the graph $G' = G - \{v_1, u_1\}$. Its first partite set is $V'_1 = V_1 - \{v_1\}$ and has order $n_1 - 1$. On deleting u_1 , a vertex of V'_1 can lose at most one incident edge. Therefore the minimum degree of the vertices of V'_1 in G' is at least $\delta - 1$.

By the induction hypothesis, G' has at least $r(\delta - 1, n_1 - 1)$ complete matchings. Each such matching, together with the edge $v_1 u_1$, gives a complete matching of G . Since there are at least δ choices for u_1 , the number of complete matchings of G is at least

$$\delta r(\delta - 1, n_1 - 1).$$

If $\delta \geq n_1$, then $\delta - 1 \geq n_1 - 1$, and hence

$$\delta r(\delta - 1, n_1 - 1) = \delta \prod_{i=1}^{n_1-1} (\delta - i) = \delta(\delta - 1) \cdots (\delta - n_1 + 1) = r(\delta, n_1).$$

If $\delta < n_1$, then $\delta - 1 < n_1 - 1$, and therefore

$$\begin{aligned} \delta \cdot \prod_{i=1}^{\min(\delta-1, n_1-1)} (\delta - i) &= \delta \cdot \prod_{i=1}^{\delta-1} (\delta - i) \\ &= \delta((\delta - 1)(\delta - 2) \cdots 1) \\ &= \delta(\delta - 1)! \\ &= \delta! \\ &= \prod_{i=1}^{\delta} (\delta - i + 1). \end{aligned}$$

Thus, in this case also, the number of complete matchings is at least $r(\delta, n_1)$.

Hence the result follows by induction. ■

Theorem 2.1.12. Every k -regular bipartite graph with $k > 0$ has a perfect matching.

Proof. Let $G = (X \cup Y, E)$ be a k -regular bipartite graph, where X and Y are the two partite sets.

Since every vertex has degree k , counting the edges from the side X gives

$$|E| = \sum_{x \in X} d(x) = k|X|.$$

Similarly, counting the same edges from the side Y gives

$$|E| = \sum_{y \in Y} d(y) = k|Y|.$$

Hence

$$k|X| = k|Y|.$$

Since $k > 0$, we obtain

$$|X| = |Y|.$$

We now verify Hall's condition. Let $S \subseteq X$, and let $N(S)$ denote the set of all vertices of Y adjacent to at least one vertex of S .

Every vertex of S has degree k , and all of its incident edges join a vertex of S to a vertex of $N(S)$. Therefore, the number of edges between S and $N(S)$ is

$$|E(S, N(S))| = k|S|.$$

On the other hand, each vertex of $N(S)$ has total degree k . Therefore, at most $k|N(S)|$ edges can join S to $N(S)$. Thus

$$|E(S, N(S))| \leq k|N(S)|.$$

Combining the two relations, we get

$$k|S| \leq k|N(S)|.$$

Since $k > 0$,

$$|S| \leq |N(S)|.$$

Thus Hall's condition holds for every subset $S \subseteq X$. By Hall's Marriage Theorem, G has a matching that saturates every vertex of X .

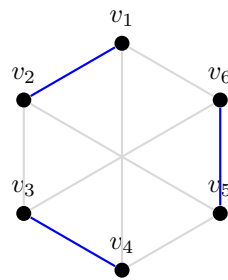
Finally, since $|X| = |Y|$, a matching that saturates all vertices of X also saturates all vertices of Y . Hence the matching is a perfect matching. ■

2.2 Factors of Graphs

Definition 2.2.1 (Factor). Let G be a graph. A **factor** F of G is a spanning subgraph of G . That is, $V(F) = V(G)$ and $E(F) \subseteq E(G)$.

Definition 2.2.2 ($\mathcal{F}$ -Factor and H -Factor). Let $\mathcal{F} = \{H_1, H_2, \dots, H_k\}$ be a collection of graphs. A factor F is called an **$\mathcal{F}$ -factor** if every connected component of F is isomorphic to some element of $\mathcal{F}$.

In the specific case where $\mathcal{F} = \{H\}$ (a singleton set), the factor F is called an **H -factor**. In this case, every component of the spanning subgraph is isomorphic to H .



K_2 -Factor

The blue edges form a spanning subgraph where every component is isomorphic to K_2 .

Figure 2.12: Visualisation of an H -factor where $H = K_2$.

Degree Constraints: f -Factors and k -Factors

Let G_1 and G_2 be graphs. Recall the *Cartesian product* of G_1 and G_2 , denoted by $G_1 \square G_2$, is the graph with vertex set

$$V(G_1 \square G_2) = V(G_1) \times V(G_2).$$

Two vertices (u_1, u_2) and (v_1, v_2) are adjacent in $G_1 \square G_2$ if and only if either $u_1 = v_1$ and $u_2 v_2 \in E(G_2)$, or $u_2 = v_2$ and $u_1 v_1 \in E(G_1)$.

Example 2.2.3. Consider the graph $G = K_3 \square K_3$ (also known as the Rook's graph on a 3×3 grid). This graph has 9 vertices and is 4-regular. It admits a K_3 -factorization consisting of two factors, F_1 and F_2 .

- **Factor 1 (Red):** Three disjoint triangles formed horizontally.
- **Factor 2 (Blue):** Three disjoint triangles formed vertically.

Since $E(G) = E(F_1) \cup E(F_2)$ and $E(F_1) \cap E(F_2) = \emptyset$, this constitutes a factorization.

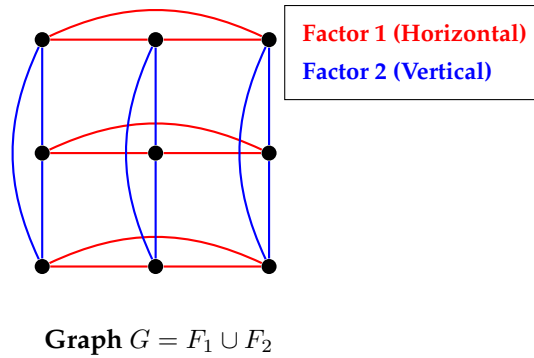


Figure 2.13: Visualisation of G decomposed into two K_3 -factors.

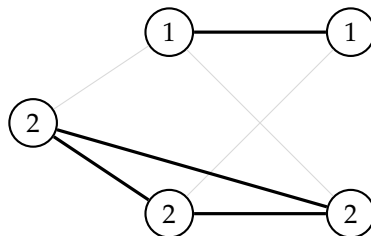
Definition 2.2.4 (Vertex Function). A **vertex function** of a graph $G(V, E)$ is a mapping $f : V \rightarrow \mathbb{N}_0$, where $\mathbb{N}_0$ denotes the set of non-negative integers.

Definition 2.2.5 (f -Factor). A spanning subgraph F of a graph G is called an **f -factor** if the degree of every vertex v in F corresponds to the value assigned by the vertex function. That is:

$$\deg_F(v) = f(v) \quad \forall v \in V.$$

Definition 2.2.6 (k -Factor). A **k -factor** is a special case of an f -factor where $f(v) = k$ for all $v \in V$. Thus, a k -factor is a k -regular spanning subgraph.

- A **1-factor** is a perfect matching (vertices have degree 1).
- A **2-factor** is a disjoint union of cycles covering all vertices.



Example of an f -factor

Numbers inside nodes represent the target degree $f(v)$. The thick edges form the subgraph satisfying these constraints.

Figure 2.14: An f -factor where some vertices require degree 1 and others degree 2.

Factorisation

Definition 2.2.7 (Factorisation / Decomposition). Let G be a graph and let $\{F_1, F_2, \dots, F_q\}$ be a set of factors of G . These factors constitute a **factorisation** (or decomposition) of G if their edge sets form a partition of $E(G)$. We write:

$$G = F_1 \oplus F_2 \oplus \dots \oplus F_q$$

This means every edge of G belongs to exactly one factor F_i .

Definition 2.2.8 (Isomorphic Factorisation). If each factor F_i in a factorisation is isomorphic to the same graph F , the factorisation is called an **isomorphic factorisation** or an **F -decomposition** of G .

Definition 2.2.9 (k -Factorisation). A factorisation is called a **k -factorisation** if every factor F_i is a k -factor. A graph admitting such a decomposition is said to be **k -factorable**.

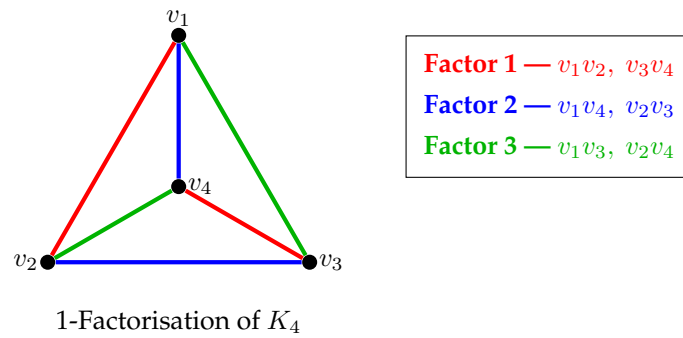


Figure 2.15: A 1-factorisation of the complete graph K_4 . The edges are partitioned into three distinct perfect matchings (Red, Blue, Green).

Definition 2.2.10. A component of a graph is odd or even according to whether it has an odd or even number of vertices. The number of odd components of a graph G is denoted by $k_0(G)$.

Tutte's One-Factor Theorem: A graph $G = (V, E)$ has a perfect matching if and only if for every subset $S \subseteq V$, the number of odd components in $G - S$ is less than or equal to $|S|$. That is, $k_0(G - S) \leq |S|$ for all $S \subseteq V$.

Proof. A Video lecture available at

<https://youtu.be/EW4oA9HZzgg?si=8QGKe7Sfpdem9Rg5>

or visit YouTube via <https://aijaz.vercel.app/>

■

Theorem 2.2.11 (Tutte). A graph G has a 1-factor if and only if

$$k_0(G - S) \leq |S|, \text{ for all } S \subseteq V. \quad (8.8.1)$$

Proof. Let G be a graph having 1-factor M . Let S be an arbitrary subset of V and let $O_1, O_2, \dots, O_k$ be the odd components of $G - S$. For each i , the vertices in O_i can be adjacent only to other vertices in O_i and to vertices in S . Since G has a 1-factor (perfect matching), at least one vertex from each O_i is to be matched with a different vertex in S . Thus the number of vertices in S is at least k , the number of odd components. Hence $|S| \geq k$, so that $k_0(G - S) \leq |S|$ (Fig. T1).

Conversely, assume that the condition (8.8.1) holds. Let G have no 1-factor. Join pairs of non-adjacent vertices of G and continue this process till we get a maximal supergraph G^* of G having no 1-factor. Since joining two odd components by an edge results in an even component, therefore

$$k_0(G^* - S) \leq k_0(G - S). \quad (8.8.2)$$

Thus for G^* also, we have $k_0(G^* - S) \leq |S|$.

First, we note that G has even order. This is because the condition (8.8.1) must hold for $S = \phi$, which gives $k_0(G) = 0$. Therefore, in this case G has no odd component, so G itself is an even-ordered graph. Also, for every pair of non-adjacent vertices u and v of G^* , $G^* + uv$ has a 1-factor, and any such 1-factor definitely contains the edge uv .

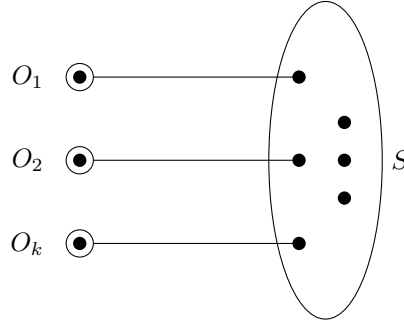


Figure T1

Now, let K be the set of those vertices of G^* which are having degree $n - 1$. Clearly, $K \neq V$, since otherwise $G^* = K_n$ has a perfect matching.

Claim that each component of $G^* - K$ is complete. Assume, on the contrary, that in $G^* - K$, there exists a component G_1 which is not complete. Then in G_1 , there are vertices x, y and z such that $xy, yz \in E(G^*)$ and $xz \notin E(G^*)$. Also, $y \in V(G_1)$ so $y \notin K$ and so $d(y|G^*) < n - 1$ and thus there exists a vertex w of G^* with $yw \notin E(G^*)$. Evidently, $w \notin K$ (Fig. T2).

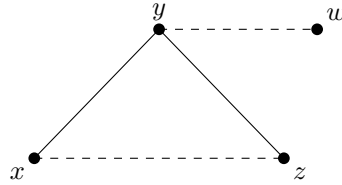
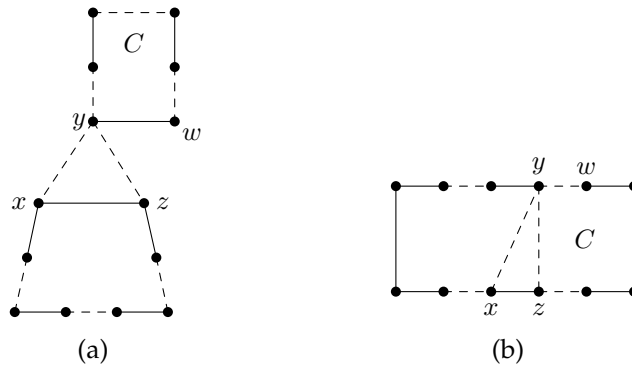


Figure T2

Now, by the choice of G^* , both $G^* + xz$ and $G^* + yw$ have 1-factors, say M_1 and M_2 respectively. M_1 must contain xz and M_2 must contain yw . Let H be the subgraph of $G^* + \{xz, yw\}$ induced by the edges that are in M_1 and M_2 but not in both (that is, in the symmetric difference of M_1 and M_2). Since M_1 and M_2 are 1-factors, each vertex of G^* is saturated by both M_1 and M_2 , and H is a disjoint union of even cycles in which the edges alternate between M_1 and M_2 . We have two cases to consider.

Case 1 Let xz and yw belong to different components of H , as shown in Figure T3 (a). If yw belongs to the cycle C , then the edge of M_1 in C together with the edges of M_2 not belonging to C form a 1-factor of G^* , contradicting the choice of G^* .



Ordinary lines correspond to the edges of M_1 and bold lines to the edges of M_2 .

Figure T3

Case 2 Let xz and yw belong to the same component C of H . Since each component of H is a cycle, C is a cycle (Fig. T3 (b)). By symmetry of x and z , it can be assumed that the vertices x, y, w and z appear in that order on C . Then the edges of M_1 belonging to the $yw \dots x$ section of C , together with the edge yz , and the edges of M_2 not in the $yw \dots z$ section of C form a 1-factor of G^* , again contradicting the choice of G^* .

Thus each component of $G^* - K$ is complete.

By conditions (8.8.2), $k_0(G^* - K) \leq |K|$. Therefore one vertex of each of the odd components of $G^* - K$ is matched to a vertex of K . This is because each vertex of K is adjacent to every other vertex of G^* . Also the remaining vertices in each of the odd and even components of $G^* - K$ can be matched among themselves (Fig. T4). The total number of vertices thus matched is even. As $|V(G^*)|$ is even, the remaining vertices of K can be matched among themselves, thus giving a 1-factor of G^* , which is a contradiction.

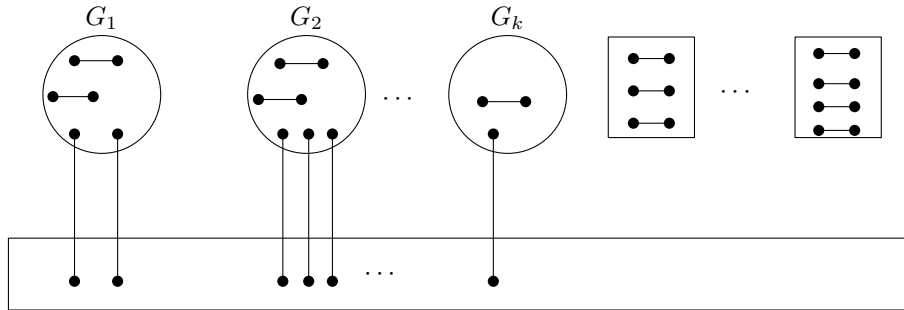


Figure T4

Remark 2.2.12. Among the various proofs of Theorem 8.8, one proof due to Anderson [4] uses Hall's theorem.

Tutte's characterisation of graphs admitting 1-factors has been used to obtain several results including those for regular graphs and many of these are reported in the review article by Akiyama and Kano [1]. We present some of these results. The first is due to Peterson [181].

Theorem 2.2.13 (Peterson). *Every cubic graph without cut edges has a 1-factor*

Proof. Let G be a cubic graph with vertex set $V(G)$. Let G have no cut edges and let $S \subseteq V$. Let $O_1, O_2, \dots, O_r$ be the odd components in $G - S$. Note that the vertices of O_i are either adjacent within O_i or adjacent to some vertex of S . Let $|O_i| = n_i$, m_i be the number of edges in G having both ends in O_i and let k_i be the number of edges in G having one end in O_i and the other end in S . Then we have;

$$\sum_{v \in V(O_i)} \deg(v) = 3n_i = 2m_i + k_i.$$

This gives, $k_i = 3n_i - 2m_i$, $1 \leq i \leq r$. As n_i is odd, therefore, k_i must be odd. Moreover, since there are no cut edges in the graph, we have $k_i \neq 1$, that is, $k_i \geq 3$, that is, there are at least 3 edges from each O_i to S . Hence, we have

$$3r \leq \sum_{s \in S} \deg(s) = 3|S|.$$

Therefore,

$$k_o(G - S) = r \leq |S|$$

for each $S \subseteq V(G)$. The proof then follows by Tutte's theorem.

Theorem 2.2.14 (Generalization of Theorem 2.2.13). *Let G be a k -regular graph having even number of vertices and is $(k - 1)$ -edge connected. Then G has a 1-factor.*

Proof. Let G be a k -regular, $(k - 1)$ -edge-connected graph with even order and let S be any subset of V . Let $O_1, O_2, \dots, O_r$ be the odd components of $G - S$. Note that the vertices of O_i are either adjacent within O_i or adjacent to some vertex of S . Let $|O_i| = n_i$, m_i be the number of edges in G having both ends in O_i and let k_i be the number of edges in G having one end in O_i and the other end in S for $1 \leq i \leq r$. Then as G is $(k - 1)$ -connected, we have

$$k_i \geq k - 1, \tag{2.1}$$

Now, G is k -regular and vertices of any O_i are either adjacent within O_i or adjacent to some vertex of S , therefore,

$$\sum_{v \in V(O_i)} d(v) = kn_i = 2m_i + k_i \quad (2.2)$$

and vertices in S can be adjacent within S or adjacent to some vertex of O_i , $1 \leq i \leq r$ or may be adjacent to some other vertex of G , so

$$\sum_{v \in S} d(v) = k|S| \geq \sum_{i=1}^r k_i + 2|E(S)|. \quad (2.3)$$

Now, (2.2) gives $k_i = kn_i - 2m_i$, so that if k and k_i have the same parity, therefore, $k_i \geq k - 1$ implies $k_i \geq k$. Summing this over r odd components, we obtain by using (2.3)

$$rk \leq \sum_{i=1}^r k_i \leq k|S| + 2|E(S)| \leq k|S|.$$

Therefore, $k_0(G - S) = r \leq |S|$ and by Tutte's theorem, G has a 1-factor. ■

Definition 2.2.15 (Anti-factors). A subset $S \subseteq V$ of a graph $G(V, E)$ such that $k_0(G - S) > |S|$ is called an antifactor set. An antifactor set S such that no proper subset of S is an antifactor set is called a minimal antifactor set.

Theorem 2.2.16. Let G be a k -regular graph of even order that is also $(k - 1)$ -edge connected. If G_1 is obtained from G by deleting any $k - 1$ edges, then G_1 has a 1-factor.

Proof. Let G be a k -regular, $(k - 1)$ -edge-connected graph with even order. Let $K \subseteq V(G)$ and let $V(G_1) = V(G) - K$ and let S be any subset of V . Let $O_1, O_2, \dots, O_r$ be the odd components of $G - S$. Note that the vertices of O_i are either adjacent within O_i or adjacent to some vertex of S . Let $|O_i| = n_i$, m_i be the number of edges in G having both ends in O_i and let k_i be the number of edges in G having one end in O_i and the other end in S for $1 \leq i \leq r$. Then as G is $(k - 1)$ -connected, we have

a_i = number of edges from K to S

b_i = number of edges within O_i

m_i = number of edges from O_i to S

$$a_i + m_i \geq k - 1, \text{ then } \sum_{i=1}^r a_i + \sum_{i=1}^r m_i \geq k$$

Moreover, $\sum_{i=1}^r a_i + \frac{1}{2} \sum_{i=1}^r b_i \leq k - 1$, which gives, $2 \sum_{i=1}^r a_i + \sum_{i=1}^r b_i \leq 2(k - 1)$.

Also, $\sum_{i=1}^r m_i \leq k|S|$. Adding, we get,

$$2 \sum_{i=1}^r a_i + \sum_{i=1}^r b_i + \sum_{i=1}^r m_i \leq 2(k - 1) + k|S| = k(|S| + 2) - 2.$$

Finally,

$$\begin{aligned} kr &\leq \sum_{i=1}^r a_i + \sum_{i=1}^r m_i \\ &\leq 2 \sum_{i=1}^r a_i + \sum_{i=1}^r b_i + \sum_{i=1}^r m_i \\ &\leq k(|S| + 2) - 2 \\ &< k(|S| + 2). \end{aligned}$$

Thus, $r < |S| + 2$, a contradiction. ■

Corollary 2.2.17. *A $(k - 1)$ -regular simple graph on $2k$ vertices has a 1-factor.*

Proof. Assume G is a $(k - 1)$ -regular simple graph with $2k$ vertices having no 1-factor. Then G has an antifactor set S and by Lemma 8.4, $k_0(G - S) \geq |S| + 2$. Therefore,

$$|S| + (|S| + 2) \leq 2k \implies |S| \leq k - 1.$$

Let $|S| = k - r$. Then $r \neq 1$. For if $r = 1$, then $|S| = k - 1$ and therefore $k_0(G - S) = k + 1$. Thus each odd component of $G - S$ is a singleton and therefore each such vertex is adjacent to all the $k - 1$ vertices of S , as G is $(k - 1)$ -regular. This implies that every vertex of S is of degree at least $k + 1$, a contradiction.

Thus, $|S| = k - r$, with $2 \leq r \leq k - 1$.

If G_1 is any component of $G - S$ and $v \in V(G_1)$, then v can be adjacent to at most $|S|$ vertices of S . Since G is $(k - 1)$ -regular, v is adjacent to at least $(k - 1) - (k - r) = r - 1$ vertices of G_1 . So, $|V(G_1)| \geq r$.

Counting the vertices of S and the components, we obtain:

$$(|S| + 2)r + |S| \leq 2k$$

Substituting $|S| = k - r$:

$$(k - r + 2)r + (k - r) \leq 2k$$

Expanding and simplifying:

$$kr - r^2 + 2r + k - r \leq 2k$$

$$kr - r^2 + r - k \leq 0$$

Multiplying by -1 and factoring:

$$r^2 - r(k + 1) + k \geq 0$$

$$(r - 1)(r - k) \geq 0$$

Since $2 \leq r \leq k - 1$, the term $(r - 1)$ is positive and $(r - k)$ is negative. Their product is negative, which violates the condition $(r - 1)(r - k) \geq 0$. ■

Definition 2.2.18. A *vertex function* for a general graph $G(V, E)$ is a function $f : V \rightarrow \mathbb{N}_0$ such that $0 \leq f(v) \leq d(v)$, for every $v \in V$.

For any function $f : V \rightarrow \mathbb{N}_0$ and for any $U \subseteq V$, we set

$$f(U) = \sum_{v \in U} f(v).$$

Definition 2.2.19. Three pair-wise disjoint (possibly empty) subsets S, T and U of the vertex set V of a general graph G , such that $V = S \cup T \cup U$, is called a *decomposition* $D = (S, T, U)$ of G . The components, say $C_1, C_2, \dots, C_k$, of the vertex-induced subgraph $\langle U \rangle$ of G are called the components of U . For each C_i , let

$$h_i = f(C_i) + q_G[T, C_i],$$

where f is a given vertex function of G , and $q_G[T, C_i]$ denotes the number of edges of G , whose one end is in T and the other end is in C_i . Then C_i is said to be an odd or even component of U , depending upon h_i being odd or even, and the number of odd components of U is denoted by $k'_0(D, f)$.

The *f-deficiency* of D for the given G is defined as

$$\delta(D, f) = k'_0(D, f) - f(S) + f(T) - d(T) + q[S, T].$$

Construction 1 (Construction of Graph G' for f -factor Theorem). Let G be a general graph with the vertex function f . This defines for each vertex v_i , its degree $d_i = d(v_i|G)$, $f_i = f(v_i)$ and $k_i = d_i - f_i$. For each vertex v_i of G , take two subsets X_i and Y_i of vertices of G' , where $|X_i| = d_i$ and $|Y_i| = k_i$. Let the vertices of Y_i be labelled y_j^i , $1 \leq j \leq k_i$, arbitrarily. Now, label the vertices of X_i in the following manner.

Label the edges of G arbitrarily as $1, 2, \dots, m$. Since $|X_i| = \text{number of link edges } l_i \text{ incident with } v_i \text{ plus twice the number of loops } \ell_i^0 \text{ incident with } v_i$, a vertex $x_j^i \in X_i$ is taken corresponding to each link edge j incident with v_i and two vertices $x_h^i, x_{h'}^i$ are taken corresponding to each loop h incident at v_i . As for edges, if link j joins v_i and v_k , then $x_j^i x_j^k$ is made an edge. Corresponding to loop h incident with x_h^i , the edge $x_h^i x_{h'}^i$ is taken. In addition, all the edges of the complete bipartite subgraph with vertex partition $\{X_i, Y_i\}$ are taken for each i . These constitute the graph G' . The induced subgraph $\langle X_i \cup Y_i \rangle$ of G' is called the *star graph* at i and is denoted by $St(i)$. For multi graph G , it is a complete bipartite graph, but for general graphs it will contain edges in $\langle X_i \rangle$ which we call *loop edges*. The other edges of $St(i)$ are called *star edges* and the remaining edges of G' are called as *link edges*.

Illustration: This construction is illustrated in Figure 2.16. In the general graph G , $d(v_1) = 5$, $d(v_2) = 3$ and $d(v_3) = 2$. Let $f = (3, 2, 1)$ be the vertex function, so that $f(v_1) = 3$, $f(v_2) = 2$ and $f(v_3) = 1$. Therefore, $k_1 = 2$, $k_2 = 1$ and $k_3 = 1$. Here $|X_1| = 5$, $|X_2| = 3$ and $|X_3| = 2$, and $|Y_1| = 2$, $|Y_2| = 1$ and $|Y_3| = 1$.

Label the edges of G as 1, 2, 3, 4 and 5 as shown. Then $Y_1 = \{y_1^1, y_2^1\}$, $Y_2 = \{y_1^2\}$ and $Y_3 = \{y_1^3\}$, and we have $X_1 = \{x_1^1, x_{1'}^1, x_2^1, x_3^1, x_4^1\}$, $X_2 = \{x_2^2, x_3^2, x_5^2\}$ and $X_3 = \{x_4^3, x_5^3\}$. The vertex set of G' is $X_i \cup Y_i$, $1 \leq i \leq 3$. The edges of G' are $x_1^1 x_{1'}^1, x_2^1 x_2^2, x_3^1 x_3^2, x_4^1 x_4^3, x_5^2 x_5^3$ (shown by dotted lines) and the star edges (shown by bold lines). The dotted lines in G form an f -factor with $f = (3, 2, 1)$.

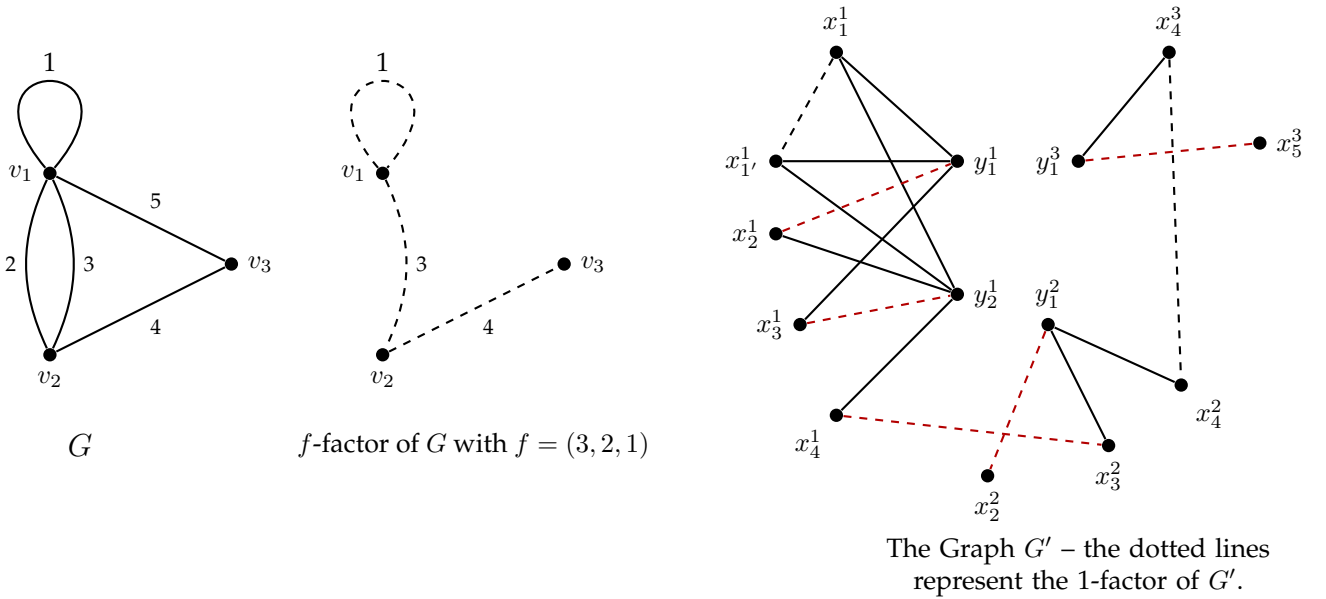


Figure 2.16: Dark dotted edges in G' represent edges induced by f -factor of G for a 1-factor of G' and red dotted edges are additional edges required to produce a 1-factor of G' .

Example 2.2.20 (Construction of G' for a Cycle C_4). Let G be a cycle C_4 with vertices v_1, v_2, v_3, v_4 . Let $f(v_i) = 1$ for all i . Thus, for every vertex v_i :

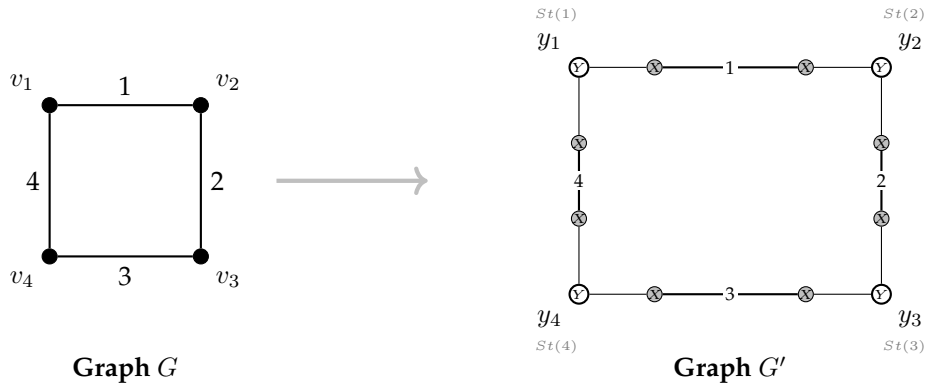
$$|X_i| = d(v_i) = 2 \quad \text{and} \quad |Y_i| = d(v_i) - f(v_i) = 1.$$

The construction is visualized below (Figure 2.17) The graph G' consists of 4 "corner" gadgets. Finding a 1-factor in G' corresponds to finding a 1-factor in G .

Theorem 2.2.21. A general graph G has an f -factor if and only if G' has a 1-factor

Proof. We prove the necessary and sufficient conditions separately.

($\Rightarrow$) **Necessity** Assume G has an f -factor F with link edges L and loop edges L^0 . Let g_i denote the number of link edges and f_i^0 the number of loop edges of F incident at vertex v_i . Consequently, the degree in the factor is $f_i = g_i + 2f_i^0$.

Figure 2.17: Transformation of C_4 into G' with $f(v) = 1$.

Recall that for the constructed graph G' , the size of the sets X_i and Y_i are defined as:

$$|X_i| = d_i = l_i + 2l_i^0 \quad \text{and} \quad |Y_i| = k_i = d_i - f_i,$$

where l_i is the number of link edges incident with v_i and l_i^0 is the number of loops incident with v_i in the original graph G . We can expand the expression for $|Y_i|$ as follows:

$$\begin{aligned} |Y_i| &= (l_i + 2l_i^0) - (g_i + 2f_i^0) \\ &= (l_i - g_i) + 2(l_i^0 - f_i^0). \end{aligned}$$

Let L' and $(L^0)'$ be the set of link and loop edges in G' corresponding to L and L^0 . The matching consisting of these edges keeps all vertices of Y_i unsaturated. Furthermore, it leaves unsaturated a set of vertices in X_i with cardinality:

$$(l_i - g_i) + 2(l_i^0 - f_i^0) = k_i = |Y_i|.$$

We can now match the vertices of Y_i with these remaining unsaturated vertices of X_i using the star edges $St(i)$. Let S' be this set of edges. Clearly, the union $L' \cup (L^0)' \cup S'$ constitutes a 1-factor in G' .

($\Leftarrow$) **Sufficiency** Conversely, assume G' has a 1-factor H . Since the vertices in Y_i can only be matched to vertices in X_i (by the construction of the star graphs), there are exactly

$$d_i - |Y_i| = d_i - (d_i - f_i) = f_i$$

vertices of X_i that are *not* matched by H to vertices in Y_i . Let us denote this set of vertices as X_i^* . Note that $f_i = g_i + 2f_i^0$.

Clearly, $2f_i^0$ vertices from X_i^* correspond to loop edges of the induced subgraph $\langle X_i \rangle$ and are thus matched in pairs amongst themselves. The remaining g_i vertices in X_i^* are matched by the link edges of H to vertices in X_j belonging to other star graphs $St(j)$ of G' .

If F is the set of edges in G corresponding to these loop edges and link edges selected by H , then the degree of v_i in the subgraph induced by F is:

$$d_{\langle F \rangle}(v_i) = g_i + 2f_i^0 = f_i.$$

Thus, F is an f -factor of G . ■

Theorem 2.2.22 (f -factor theorem). *A general graph G has an f -factor if and only if it has no decomposition D with $\delta(D, f) > 0$.*

OR

If f is a vertex function for a graph G , then either G has an f -factor or an f -barrier but not both.

Proof. Refer to "Introduction to Graph Theory" by S Pirzada. ■

Definition 2.2.23. Suppose $D = (S, T, U)$ is a decomposition of V and $x \in S, y \in T$. The decomposition $D_1 = (S \setminus \{x\}, T, U \cup \{x\})$ is said to be obtained by an x -**transfer from** S , and the decomposition $D_2 = (S, T \setminus \{y\}, U \cup \{y\})$ is said to be obtained by a y -**transfer from** T .

Lemma 2.2.24. Let $D = (S, T, U)$ be a decomposition of $V(G)$. An x -transfer from S does not reduce the deficiency of D if $f(x) = d(x)$ or $d(x) - 1$, and a y -transfer from T does not reduce the deficiency of D if $f(y) = 0$ or 1 .

Proof. Let $D' = (S - \{x\}, T, U \cup \{x\})$ be obtained from D by an x -transfer. Then

$$\delta(D', f) - \delta(D, f) \geq f(x) - d(x).$$

Thus the difference is at least 0 when $f(x) = d(x)$, and at least -1 when $f(x) = d(x) - 1$. Since the deficiencies of any two decompositions have the same parity, the latter difference cannot be -1 . Hence

$$\delta(D', f) \geq \delta(D, f).$$

Similarly, if $D'' = (S, T - \{y\}, U \cup \{y\})$ is obtained by a y -transfer, then

$$\delta(D'', f) - \delta(D, f) \geq -f(y).$$

If $f(y) = 0$, this is nonnegative, while if $f(y) = 1$, it is at least -1 . Again parity excludes the value -1 . Therefore

$$\delta(D'', f) \geq \delta(D, f).$$

Hence neither transfer reduces the deficiency under the stated hypotheses. ■

Theorem 2.2.25. The f -factor theorem implies the 1-factor theorem of Tutte.

Proof. Take $f(v) = 1$ for every $v \in V(G)$. Then an f -factor is exactly a 1-factor.

Assume that $k_0(G - S) \leq |S|$ for every $S \subseteq V(G)$. Suppose, to the contrary, that G has no 1-factor. By the f -factor theorem, there exists a decomposition $D = (S, T, U)$ with positive deficiency.

Since $f(y) = 1$ for every $y \in T$, Lemma 2.2.24 allows us to transfer every vertex of T into U without decreasing the deficiency. Hence we obtain the decomposition

$$(S, \emptyset, V(G) \setminus S)$$

with positive deficiency.

For this decomposition, since $f(S) = |S|$, we have

$$\delta(D, 1) = k_0(G - S) - |S|.$$

Thus $k_0(G - S) > |S|$, contrary to the assumed condition. Therefore G has a 1-factor. Hence the f -factor theorem implies Tutte's 1-factor theorem. ■

Theorem 2.2.26 (Erdős–Gallai Theorem). Let $d_1 \geq d_2 \geq \dots \geq d_n \geq 0$ be a nonincreasing sequence of integers. Then $(d_1, d_2, \dots, d_n)$ is graphical if and only if $\sum_{i=1}^n d_i$ is even and, for every $k = 1, 2, \dots, n$,

$$\sum_{i=1}^k d_i \leq k(k-1) + \sum_{i=k+1}^n \min\{d_i, k\}.$$

Corollary 2.2.27. The Erdős–Gallai theorem on degree sequences follows from the f -factor theorem.

Proof. Let $d_1 \geq d_2 \geq \dots \geq d_n \geq 0$ and suppose that $\sum_{i=1}^n d_i \equiv 0 \pmod{2}$. To obtain this condition, we observe that d is graphic if and only if the complete graph K_n has a d -factor. Let $V(K_n) =$

$\{v_1, \dots, v_n\}$ and define $f(v_i) = d_i$ for $1 \leq i \leq n$. Then $(d_1, \dots, d_n)$ is graphic if and only if K_n has an f -factor.

By the f -factor theorem, K_n has an f -factor if and only if $\delta(D, f) \leq 0$ for every decomposition $D = (S, T, U)$ of $V(K_n)$, where

$$\delta(D, f) = k'_0(D, f) - f(S) + f(T) - d(T) + q[S, T].$$

Let $r = |T|$. Since every vertex of K_n has degree $n - 1$, we have $d(T) = r(n - 1)$. Also, every vertex of S is adjacent to every vertex of T , so $q[S, T] = |T||S| = r|S|$. Hence

$$\begin{aligned} \delta(D, f) &= k'_0(D, f) - f(S) + f(T) - d(T) + q[S, T] \\ &= k'_0(D, f) - f(S) + f(T) - r(n - 1) + r|S| \\ &= k'_0(D, f) + f(T) - f(S) - r(n - 1 - |S|) \end{aligned}$$

Now $K_n[U]$ is complete. Thus, if $U \neq \emptyset$, it has only one component, and consequently $k'_0(D, f) \in \{0, 1\}$. Moreover, $f(V(K_n)) = \sum_{i=1}^n d_i$ is even, and therefore $\delta(D, f)$ is even because $\delta(D, f)$ and $\sum f(V(G))$ always have the same parity. Hence the condition $\delta(D, f) \leq 0$ is equivalent to

$$f(T) \leq f(S) + r(n - 1 - |S|).$$

Since $U = V(K_n) \setminus (S \cup T)$, we have $n - 1 - |S| = (r - 1) + |U|$. Therefore,

$$f(T) \leq r(r - 1) + f(S) + r|U|.$$

For fixed T , the sets S and U partition $V(K_n) \setminus T$. For each $v \notin T$, placing v in S contributes $f(v)$, whereas placing v in U contributes r . Hence the minimum possible value of $f(S) + r|U|$ is $\sum_{v \notin T} \min\{f(v), r\}$. Thus the f -factor condition is equivalent to

$$f(T) \leq r(r - 1) + \sum_{v \notin T} \min\{f(v), r\}$$

for every $T \subseteq V(K_n)$ with $|T| = r$.

Taking $T = \{v_1, \dots, v_r\}$ gives $f(T) = \sum_{i=1}^r d_i$ and $\sum_{v \notin T} \min\{f(v), r\} = \sum_{i=r+1}^n \min\{d_i, r\}$. Hence

$$\sum_{i=1}^r d_i \leq r(r - 1) + \sum_{i=r+1}^n \min\{d_i, r\},$$

which is the Erdős–Gallai inequality. ■

Theorem 2.2.28 (k -factor Theorem). *Let $d = [d_i]_1^n$ be a graphical sequence. If the sequence $d' = [d_i - k]_1^n$ is also graphical, then there exists a realization of d that contains a k -factor.*

Proof. Let G_1 be a realization of $d = [d_i]_1^n$ and let G_2 be a realization of $d' = [d_i - k]_1^n$, both on the same vertex set $V = \{v_1, \dots, v_n\}$. Thus $d_{G_1}(v_i) = d_i$ and $d_{G_2}(v_i) = d_i - k$ for every i . Choose G_1 and G_2 so that $|E(G_1) \cap E(G_2)|$ is as large as possible. Put $E_1 = E(G_1)$ and $E_2 = E(G_2)$. We claim that $E_2 \subseteq E_1$. Once this is proved, the graph with edge set $E_1 \setminus E_2$ has degree $d_i - (d_i - k) = k$ at every vertex, and hence is a k -factor of G_1 .

We first record a switching observation. Suppose that $v_i v_j \in E_2 \setminus E_1$ and $v_i v_r, v_j v_s \in E_1 \setminus E_2$. Then $v_r v_s \in E_1 \setminus E_2$. Indeed, if $v_r v_s \in E_2$, then in G_2 we may replace the edges $v_i v_j$ and $v_r v_s$ by $v_i v_r$ and $v_j v_s$. This preserves every vertex degree and increases $|E_1 \cap E_2|$, a contradiction. If $v_r v_s \notin E_1$, then in G_1 we may replace $v_i v_r$ and $v_j v_s$ by $v_i v_j$ and $v_r v_s$. Again all degrees are preserved and the number of common edges increases. Hence $v_r v_s \in E_1 \setminus E_2$.

Now suppose, to the contrary, that $E_2 \setminus E_1 \neq \emptyset$. For each vertex u , let $t(u) = d_{E_2 \setminus E_1}(u)$. Since $d_{G_1}(u) - d_{G_2}(u) = k$, we have $d_{E_1 \setminus E_2}(u) = t(u) + k$. Choose a vertex v_1 for which $t(v_1)$ is maximum, and write $t = t(v_1) > 0$. Choose v_j such that $v_1 v_j \in E_2 \setminus E_1$. Since $t(v_j) \geq 1$, we have $d_{E_1 \setminus E_2}(v_j) = t(v_j) + k \geq k + 1$. Hence there exists a vertex w such that $v_j w \in E_1 \setminus E_2$.

Let $A = N_{E_1 \setminus E_2}(v_1)$. Since $d_{E_1 \setminus E_2}(v_1) = t + k$, we have $|A| = t + k$. First suppose that $w \notin A$. For every $a \in A$, the switching observation applied to $v_1 v_j \in E_2 \setminus E_1$ and $v_1 a, v_j w \in E_1 \setminus E_2$ gives $aw \in E_1 \setminus E_2$. Thus w is incident with the $t + k$ edges joining it to the vertices of A , as well as the edge $v_j w$. Therefore $d_{E_1 \setminus E_2}(w) \geq t + k + 1$, and hence $t(w) \geq t + 1$, contradicting the maximality of t .

Now suppose that $w \in A$. For every $a \in A \setminus \{w\}$, the same switching observation gives $aw \in E_1 \setminus E_2$. In addition, both $v_1 w$ and $v_j w$ belong to $E_1 \setminus E_2$. Hence $d_{E_1 \setminus E_2}(w) \geq (t + k - 1) + 2 = t + k + 1$, so again $t(w) \geq t + 1$, contradicting the maximality of t .

Therefore $E_2 \setminus E_1 = \emptyset$, and hence $E_2 \subseteq E_1$. Consequently, the graph $H = (V, E_1 \setminus E_2)$ satisfies $d_H(v_i) = d_i - (d_i - k) = k$ for every i . Hence H is a k -factor of G_1 . ■

2.3 Factorization of Complete Graphs

Definition 2.3.1. A **factorization** of a graph G is a decomposition of the edges of G into edge-disjoint spanning subgraphs, known as factors. If the graph is decomposed into k -factors (where every vertex has degree k), it is called a **k -factorization**.

For the complete graph K_n , the type of factorization possible depends on the parity of n .

Case 1: n is Even (1-Factorization)

Consider a complete graph K_{2n} . It has $2n$ vertices and is $(2n - 1)$ -regular.

- A 1-factor is a perfect matching (regular of degree 1).
- Since the total degree of each vertex is $2n - 1$, K_{2n} can be decomposed into exactly $2n - 1$ perfect matchings.

Case 2: n is Odd (2-Factorization)

Consider a complete graph K_{2n+1} . It has $2n + 1$ vertices and is $2n$ -regular.

- A 2-factor is a spanning subgraph where every vertex has degree 2 (a disjoint union of cycles).
- In the context of K_n , we are often interested in decomposing it into Hamiltonian cycles.
- Since the total degree is $2n$, K_{2n+1} can be decomposed into n Hamiltonian cycles.

Theorem 2.3.2. For every positive integer n , the complete graph K_{2n} is 1-factorable.

Proof. Let the vertex set be $V = \{0, 1, \dots, 2n - 2, 2n - 1\}$. We designate the vertex $2n - 1$ as the fixed center and treat the remaining vertices $\{0, 1, \dots, 2n - 2\}$ as elements of $\mathbb{Z}_{2n-1}$ (the integers modulo $2n - 1$).

We construct $2n - 1$ factors, denoted $F_0, F_1, \dots, F_{2n-2}$. For each $i \in \{0, \dots, 2n - 2\}$, define the set of edges F_i as:

$$F_i = \{\{2n - 1, i\}\} \cup \{\{i - k, i + k\} \pmod{2n - 1} \mid k = 1, \dots, n - 1\}.$$

1. F_i is a 1-factor: The fixed vertex $2n - 1$ is matched to vertex i . The remaining $2n - 2$ vertices form the polygon and are paired distinctly. The relation $i - k \equiv i + k \pmod{2n - 1}$ implies $2k \equiv 0$, which is impossible for $1 \leq k \leq n - 1$ since $2n - 1$ is odd. Thus, every vertex has degree exactly 1 in F_i .

2. Decomposition:

- Every edge incident to the center $2n - 1$ is of the form $\{2n - 1, x\}$ and appears uniquely in factor F_x .
- For any edge $\{u, v\}$ not incident to $2n - 1$, the sum of the indices modulo $2n - 1$ is invariant for a fixed factor. Specifically, $u + v \equiv (i - k) + (i + k) \equiv 2i \pmod{2n - 1}$. Since $2n - 1$ is odd, $\gcd(2, 2n - 1) = 1$, meaning 2 is invertible. Thus, $i \equiv (u + v) \cdot 2^{-1} \pmod{2n - 1}$ is uniquely determined.

Therefore, every edge of K_{2n} belongs to exactly one factor, constituting a 1-factorization. ■

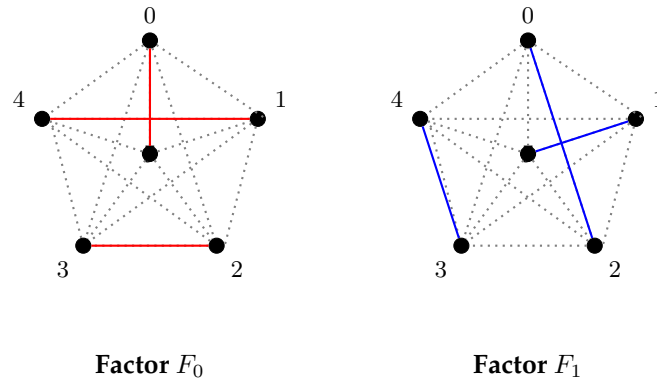


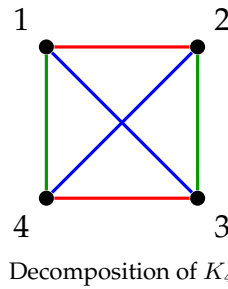
Figure 2.18: 1-Factorization of K_6 ($n = 3$). The solid colored lines represent the specific 1-factor, while the dotted lines represent the remaining edges of the complete graph.

Theorem 2.3.3. *For every positive integer n , the complete graph K_{2n+1} is 2-factorable into Hamiltonian cycles.*

Example 2.3.4 (1-Factorization of K_4). Let $n = 4$. The graph K_4 has 4 vertices and degree 3. It can be decomposed into $4 - 1 = 3$ perfect matchings (1-factors).

The Factors:

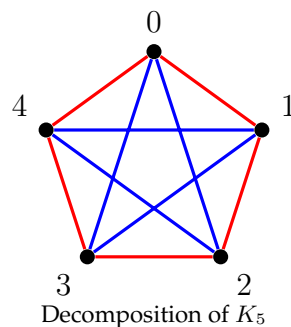
- F_1 (Red): Matches $\{(1, 2), (3, 4)\}$
- F_2 (Blue): Matches $\{(1, 3), (2, 4)\}$
- F_3 (Green): Matches $\{(1, 4), (2, 3)\}$



Example 2.3.5 (2-Factorization of K_5). Let $n = 5$. The graph K_5 has 5 vertices and degree 4. It can be decomposed into $(5 - 1)/2 = 2$ Hamiltonian cycles.

The Factors:

- H_1 (Red): The outer pentagon cycle $\{(0, 1), (1, 2), (2, 3), (3, 4), (4, 0)\}$.
- H_2 (Blue): The inner star cycle $\{(0, 2), (2, 4), (4, 1), (1, 3), (3, 0)\}$.



Theorem 2.3.6. *For every positive integer n , the complete graph K_{2n+1} is 2-factorable into Hamiltonian cycles.*

Proof. Let the vertex set be $V = \{0, 1, \dots, 2n\}$. We designate the vertex $2n$ as the fixed center and treat the remaining set $\{0, \dots, 2n-1\}$ as elements of $\mathbb{Z}_{2n}$.

We construct n Hamiltonian cycles $C_0, C_1, \dots, C_{n-1}$ as follows:

1. The Base Cycle C_0 : Define the cycle C_0 by the sequence of vertices:

$$C_0 = (2n, 0, 1, 2n-1, 2, 2n-2, \dots, n, 2n).$$

The path within $\mathbb{Z}_{2n}$ (excluding the fixed vertex $2n$) zigzags through the vertices. The lengths of the edges in this path correspond to the distances $1, 2, \dots, n$ in the modular arithmetic of $\mathbb{Z}_{2n}$.

2. The General Cycles: For each $i \in \{0, \dots, n-1\}$, let C_i be the cycle obtained by adding $i \pmod{2n}$ to every vertex in $\{0, \dots, 2n-1\}$, while keeping the vertex $2n$ fixed.

3. Verification:

- **Edges incident to $2n$:** In cycle C_i , the vertex $2n$ connects to i and $n+i$. As i ranges from 0 to $n-1$, these pairs $\{i, n+i\}$ are disjoint and cover all vertices in $\mathbb{Z}_{2n}$. Thus, every edge incident to $2n$ appears exactly once.
- **Edges within $\mathbb{Z}_{2n}$:** The base path in C_0 contains exactly one edge of each "length" class $d \in \{1, \dots, n\}$. By the properties of cyclic groups, rotating a set of edges that contains one representative of each length generates all possible edges within the complete graph induced by $\mathbb{Z}_{2n}$.

Thus, the collection $\{C_0, \dots, C_{n-1}\}$ forms a 2-factorization of K_{2n+1} . ■

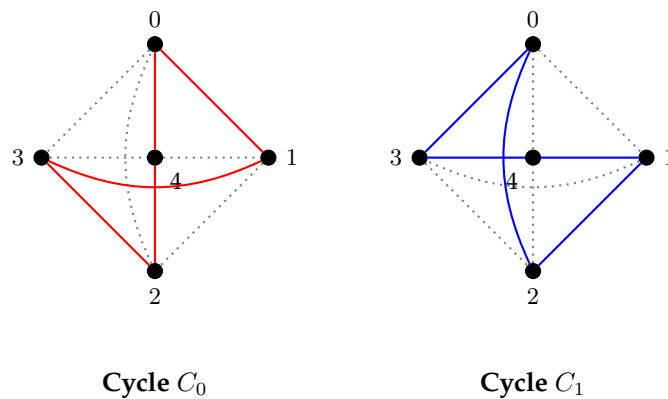


Figure 2.19: Decomposition of K_5 . We curve the diagonal edges (0-2 and 1-3) so they do not overlap with the central vertex 4. All 10 edges of K_5 are present in both figures (either solid or dotted).

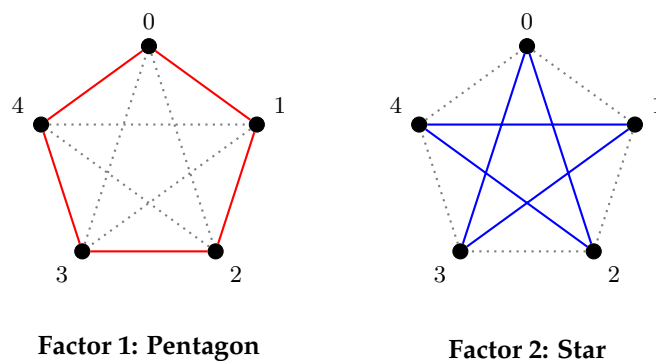


Figure 2.20: Decomposition of K_5 into 2 Hamiltonian Cycles. The dotted lines in each figure represent the edges belonging to the other factor.

Chapter 3

Edge Graphs

Please refer to Chapter 9 of the Book "An Introduction to Graph Theory" by S Pirzada.

Chapter 4

Spectral Theory of Graphs

INTRODUCTION

The algebraic study of graphs has two distinct but powerful roots: group theory and linear algebra. The study of symmetry began with **Arthur Cayley** (1878), who introduced graphical representations of groups. This connection was cemented by **Robert Frucht** in 1939, who proved the surprising result that any finite group is the automorphism group of some graph. Parallel to this, **Spectral Graph Theory** emerged from the intersection of mathematics and quantum chemistry. While early connections were made by Hückel in the 1930s regarding molecular orbitals, the mathematical foundations were solidified by Collatz and Sinogowitz (1957). The concept of **Graph Energy** was formally introduced by **Ivan Gutman** in 1978, bridging the gap between the sum of eigenvalues and the total π -electron energy of molecules. Why translate graph theory into the language of algebra? The answer lies in the power of invariants. **Automorphism groups** allow us to quantify the symmetry and indistinguishability of vertices, while **Cayley graphs** provide a geometric way to visualize abstract algebraic structures.

Simultaneously, the **Spectrum** (eigenvalues of the Adjacency or Laplacian matrix) acts as a "fingerprint" of the graph, revealing structural properties—such as connectivity, bipartiteness, and expansion—that are not immediately visible in a drawing. By studying invariants like **Graph Energy**, we gain precise tools to compare graphs, measure their structural complexity, and predict their behavior in physical and chemical systems.

This chapter explores the intersection of combinatorics and algebra, using group theory to analyze graph symmetries through automorphisms and Cayley digraphs. We examine fundamental connections between abstract groups and topological structures before transitioning to spectral graph theory, where linear algebra reveals hidden graph properties. By investigating the eigenvalues of adjacency and Laplacian matrices, we characterize standard graph classes and introduce the concept of graph energy. Finally, we demonstrate how these algebraic tools provide a quantitative measure of structural complexity and stability.

Video Resources

A detailed lecture series for this chapter is available at

<https://www.youtube.com/@Prof.DrAijaz/playlists>

or visit YouTube via <https://aijaz.vercel.app/>

4.1 Spectrum of Graphs

Let $G = (V, E)$ be a simple undirected graph with n vertices $\{v_1, \dots, v_n\}$ and m edges.

Definition 4.1.1 (Adjacency Matrix). The adjacency matrix $A(G)$ is an $n \times n$ matrix where:

$$A_{ij} = \begin{cases} 1 & \text{if } v_i \text{ is adjacent to } v_j \\ 0 & \text{otherwise} \end{cases}$$

For simple graphs, $A_{ii} = 0$ for all i (no self-loops).

Definition 4.1.2 (Spectrum). The spectrum of G , denoted $\text{Spec}(G)$, is the set of eigenvalues $\lambda_1, \lambda_2, \dots, \lambda_n$ of $A(G)$. These are the roots of the characteristic polynomial $\phi(G, \lambda) = \det(\lambda I - A)$. We typically order them as $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_n$.

Theorem 4.1.3 (Real Eigenvalues). The eigenvalues of the adjacency matrix $A(G)$ are all real numbers.

Proof. $A(G)$ is a real symmetric matrix (since the graph is undirected, $v_i \sim v_j \implies v_j \sim v_i$, so $A_{ij} = A_{ji}$). By the Spectral Theorem of linear algebra, all eigenvalues of a real symmetric matrix are real, and there exists an orthonormal basis of eigenvectors. ■

Theorem 4.1.4 (Sum of Eigenvalues). The sum of the eigenvalues of a simple graph is zero.

$$\sum_{i=1}^n \lambda_i = 0$$

Proof. For any matrix, the sum of the eigenvalues equals the trace (the sum of diagonal elements).

$$\sum_{i=1}^n \lambda_i = \text{tr}(A) = \sum_{i=1}^n A_{ii}$$

Since G is simple, it has no self-loops, meaning $A_{ii} = 0$ for all i . Thus, the sum is 0. ■

Lemma 4.1.5. Let A be the adjacency matrix of a graph G .

(A) The sum of all 2×2 principal minors of A equals $-|E(G)|$.

(B) The sum of all 3×3 principal minors of A equals twice the number of triangles of G .

Proof. Let $A = [a_{ij}]$ be the adjacency matrix. Since G is simple, $a_{ii} = 0$ and $a_{ij} = a_{ji} \in \{0, 1\}$.

(A) A 2×2 principal minor for indices $i < j$ is $\det \begin{pmatrix} 0 & a_{ij} \\ a_{ji} & 0 \end{pmatrix} = -a_{ij}^2 = -a_{ij}$. This value is -1 if $\{i, j\} \in E(G)$ and 0 otherwise. Summing over all pairs $\{i, j\}$ yields $-|E(G)|$.

(B) A 3×3 principal minor for indices $i < j < k$ is

$$\det \begin{pmatrix} 0 & a_{ij} & a_{ik} \\ a_{ji} & 0 & a_{jk} \\ a_{ki} & a_{kj} & 0 \end{pmatrix} = a_{ij}a_{jk}a_{ki} + a_{ik}a_{ji}a_{kj} = 2a_{ij}a_{jk}a_{ki}.$$

This product is 1 if edges $\{i, j\}, \{j, k\}, \{k, i\}$ all exist (forming a triangle), and 0 otherwise. Thus, the minor equals 2 if the vertices form a triangle and 0 otherwise. Summing over all vertex triplets gives twice the number of triangles. ■

Theorem 4.1.6 (Number of Walks and Second Moment). Let N_k be the total number of closed walks of length k in G . Then $\sum_{i=1}^n \lambda_i^k = N_k$. Specifically for $k = 2$:

$$\sum_{i=1}^n \lambda_i^2 = 2m$$

Proof. A standard result in graph theory states that the (i, j) -entry of the matrix A^k represents the number of walks of length k from vertex v_i to vertex v_j . A closed walk starts and ends at the same vertex. Thus, the number of closed walks is the sum of the diagonal entries of A^k :

$$N_k = \sum_{i=1}^n (A^k)_{ii} = \text{tr}(A^k)$$

If λ_i are eigenvalues of A , then λ_i^k are eigenvalues of A^k . The trace is the sum of eigenvalues:

$$\text{tr}(A^k) = \sum_{i=1}^n \lambda_i^k$$

For $k = 2$, a closed walk of length 2 starting at v_i must go to a neighbor v_j and return to v_i . This corresponds to traversing an edge back and forth. Each edge contributes 2 to the total count (once for each endpoint). Thus $N_2 = 2m$. ■

Theorem 4.1.7 (Spectrum of Bipartite Graphs). *A graph G is bipartite if and only if its spectrum is symmetric with respect to 0. That is, if λ is an eigenvalue, then $-\lambda$ is also an eigenvalue with the same multiplicity.*

Proof. Let G be bipartite with parts V_1 and V_2 . We can index vertices such that the adjacency matrix has the block form:

$$A = \begin{pmatrix} 0 & B \\ B^T & 0 \end{pmatrix}$$

Let λ be an eigenvalue with eigenvector $\mathbf{x} = \begin{pmatrix} \mathbf{u} \\ \mathbf{v} \end{pmatrix}$, where $\mathbf{u}$ corresponds to V_1 and $\mathbf{v}$ to V_2 . The equation $A\mathbf{x} = \lambda\mathbf{x}$ yields:

$$B\mathbf{v} = \lambda\mathbf{u} \quad \text{and} \quad B^T\mathbf{u} = \lambda\mathbf{v}$$

Now consider the vector $\mathbf{y} = \begin{pmatrix} \mathbf{u} \\ -\mathbf{v} \end{pmatrix}$.

$$A\mathbf{y} = \begin{pmatrix} 0 & B \\ B^T & 0 \end{pmatrix} \begin{pmatrix} \mathbf{u} \\ -\mathbf{v} \end{pmatrix} = \begin{pmatrix} -B\mathbf{v} \\ B^T\mathbf{u} \end{pmatrix} = \begin{pmatrix} -\lambda\mathbf{u} \\ \lambda\mathbf{v} \end{pmatrix} = -\lambda \begin{pmatrix} \mathbf{u} \\ -\mathbf{v} \end{pmatrix} = -\lambda\mathbf{y}$$

Thus, $-\lambda$ is also an eigenvalue. ■

Theorem 4.1.8. *The adjacency spectrum of the complete graph K_n is $\{(n-1)^{(1)}, (-1)^{(n-1)}\}$.*

Proof. The adjacency matrix is $A = J - I$, where J is the all-ones matrix. The matrix J has rank 1 with a non-zero eigenvalue n (corresponding to eigenvector $\mathbf{1}$) and eigenvalue 0 with multiplicity $n-1$. Since A is a polynomial of J , specifically $A = J - I$, its eigenvalues are obtained by shifting the eigenvalues of J by -1 . Thus, the spectrum is $n-1$ (multiplicity 1) and $0-1 = -1$ (multiplicity $n-1$). ■

Theorem 4.1.9. *The adjacency spectrum of the complete bipartite graph $K_{r,s}$ is $\{\sqrt{rs}^{(1)}, -\sqrt{rs}^{(1)}, 0^{(n-2)}\}$.*

Proof. The adjacency matrix of $K_{r,s}$ has only two linearly independent rows so its rank is 2. Consequently, 0 is an eigenvalue with multiplicity $n-2$. Since the trace is zero, implying the two non-zero eigenvalues are λ and $-\lambda$. Using the property that the sum of squares of eigenvalues equals twice the number of edges, we have $\lambda^2 + (-\lambda)^2 = 2rs$. This simplifies to $2\lambda^2 = 2rs$, yielding $\lambda = \sqrt{rs}$. ■

Lemma 4.1.10. *For $n \geq 2$, the eigenvalues of Q_n are $1, \omega, \omega^2, \dots, \omega^{n-1}$, where $\omega = e^{\frac{2\pi i}{n}}$, is the primitive n th root of unity.*

Proof. The characteristic polynomial of Q_n is $\det(Q_n - \lambda I) = (-1)^n(\lambda^n - 1)$. Clearly, the roots of the characteristic polynomial are the n roots of unity. □ ■

Theorem 4.1.11. For $n \geq 2$, the eigenvalues of C_n are $2 \cos \frac{2\pi k}{n}$, $k = 1, \dots, n$.

Proof. Note that $A(C_n) = Q_n + Q'_n = Q_n + Q_n^{n-1}$ is a polynomial in Q_n . Thus, the eigenvalues of $A(C_n)$ are obtained by evaluating the same polynomial at each of the eigenvalues of Q_n . Thus, by Lemma 3.5, the eigenvalues of $A(C_n)$ are $\omega^k + \omega^{n-k}$, $k = 1, \dots, n$. Note that

$$\begin{aligned}\omega^k + \omega^{n-k} &= \omega^k + \omega^{-k} \\ &= e^{\frac{2\pi i k}{n}} + e^{-\frac{2\pi i k}{n}} \\ &= 2 \cos \frac{2\pi k}{n},\end{aligned}$$

$k = 1, \dots, n$, and the proof is complete. ■

2nd Proof. Let the vertices of C_n be labeled $0, 1, \dots, n-1$ such that vertex j is adjacent to $(j-1) \pmod{n}$ and $(j+1) \pmod{n}$. The adjacency matrix A is a circulant matrix where the rows are cyclic permutations of the vector $(0, 1, 0, \dots, 0, 1)$.

Let $\omega = e^{i\frac{2\pi}{n}}$ be the primitive n -th root of unity. Consider the vector $\mathbf{x}_k = (1, \omega^k, \omega^{2k}, \dots, \omega^{(n-1)k})^T$. The r -th entry of $A\mathbf{x}_k$ is the sum of the neighbors of vertex r :

$$(A\mathbf{x}_k)_r = (\mathbf{x}_k)_{r-1} + (\mathbf{x}_k)_{r+1} = \omega^{k(r-1)} + \omega^{k(r+1)}$$

Factoring out ω^{kr} , we get:

$$(A\mathbf{x}_k)_r = \omega^{kr}(\omega^{-k} + \omega^k) = (\mathbf{x}_k)_r \cdot (\omega^{-k} + \omega^k)$$

Thus, $\mathbf{x}_k$ is an eigenvector with eigenvalue $\lambda_k = \omega^k + \omega^{-k}$. Using Euler's formula ($e^{ix} + e^{-ix} = 2 \cos x$), we conclude:

$$\lambda_k = 2 \cos \left(\frac{2\pi k}{n} \right)$$
■

Lemma 4.1.12. If $X = (x_1, x_2, \dots, x_n)$ is an eigenvector of the adjacency matrix A of the path P_n corresponding to the eigenvalue λ , then λ is an eigenvalue of C_{2n+2} with

$$Y = (x_1, \dots, x_n, 0, -x_n, \dots, -x_1, 0) \text{ and}$$

$$Z = (0, x_1, \dots, x_n, 0, -x_n, \dots, -x_1)$$

as the corresponding eigenvectors.

Proof. Given (λ, X) is an eigenpair of $A(P_n)$ where $X = (x_1, x_2, \dots, x_n)$. This gives

$$A(P_n) = \lambda X.$$

Let $P_n = v_1 - v_2 - \dots - v_{n-1} - v_n$. Then

$$A(P_n) \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = \lambda \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix},$$

which gives

$$\begin{aligned}x_2 &= \lambda x_1 \\ x_1 + x_3 &= \lambda x_2 \\ x_2 + x_4 &= \lambda x_3 \\ &\vdots \\ x_{n-2} + x_n &= \lambda x_{n-1} \\ x_{n-1} &= \lambda x_n\end{aligned}$$

Now,

$$A(C_{2n+2})Y = A(C_{2n+2}) \begin{bmatrix} x_1 \\ \vdots \\ x_n \\ 0 \\ -x_n \\ \vdots \\ -x_1 \\ 0 \end{bmatrix} = \begin{bmatrix} x_2 + 0 \\ x_1 + x_3 \\ \vdots \\ x_{n-1} + 0 \\ x_n + (-x_n) \\ 0 - x_{n-1} \\ \vdots \\ -x_2 + 0 \\ -x_1 + x_1 \end{bmatrix} = \begin{bmatrix} \lambda x_1 \\ \lambda x_2 \\ \vdots \\ \lambda x_n \\ \lambda \cdot 0 \\ -\lambda x_n \\ \vdots \\ -\lambda x_1 \\ -\lambda \cdot 0 \end{bmatrix} = \lambda Y.$$

Therefore, Y is an eigenvector of $A(C_{2n+2})$ corresponding to the eigenvalue λ . On similar lines. we have

$$A(C_{2n+2})Z = A(C_{2n+2}) \begin{bmatrix} 0 \\ x_1 \\ x_2 \\ \vdots \\ x_n \\ 0 \\ -x_n \\ \vdots \\ -x_1 \end{bmatrix} = \lambda Z$$

so that Z is an eigenvector of $A(C_{2n+2})$ corresponding to the eigenvalue λ . Thus, λ is an eigenvalue of C_{2n+2} having two linearly independent eigenvectors corresponding to it. ■

Theorem 4.1.13. *The eigenvalues of the path P_n are $2 \cos\left(\frac{\pi k}{n+1}\right)$, $k = 1, 2, \dots, n$.*

Proof. Let λ be an eigenvalue of $A(P_n)$ and let $X = (x_1, x_2, \dots, x_n)$ be the corresponding eigenvector. Then by previous theorem, we have

$$Y = (x_1, x_2, \dots, x_n, 0, -x_n, x_{n-1}, \dots, -x_1, 0) \text{ and} \\ Z = (0, x_1, x_2, \dots, x_n, 0, -x_n, -x_{n-1}, \dots, -x_1)$$

are the eigenvectors of C_{2n+2} corresponding to λ . Clearly Y and Z are linearly independent eigenvectors of C_{2n+2} corresponding to eigenvector λ . This means each eigenvalue λ of P_n is also an eigenvalue of C_{2n+2} with geometric multiplicity 2. But as adjacency matrix is symmetric (hence diagonalizable), thus arithmetic multiplicity of λ as an eigenvalue of C_{2n+2} is also 2. Hence the eigenvalues of path P_n are precisely those eigenvalues of C_{2n+2} whose arithmetic multiplicity is 2.

Now, the eigenvalues of C_{2n+2} are $2 \cos\left(\frac{2\pi k}{2n+2}\right) = 2 \cos\left(\frac{\pi k}{n+1}\right)$; $k = 1, 2, \dots, 2n+2$. Using values of $k = 1, 2, \dots, 2n+2$, these can be written as

$$2 \cos\left(\frac{\pi}{n+1}\right), 2 \cos\left(\frac{2\pi}{n+1}\right), \dots, 2 \cos\left(\frac{n\pi}{n+1}\right), 2 \cos\left(\frac{(n+1)\pi}{n+1}\right)$$

for $k = 1, 2, \dots, 2n+1$ and for $k = 2n+1, 2n, 2n-1, \dots, n+2$ and $2n+2$.

$$2 \cos\left(\frac{(2n+1)\pi}{n+1}\right), 2 \cos\left(\frac{2n\pi}{n+1}\right), \dots, 2 \cos\left(\frac{(n+2)\pi}{n+1}\right), 2 \cos\left(\frac{(2n+2)\pi}{n+1}\right)$$

These two lists each containing $n+1$ eigenvalues give the $2n+2$ eigenvalue of C_{2n+2} . Now, we have $\cos\left(\frac{(2n+2-p)\pi}{n+1}\right) = \cos\left(2\pi - \frac{p\pi}{n+1}\right) = \cos\left(\frac{p\pi}{n+1}\right)$. Therefore, for different values of p we see that the terms in the above lists are identical except the last term in each list. Thus the eigenvalues of P_n are

$$2 \cos\left(\frac{k\pi}{n+1}\right); \quad k = 1, 2, \dots, n.$$

Definition 4.1.14. Determinant of a matrix $A = [a_{ij}]_{n \times n}$ is defined as $|A| = \sum_{\sigma \in S_n} \text{sgn}(\sigma) \cdot \prod_{i=1}^n a_{i\sigma(i)}$ where

$$\text{sgn}(\sigma) = \begin{cases} +1 & \text{if } \sigma \text{ is even permutation} \\ -1 & \text{if } \sigma \text{ is an odd permutation} \end{cases}$$

Note: If a permutation can be decomposed into d -disjoint cycles of length ≥ 2 , then

$$\text{sgn}(\sigma) = (-1)^{n-d}.$$

Example 4.1.15. Evaluate $\begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix}$ using the above definition.

Solution. $|A| = \sum_{\sigma \in S_n} \text{sgn}(\sigma) \prod_{i=1}^n a_{i\sigma(i)}$. Here $S_2 = \{\sigma_1 = I, \sigma_2 = (12)\}$. Therefore,

$$\begin{aligned} |A| &= \text{sgn}(\sigma_1) \prod_{i=1}^2 a_{i\sigma_1(i)} + \text{sgn}(\sigma_2) \prod_{i=1}^2 a_{i\sigma_2(i)} \\ &= \text{sgn}(\sigma_1) a_{1\sigma_1(1)} a_{2\sigma_1(2)} + \text{sgn}(\sigma_2) a_{1\sigma_2(1)} a_{2\sigma_2(2)} \\ &= 1 \cdot a_{11} a_{22} + (-1) a_{12} a_{21} \\ &= a_{11} a_{22} - a_{12} a_{21}. \end{aligned}$$

Definition 4.1.16 (Elementary Spanning Subgraph). A spanning subgraph $H \subseteq G$ is *elementary* if every component of H is either:

1. An edge (K_2), or
2. A cycle (C_k with $k \geq 3$).

The relationship between graph structure and the characteristic polynomial $P(G, \lambda) = \det(\lambda I - A)$ is given by the following theorem, often attributed to Harary and Sachs.

Theorem 4.1.17 (Determinant Expansion). Let A be the adjacency matrix of a simple graph G . Then:

$$\det(A) = \sum_H (-1)^{n-c_1(H)-c(H)} 2^{c(H)}$$

where the sum runs over all elementary spanning subgraphs H . Here, $c_1(H)$ is the number of K_2 components and $c(H)$ is the number of cycle components ($C_k, k \geq 3$).

Example 4.1.18. Find the determinant of the adjacency matrix of the graph given in figure below.

Sol: We have

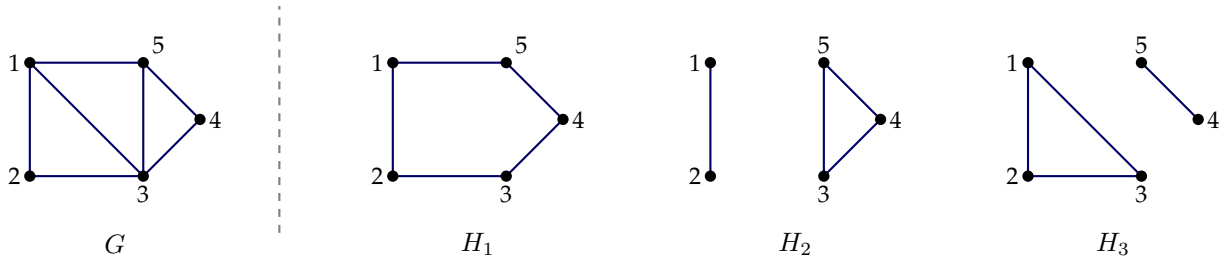
$$|A| = \sum_{H \in \mathcal{H}_G} (-1)^{n-c_1(H)-c(H)} \cdot 2^{c(H)}.$$

Here we have

$$\begin{aligned} H_1 &= \{(1, 2, 3, 4, 5)\}, & H_2 &= \{(1, 2), (3, 4, 5)\}, \\ H_3 &= \{(1, 2, 3), (4, 5)\} \end{aligned}$$

Hence,

$$|A| = (-1)^{5-0-1} \cdot 2^1 + (-1)^{5-1-1} \cdot 2^1 + (-1)^{5-1-1} \cdot 2^1.$$

Figure 4.1: Elementary spanning subgraphs of G .

Proof of Theorem 4.1.17. Let $A = [a_{ij}]$ be the adjacency matrix of G . By the definition of the determinant:

$$|A| = \sum_{\sigma \in S_n} \text{sgn}(\sigma) \prod_{i=1}^n a_{i\sigma(i)}$$

Consider a permutation $\sigma \in S_n$. Any permutation can be written as a product of disjoint cycles. We are only interested in permutations σ that contribute a non-zero quantity to the sum. Since G is a simple graph, $a_{ii} = 0$ for all i . Thus, for the product $\prod a_{i\sigma(i)}$ to be non-zero, $\sigma(i) \neq i$ for all i . Consequently, the disjoint cycle decomposition of such a σ contains no cycles of length 1. Every cycle must have length ≥ 2 .

These non-zero contributions correspond to spanning subgraphs H where every vertex has degree exactly 1 (if part of a transposition) or degree 2 (if part of a longer cycle). Specifically:

- A cycle of length 2 in σ corresponds to a K_2 component (an edge) in H .
- A cycle of length $k \geq 3$ in σ corresponds to a cycle subgraph C_k in H .

Thus, such a subgraph H is an elementary spanning subgraph.

The sign of the permutation is given by $\text{sgn}(\sigma) = (-1)^{n-d}$, where d is the number of disjoint cycles in the decomposition of σ . Here, $d = c_1(H) + c(H)$, where $c_1(H)$ is the number of cycles of length 2 (K_2) and $c(H)$ is the number of cycles of length ≥ 3 .

Regarding the magnitude of the contribution:

- For a K_2 component (transposition (ij)), the term is $a_{ij}a_{ji} = 1 \cdot 1 = 1$. There is only one way to traverse an edge.
- For a cycle component C_k ($k \geq 3$) with vertices $v_1, v_2, \dots, v_k$, the permutation can correspond to the cycle $(v_1v_2 \dots v_k)$ or the reverse $(v_kv_{k-1} \dots v_1)$. Both result in the product of adjacency entries being 1. Thus, every cycle component of length ≥ 3 contributes a factor of 2 to the sum (accounting for the two possible directions of traversal).

Therefore, combining the sign and the multiplicity of cycle traversals:

$$|A| = \sum_H (-1)^{n-(c_1(H)+c(H))} \cdot 2^{c(H)}.$$

■

Theorem 4.1.19 (Sachs Theorem). Let $P(G, \lambda) = \lambda^n + a_1\lambda^{n-1} + \dots + a_n$ be the characteristic polynomial of G . The coefficient a_i is given by:

$$a_i = \sum_{H_i \in \mathcal{H}_i} (-1)^{p(H_i)} 2^{c(H_i)}$$

where: H_i is the set of elementary subgraphs of G with exactly i vertices, $p(H_i)$ is the number of components in H_i , and $c(H_i)$ is the number of cycle components in H_i .

Proof. Let G be a graph with adjacency matrix $A(G)$. From Matrix Theory, the coefficient a_i of the characteristic polynomial is given by:

$$a_i = (-1)^i [\text{Sum of all } i \times i \text{ principal minors of } A]$$

An $i \times i$ principal minor of A corresponds to the determinant of the adjacency matrix of an induced subgraph of G with i vertices. Let such a subgraph be denoted by G' . Applying the previous **Determinant Expansion Theorem** to this subgraph G' of order i :

$$\det(A(G')) = \sum_{H \subseteq G'} (-1)^{i-c_1(H)-c(H)} 2^{c(H)}$$

where H represents an elementary spanning subgraph of the induced subgraph G' (which is simply an elementary subgraph of G with i vertices).

Substituting this back into the expression for a_i :

$$\begin{aligned} a_i &= (-1)^i \sum_{H_i \in \mathcal{H}_i} (-1)^{i-c_1(H_i)-c(H_i)} 2^{c(H_i)} \\ &= \sum_{H_i \in \mathcal{H}_i} (-1)^{2i-c_1(H_i)-c(H_i)} 2^{c(H_i)} \end{aligned}$$

Since $2i$ is even, $(-1)^{2i} = 1$. Also, let $p(H_i)$ be the total number of components in H_i , so $p(H_i) = c_1(H_i) + c(H_i)$. The exponent of -1 becomes $-(c_1(H_i) + c(H_i)) = -p(H_i)$. Since $(-1)^{-x} = (-1)^x$, we have:

$$a_i = \sum_{H_i \in \mathcal{H}_i} (-1)^{p(H_i)} 2^{c(H_i)}.$$

■

4.2 Isomorphism and Co-spectral Graphs

Definition 4.2.1. Two graphs G_1 and G_2 are **co-spectral** if $\text{Spec}(G_1) = \text{Spec}(G_2)$.

While isomorphic graphs are always co-spectral, the converse is not true.

Example 4.2.2. The smallest non-isomorphic co-spectral pair consists of:

- $G_1 = K_{1,4}$ (The Star graph)
- $G_2 = C_4 \cup K_1$ (The union of a cycle and an isolated vertex) — (See Figure 4.2)

Moreover, the connected non-isomorphic graphs of smallest order that are cospectral are given in Figure 4.3 having the characteristic polynomial

$$\phi(G, \lambda) = \lambda^6 - 7\lambda^4 - 4\lambda^3 + 7\lambda^2 + 4\lambda - 1.$$



Figure 4.2: Non-isomorphic co-spectral graphs of smallest orders

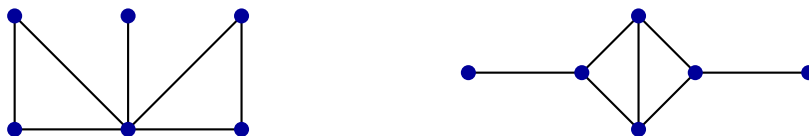


Figure 4.3: Smallest connected co-spectral non-isomorphic graphs

4.3 Regular Graphs

Definition 4.3.1. A graph G is said to be k -regular if each vertex of G has degree k .

Theorem 4.3.2. Let G be a k -regular graph of order n . Then:

1. k is an eigenvalue of $A(G)$ with multiplicity at least 1, and $\mathbf{1} = (1, 1, \dots, 1)^\top$ is a corresponding eigenvector.
2. For any eigenvalue λ of G , $|\lambda| \leq k$.
3. The multiplicity of k is 1 if and only if G is connected.

Proof. Let $A(G)$ be the adjacency matrix of G . Since G is k -regular, every row sum and column sum of $A(G)$ is equal to k . Therefore, k is an eigenvalue of $A(G)$.

Observing the matrix-vector product:

$$A(G)\mathbf{1} = \begin{bmatrix} \dots & \text{row 1} & \dots \\ \vdots & \vdots & \vdots \\ \dots & \text{row } n & \dots \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{bmatrix} = \begin{bmatrix} k \\ k \\ \vdots \\ k \end{bmatrix} = k \begin{bmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{bmatrix}.$$

Thus, $\mathbf{1} = (1, 1, \dots, 1)^\top$ is an eigenvector of $A(G)$ corresponding to k .

Next, we consider the multiplicity. We show that if G is connected, the geometric multiplicity $GM(k) = 1$. Since $A(G)$ is real and symmetric, the algebraic multiplicity $AM(k)$ equals $GM(k)$.

Let $X = (x_1, x_2, \dots, x_n)^\top$ be an eigenvector of $A(G)$ corresponding to k . By definition, $AX = kX$. Let x_j be an entry in X with the largest absolute value. Consider the j -th row of the eigenvalue equation:

$$(A(G)X)_j = kx_j \implies \sum_{v_i \sim v_j} x_i = kx_j,$$

where the sum runs over the k neighbors v_i adjacent to v_j . Since x_j has the maximum absolute value, $x_i \leq x_j$. For the average of the neighbors to equal the maximum value x_j , every neighbor must also equal x_j :

$$\sum_{v_i \sim v_j} x_i = \underbrace{x_j + x_j + \dots + x_j}_{k \text{ times}}.$$

This implies $x_i = x_j$ for all neighbors v_i of v_j . Propagating this argument through the graph (assuming G is connected), we conclude that $x_i = x_j$ for all $i = 1, 2, \dots, n$.

Therefore, any eigenvector for k is a scalar multiple of $\mathbf{1}$:

$$X = c \begin{bmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{bmatrix}.$$

This implies the eigenspace has dimension 1, so $GM(k) = 1$, and consequently $AM(k) = 1$. ■

Short Proof for Bound. Since G is k -regular, $A\mathbf{1} = k\mathbf{1}$. To show $|\lambda| \leq k$, let X be an eigenvector for an eigenvalue λ . Let x_j be the entry of X with the largest absolute value ($|x_j| > 0$). The eigenvalue equation implies:

$$|\lambda x_j| = \left| \sum_{v_i \sim v_j} x_i \right| \leq \sum_{v_i \sim v_j} |x_i|.$$

Since $|x_i| \leq |x_j|$, we have:

$$|\lambda| |x_j| \leq \sum_{v_i \sim v_j} |x_j| = k |x_j|.$$

Dividing by $|x_j|$, we obtain $|\lambda| \leq k$. ■

4.3.1 Spectrum of the Complement

Theorem 4.3.3. *If G is a k -regular graph with spectrum $\{k, \lambda_2, \dots, \lambda_n\}$, then the spectrum of the complement graph $\bar{G}$ is:*

$$\text{Spec}(\bar{G}) = \{n - k - 1, -\lambda_2 - 1, \dots, -\lambda_n - 1\}.$$

Proof. Since G is k -regular, k is an eigenvalue of G with the corresponding eigenvector $u = (1, 1, \dots, 1)^\top$. Let A and $\bar{A}$ be the adjacency matrices of G and $\bar{G}$ respectively. We have the relation:

$$A + \bar{A} = J_n - I_n,$$

where J_n is the all-ones matrix and I_n is the identity matrix.

First, apply this to the eigenvector u :

$$(A + \bar{A})u = (J_n - I_n)u$$

$$Au + \bar{A}u = J_n u - I_n u.$$

Using the facts that $Au = ku$, $I_n u = u$, and $J_n u = nu$:

$$ku + \bar{A}u = nu - u$$

$$\bar{A}u = (n - 1)u - ku$$

$$\bar{A}u = (n - k - 1)u.$$

Hence, $n - k - 1$ is an eigenvalue of $\bar{A}$ with eigenvector u .

Now, let X be any other eigenvector of A corresponding to an eigenvalue λ_i (for $i = 2, \dots, n$). Since A is symmetric, eigenvectors corresponding to distinct eigenvalues are orthogonal. Thus, X is orthogonal to u , which implies the sum of entries of X is zero. Consequently, $J_n X = 0$. (Note that $J_n X = 0 \cdot X$ because any vector in $\mathbb{R}^n$, not a scalar multiple of $(1, 1, \dots, 1)$, is an eigenvector of J_n corresponding to the eigenvalue 0).

Consider the action of $\bar{A}$ on X :

$$(A + \bar{A})X = (J_n - I_n)X$$

$$AX + \bar{A}X = J_n X - I_n X$$

$$\lambda_i X + \bar{A}X = 0 - X$$

$$\bar{A}X = -X - \lambda_i X$$

$$\bar{A}X = (-\lambda_i - 1)X.$$

Hence, for each $\lambda_i \in \text{Spec}(G) \setminus \{k\}$, the value $(-\lambda_i - 1)$ is an eigenvalue of $\bar{A}$. ■

We now turn to line graphs. Let G be a graph with $V(G) = \{1, \dots, n\}$ and $E(G) = \{e_1, \dots, e_m\}$. Recall that the line graph G_ℓ of G has vertex set $E(G)$. For $i \neq j$, e_i and e_j are said to be adjacent if they have a common vertex. If G is k -regular then G_ℓ is $(2k - 2)$ -regular. We first prove a preliminary result.

Lemma 4.3.4. *Let G be a graph with n vertices. Let A and B be the adjacency matrices of G and of G_ℓ , respectively. If M is the incidence matrix of G , then $M'M = B + 2I$. Furthermore, if G is k -regular then $MM' = A + kI$.*

Proof. Any diagonal entry of $M'M$ clearly equals 2. If e_i and e_j are edges of G then the (i, j) -element of $M'M$ is 1 if e_i and e_j have a common vertex, and 0 otherwise. Hence, $M'M = B + 2I$. To prove the second part, note that for a k -regular graph, $MM' = -L + 2kI$, where L is the Laplacian of G . Hence, $A = kI - L = MM' - kI$. Therefore, $MM' = A + kI$. ■

Corollary 4.3.5. *Let G be a graph. If μ is an eigenvalue of G_ℓ then $\mu \geq -2$.*

Proof. Let B be the adjacency matrix of G_ℓ . By Lemma 4.3.4, $B + 2I = M'M$ is positive semidefinite. If μ is an eigenvalue of B then $\mu + 2$, being an eigenvalue of a positive semidefinite matrix, must be nonnegative. Hence, $\mu \geq -2$. ■

Theorem 4.3.6. *Let G be a k -regular graph with n vertices. Then*

$$\phi(G_\ell, \lambda) = (\lambda + 2)^{\frac{n}{2}(k-2)} \phi(G, \lambda + 2 - k).$$

Proof. Let A and B be the adjacency matrices of G and of G_ℓ , respectively. Let M be the incidence matrix of G . If G has m edges then M is of order $n \times m$. Let $k = \lambda_1, \lambda_2, \dots, \lambda_n$ be the eigenvalues of A . By Lemma 4.3.4 the eigenvalues of MM' are $2k, \lambda_2 + k, \dots, \lambda_n + k$. Note that the eigenvalues of $M'M$ are given by the eigenvalues of MM' , together with 0 with multiplicity $m - n$. Therefore, again by Lemma 4.3.4, the eigenvalues of B are $2k - 2, \lambda_2 + k - 2, \dots, \lambda_n + k - 2$, and -2 with multiplicity $m - n$. Since

$$\phi(G, \lambda) = (\lambda - k)(\lambda - \lambda_2) \cdots (\lambda - \lambda_n),$$

then

$$\phi(G, \lambda + 2 - k) = (\lambda + 2 - 2k)(\lambda + 2 - k - \lambda_2) \cdots (\lambda + 2 - k - \lambda_n).$$

Also,

$$\phi(G_\ell, \lambda) = (\lambda + 2 - 2k)(\lambda + 2 - k - \lambda_2) \cdots (\lambda + 2 - k - \lambda_n)(\lambda + 2)^{m-n}.$$

$$\frac{\phi(G_\ell, \lambda)}{\phi(G, \lambda + 2 - k)} = (\lambda + 2)^{m-n}.$$

Since G is k -regular then $2m = nk$ and hence $m - n = \frac{n}{2}(k - 2)$. ■

4.4 The Laplacian Spectrum of a Graph

4.4.1 Definitions

Let $G = (V, E)$ be a simple undirected graph with n vertices and m edges.

- Let $A(G)$ be the **adjacency matrix** of G .
- Let $D(G)$ be the **degree matrix** of G , a diagonal matrix where $D_{ii} = \deg(v_i)$.

Definition 4.4.1 (Laplacian Matrix). The Laplacian matrix $L(G)$ is defined as:

$$L(G) = D(G) - A(G)$$

Explicitly, the entries are given by:

$$L_{ij} = \begin{cases} \deg(v_i) & \text{if } i = j \\ -1 & \text{if } i \neq j \text{ and } v_i \sim v_j \\ 0 & \text{otherwise} \end{cases}$$

Definition 4.4.2 (Laplacian Spectrum). Since $L(G)$ is a real symmetric matrix, its eigenvalues are real. The Laplacian spectrum is the set of eigenvalues of $L(G)$, usually denoted and ordered as:

$$\mu_1 \geq \mu_2 \geq \cdots \geq \mu_n$$

Theorem 4.4.3 (Positive Semidefinite Property). *The Laplacian matrix $L(G)$ is positive semidefinite. Consequently, $\mu_i \geq 0$ for all $i = 1, \dots, n$.*

Proof. Let $\mathbf{x} = (x_1, x_2, \dots, x_n)^T \in \mathbb{R}^n$. Consider the quadratic form associated with L :

$$\begin{aligned}
 \mathbf{x}^T L \mathbf{x} &= \mathbf{x}^T (D - A) \mathbf{x} \\
 &= \mathbf{x}^T D \mathbf{x} - \mathbf{x}^T A \mathbf{x} \\
 &= \sum_{i=1}^n \deg(v_i) x_i^2 - \sum_{i=1}^n \sum_{j=1}^n A_{ij} x_i x_j \\
 &= \sum_{i=1}^n \left(\sum_{j \sim i} 1 \right) x_i^2 - \sum_{(i,j) \in E} 2x_i x_j \\
 &= \sum_{(i,j) \in E} (x_i^2 + x_j^2 - 2x_i x_j) \\
 &= \sum_{(i,j) \in E} (x_i - x_j)^2
 \end{aligned}$$

Since $(x_i - x_j)^2 \geq 0$ for any edge, the sum is non-negative. Thus, $\mathbf{x}^T L \mathbf{x} \geq 0$ for all $\mathbf{x} \neq \mathbf{0}$. It follows that all eigenvalues μ_i are non-negative. ■

Theorem 4.4.4 (Smallest Eigenvalue). *The smallest eigenvalue of $L(G)$ is always 0. That is, $\mu_n = 0$.*

Proof. Since the row sums of $L(G)$ are all zero (because the diagonal entry is the degree and the off-diagonal entries are -1 for each neighbor), the vector of all ones, $\mathbf{1} = (1, 1, \dots, 1)^T$, is an eigenvector.

$$(L\mathbf{1})_i = \sum_{j=1}^n L_{ij} \cdot 1 = \deg(v_i) - \sum_{j \sim i} 1 = \deg(v_i) - \deg(v_i) = 0$$

Thus, $L\mathbf{1} = 0 \cdot \mathbf{1}$. Since L is positive semidefinite, 0 is the smallest possible eigenvalue. ■

Theorem 4.4.5 (Trace Property). *The sum of the Laplacian eigenvalues is twice the number of edges:*

$$\sum_{i=1}^n \mu_i = 2m$$

Proof. The sum of the eigenvalues is equal to the trace of the matrix.

$$\sum_{i=1}^n \mu_i = \text{tr}(L) = \sum_{i=1}^n L_{ii} = \sum_{i=1}^n \deg(v_i)$$

By the Handshaking Lemma, $\sum \deg(v_i) = 2m$. ■

4.5 Laplacian Energy

The concept of graph energy $E(G)$ (sum of absolute values of adjacency eigenvalues) was introduced by Gutman in 1978. In 2006, Gutman and Zhou introduced the **Laplacian Energy**.

The average degree of the graph is $\frac{2m}{n}$. This is also the spectral mean of the Laplacian matrix (average of eigenvalues).

Definition 4.5.1 (Laplacian Energy). The Laplacian energy of a graph G , denoted by $LE(G)$, is defined as:

$$LE(G) = \sum_{i=1}^n \left| \mu_i - \frac{2m}{n} \right|$$

Theorem 4.5.2 (Relation to Graph Energy for Regular Graphs). *If G is a k -regular graph, then the Laplacian energy equals the ordinary graph energy.*

$$LE(G) = E(G)$$

Proof. Let G be a k -regular graph. The degree matrix is a scalar matrix $D = kI$. The Laplacian Matrix is $L = kI - A$. If $\lambda_1, \dots, \lambda_n$ are the eigenvalues of the adjacency matrix A , then the eigenvalues of L are $\mu_i = k - \lambda_i$. The number of edges is $m = \frac{nk}{2}$, so the average degree is $\frac{2m}{n} = k$.

Substituting these into the definition of Laplacian Energy:

$$\begin{aligned} LE(G) &= \sum_{i=1}^n \left| \mu_i - \frac{2m}{n} \right| \\ &= \sum_{i=1}^n |(k - \lambda_i) - k| \\ &= \sum_{i=1}^n |-\lambda_i| \\ &= \sum_{i=1}^n |\lambda_i| \\ &= E(G) \end{aligned}$$

■

Theorem 4.5.3 (Basic Properties of Laplacian Energy). *For any graph G :*

- (i) $LE(G) \geq 0$.
- (ii) $LE(G) = 0$ if and only if $m = 0$ (the graph has no edges).

Proof. (i). The sum of absolute values is always non-negative.

(ii). If $m = 0$, then L is the zero matrix, all $\mu_i = 0$, and $\frac{2m}{n} = 0$, so $LE(G) = 0$. Conversely, if $LE(G) = 0$, then $\mu_i = \frac{2m}{n}$ for all i . Since $\mu_n = 0$ (smallest eigenvalue), it implies $\frac{2m}{n} = 0$, which means $m = 0$.

■

Theorem 4.5.4. *The Laplacian spectrum of the complete graph K_n is $\{0^{(1)}, n^{(n-1)}\}$.*

Proof. The Laplacian matrix is given by $L(K_n) = nI - J$, where I is the identity matrix and J is the all-ones matrix. The spectrum of J is known to be $\{n^{(1)}, 0^{(n-1)}\}$, corresponding to the eigenvector $\mathbf{1}$ and the subspace orthogonal to $\mathbf{1}$, respectively. Since $L(K_n)$ is a linear combination of I and J , its eigenvalues are obtained via the transformation $\mu_i = n - \lambda_i(J)$.

- For $\lambda(J) = n$, we have $\mu = n - n = 0$ (multiplicity 1).
- For $\lambda(J) = 0$, we have $\mu = n - 0 = n$ (multiplicity $n - 1$).

■

Theorem 4.5.5. *The Laplacian spectrum of the complete bipartite graph $K_{r,s}$ is $\{0^{(1)}, (r+s)^{(1)}, r^{(s-1)}, s^{(r-1)}\}$.*

Proof. Let V_1 and V_2 be the partition sets with $|V_1| = r$ and $|V_2| = s$. The row sums of L are zero, so $\mu_1 = 0$ is an eigenvalue with eigenvector $\mathbf{1}$.

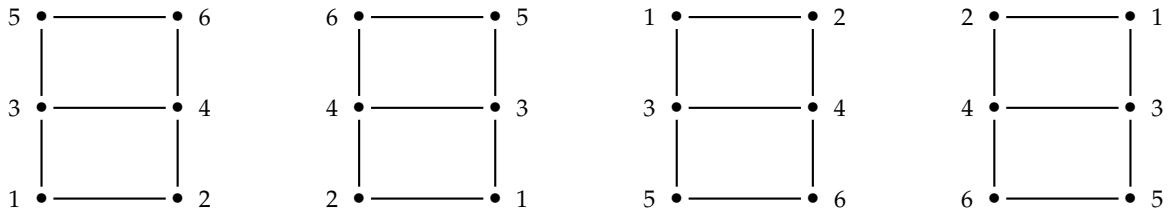
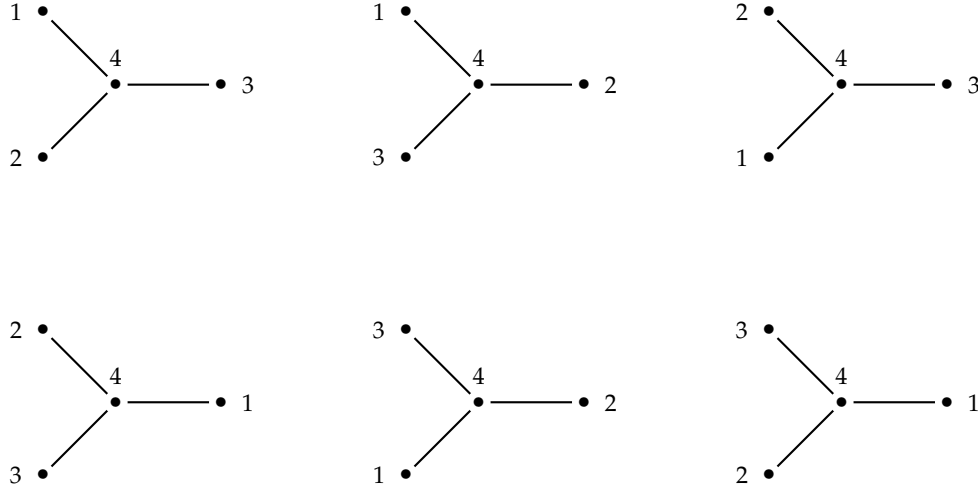
Consider vectors $\mathbf{x}$ that sum to zero on V_1 and are zero on V_2 . There are $r - 1$ such linearly independent vectors. For these vectors, $L\mathbf{x} = s\mathbf{x}$, yielding eigenvalue s with multiplicity $r - 1$. Similarly, considering vectors supported only on V_2 that sum to zero, we obtain eigenvalue r with multiplicity $s - 1$.

We find the final eigenvalue using the trace of the matrix. The trace of L is the sum of degrees: $\text{tr}(L) = r(s) + s(r) = 2rs$. Let μ_n be the unknown eigenvalue. Then:

$$\begin{aligned} 0 + s(r - 1) + r(s - 1) + \mu_n &= 2rs \\ (rs - s) + (rs - r) + \mu_n &= 2rs \\ 2rs - (r + s) + \mu_n &= 2rs \implies \mu_n = r + s \end{aligned}$$

Thus, the spectrum is established.

■

Figure 4.4: Automorphism group of $G_{2,3}$.Figure 4.5: Automorphisms of star graph S_4 .

4.6 Automorphism Groups of Graphs

Definition 4.6.1. A graph automorphism of G is a permutation φ on the set of vertices $V(G)$ that satisfies the property that $\{u_i, u_j\} \in E(G)$ if and only if $\{\varphi(u_i), \varphi(u_j)\} \in E(G)$.

Example 4.6.2. For example, the grid graph $G_{2,3}$ has four automorphisms: $(1, 2, 3, 4, 5, 6)$, $(2, 1, 4, 3, 6, 5)$, $(5, 6, 3, 4, 1, 2)$, $(6, 5, 4, 3, 2, 1)$. These correspond to the graph itself, the graph flipped from left to right, the graph flipped up down, and the graph flipped left to right and up down, respectively.

The star graph S_4 has six automorphisms: $(1, 2, 3, 4)$, $(1, 3, 2, 4)$, $(2, 1, 3, 4)$, $(2, 3, 1, 4)$, $(3, 1, 2, 4)$, $(3, 2, 1, 4)$.

Theorem 4.6.3. The set $\text{Aut}(G)$ of all graph automorphisms of a graph G forms a group under function composition.

Definition 4.6.4 (Identity Graph). A graph G whose automorphism group is $\text{Aut}(G) \cong S_1$ is called the identity graph.

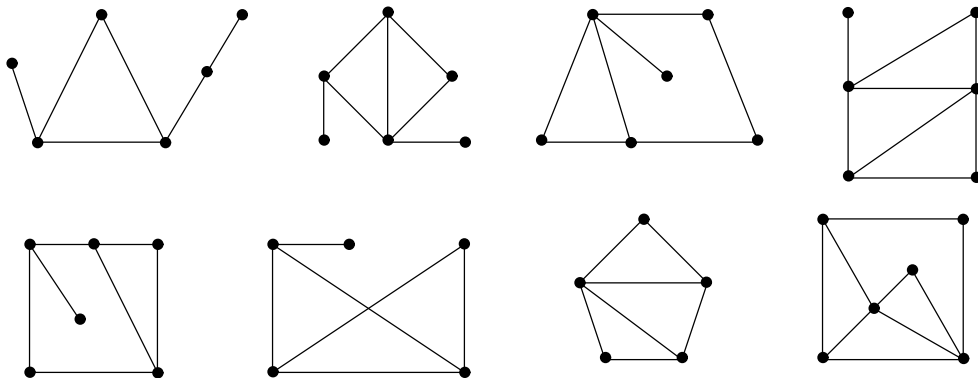


Figure 4.6: The Eight Identity Graphs of order Six.

The elements of $Aut(G)$ induce a group of permutations on the edge set E of G in the obvious way:

If $e = uv \in E$ and $\alpha \in Aut(G)$, then $\alpha_1(e) = \alpha_0(u)\alpha_0(v)$ where α_0 and α_1 denote the action of α on V and E respectively. This permutation group on E with degree $m = |E|$ and order $|Aut(G)|$ is called the induced edge group of G and is denoted by $Aut_I(G)$.

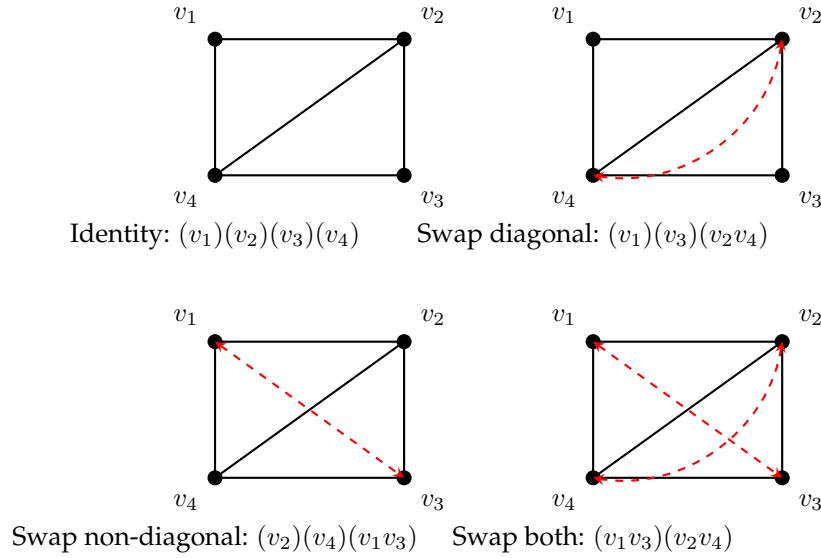


Figure 4.7: Automorphisms of $K_4 - e$

Example 4.6.5. Consider the graph $K_4 - e$ with vertices, Figure 4.7, labeled v_1, v_2, v_3, v_4 and edges e_1, e_2, e_3, e_4, e_5 . The vertex group $Aut(K_4 - e)$ consists of the four permutations:

$$(v_1)(v_2)(v_3)(v_4), \quad (v_1)(v_3)(v_2v_4), \quad (v_2)(v_4)(v_1v_3), \quad (v_1v_3)(v_2v_4)$$

or

$$\begin{pmatrix} v_1 & v_2 & v_3 & v_4 \\ v_1 & v_2 & v_3 & v_4 \end{pmatrix}, \quad \begin{pmatrix} v_1 & v_3 & v_2 & v_4 \\ v_1 & v_3 & v_4 & v_2 \end{pmatrix}, \quad \begin{pmatrix} v_2 & v_4 & v_1 & v_3 \\ v_2 & v_4 & v_3 & v_1 \end{pmatrix}, \quad \begin{pmatrix} v_1 & v_3 & v_2 & v_4 \\ v_3 & v_1 & v_4 & v_2 \end{pmatrix}.$$

The identity permutation of the vertex-group induces the identity permutation on the edges, while $(v_1)(v_3)(v_2v_4)$ induces a permutation on the edges which fixes e_5 , interchanges e_1 with e_4 , and e_2 with e_3 . In this way, the edge-group consists of the following permutations, induced respectively by the above members of the vertex-group.

$$(e_1)(e_2)(e_3)(e_4)(e_5), \quad (e_1e_4)(e_2e_3)(e_5), \quad (e_1e_2)(e_3e_4)(e_5), \quad (e_1e_3)(e_2e_4)(e_5)$$

or

$$\begin{pmatrix} e_1 & e_2 & e_3 & e_4 & e_5 \\ e_1 & e_2 & e_3 & e_4 & e_5 \end{pmatrix}, \quad \begin{pmatrix} e_1 & e_4 & e_2 & e_3 & e_5 \\ e_4 & e_1 & e_3 & e_2 & e_5 \end{pmatrix}, \quad \begin{pmatrix} e_1 & e_2 & e_3 & e_4 & e_5 \\ e_2 & e_1 & e_4 & e_3 & e_5 \end{pmatrix}, \quad \begin{pmatrix} e_1 & e_2 & e_3 & e_4 & e_5 \\ e_3 & e_1 & e_4 & e_2 & e_5 \end{pmatrix}.$$

Of course, the edge-group and the vertex-group of $K_4 - e$ are isomorphic. But they are not identical permutation groups since $Aut_I(K_4 - e)$ has degree 5 and $Aut(K_4 - e)$ has degree 4. Note that edge e_5 is fixed by every member of the edge-group. Even the permutation group obtained from $Aut_I(K_4 - e)$ by restricting its object set to e_1, e_2, e_3, e_4 is not identical with $Aut(K_4 - e)$, since these two isomorphic permutation groups of the same degree have different cycle structure.

Theorem 4.6.6. The vertex automorphism group $Aut(G)$ and the induced edge automorphism group $Aut_I(G)$ of a graph G are isomorphic if and only if G contains at most one isolated vertex and no connected component isomorphic to K_2 .

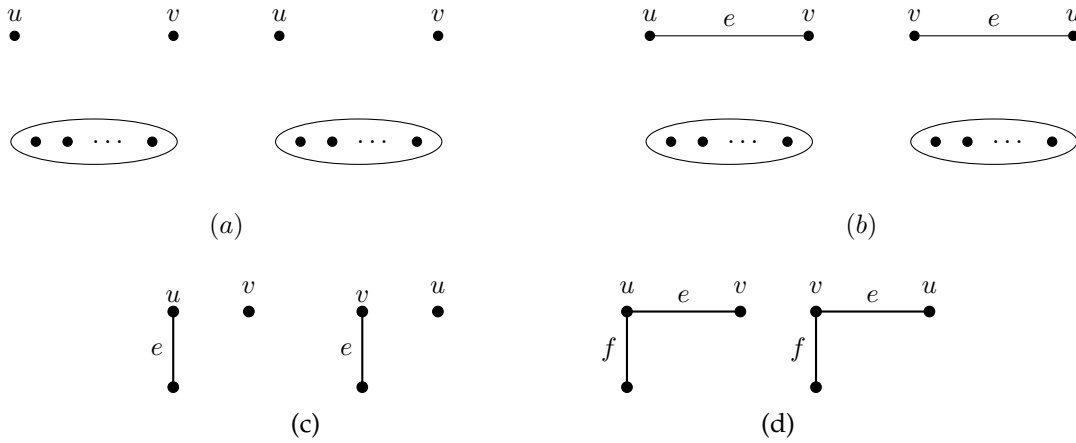


Figure 4.8: Theorem 3.4.5

Proof. Let $\phi : \text{Aut}(G) \rightarrow \text{Aut}_I(G)$ be the natural homomorphism defined by $\phi(\alpha_0) = \alpha_1$, where the edge automorphism α_1 is induced by the vertex automorphism α_0 according to the rule $\alpha_1(\{u, v\}) = \{\alpha_0(u), \alpha_0(v)\}$. Since every element of $\text{Aut}_I(G)$ arises from some vertex automorphism, ϕ is surjective. Consequently, $\text{Aut}(G) \cong \text{Aut}_I(G)$ if and only if ϕ is injective, which requires the kernel of ϕ to be trivial. We examine the conditions under which a non-identity vertex automorphism induces the identity edge automorphism.

For the necessity of the stated conditions, assume first that G contains two distinct isolated vertices, u and v . The vertex automorphism α_0 that transposes u and v while fixing all other vertices leaves the edge set entirely unchanged, as no edges are incident to u or v . Thus $\phi(\alpha_0)$ is the identity edge map, implying a non-trivial kernel. Similarly, assume G contains a component isomorphic to K_2 , consisting of vertices u and v and the single edge $e = \{u, v\}$. The automorphism that swaps u and v maps the edge e to $\{v, u\} = e$. Since this automorphism acts trivially on all other components, it fixes every edge in the graph, again resulting in a non-trivial kernel. Therefore, for the groups to be isomorphic, G must have at most one isolated vertex and no K_2 components.

For sufficiency, assume G satisfies the conditions. Let α_0 be a vertex automorphism in the kernel of ϕ , meaning α_0 maps every edge $\{x, y\}$ to the set $\{x, y\}$. Suppose, for the sake of contradiction, that α_0 is not the identity, so there exists a vertex u such that $\alpha_0(u) = v$ with $u \neq v$. Since automorphisms preserve degree, $\deg(u) = \deg(v)$. If $\deg(u) = 0$, then both u and v are isolated vertices, contradicting the assumption that G has at most one isolated vertex. Thus, $\deg(u) \geq 1$, and there exists an edge $e = \{u, w\}$. Because α_0 fixes all edges, it must map e to itself, so $\{\alpha_0(u), \alpha_0(w)\} = \{v, \alpha_0(w)\} = \{u, w\}$. Since $v \neq u$, this equality forces $v = w$, implying that u and v are adjacent via edge $e = \{u, v\}$. Furthermore, if there existed any other edge $f = \{u, z\}$ incident to u (where $z \neq v$), the condition that α_0 fixes f would similarly imply $\{v, \alpha_0(z)\} = \{u, z\}$. This requires $v \in \{u, z\}$, which is impossible since $v \neq u$ and $z \neq v$. Consequently, u has no neighbors other than v , and by symmetry, v has no neighbors other than u . This implies that u and v form an isolated K_2 component, contradicting the hypothesis. Therefore, no such vertex u exists, α_0 must be the identity, and ϕ is an isomorphism. ■

Corollary 4.6.7. If G is a connected graph with $n \geq 3$, then $\text{Aut}(G) \cong \text{Aut}_I(G)$.

Theorem 4.6.8. Given any graph G , $\text{Aut}(G) = \text{Aut}(\overline{G})$.

Proof. Let V be the vertex set of G . Recall that for the complement graph $\overline{G}$, an edge $\{u, v\}$ exists in $E(\overline{G})$ if and only if $\{u, v\} \notin E(G)$.

Let σ be a permutation of V . We show that $\sigma \in \text{Aut}(G)$ if and only if $\sigma \in \text{Aut}(\overline{G})$. By the definition of an automorphism, $\sigma \in \text{Aut}(G)$ if and only if for all distinct vertices $u, v \in V$:

$$\{u, v\} \in E(G) \iff \{\sigma(u), \sigma(v)\} \in E(G).$$

Taking the logical negation of both sides of the equivalence, we obtain:

$$\{u, v\} \notin E(G) \iff \{\sigma(u), \sigma(v)\} \notin E(G).$$

By the definition of the complement graph, $\{x, y\} \notin E(G)$ is equivalent to $\{x, y\} \in E(\overline{G})$. Substituting this into the expression above yields:

$$\{u, v\} \in E(\overline{G}) \iff \{\sigma(u), \sigma(v)\} \in E(\overline{G}).$$

This is precisely the condition for σ to be an automorphism of $\overline{G}$. Thus, $\sigma \in \text{Aut}(G) \iff \sigma \in \text{Aut}(\overline{G})$, proving that $\text{Aut}(G) = \text{Aut}(\overline{G})$. ■

Theorem 4.6.9. 1. $\text{Aut}(K_n) \cong S_n$

2. $\text{Aut}(K_n - e) \cong S_{n-2} \times S_2$

3. Automorphism group of K_n with two adjacent edges removed is $S_{n-3} \times S_2$

4. Automorphism group of K_n with two non-adjacent edges removed is $S_{n-4} \times D_4$

5. $\text{Aut}(C_n) \cong D_n$

6. For $m, n \geq 1$, $\text{Aut}(K_{m,n}) \cong S_m \times S_n$ $m \neq n$ and $\text{Aut}(K_{n,n}) \cong (S_n \times S_n) \rtimes \mathbb{Z}_2$.

7. $\text{Aut}(P_n) \cong S_2$.

Definition 4.6.10. Let S_n be the symmetric group of degree n , and let $\mathbb{Z}_2 = \{0, 1\}$ be the cyclic group of order 2 (under addition modulo 2).

The **semidirect product** $S_n \rtimes_{\varphi} \mathbb{Z}_2$ is the set of ordered pairs $S_n \times \mathbb{Z}_2$ equipped with a group operation defined by a homomorphism:

$$\varphi : \mathbb{Z}_2 \rightarrow \text{Aut}(S_n)$$

Let $\varphi_0 = \text{id}_{S_n}$ and $\varphi_1 = \theta$, where θ is an automorphism of S_n such that $\theta^2 = \text{id}$ (an involution).

The group operation for two elements (σ, a) and (τ, b) in $S_n \rtimes_{\varphi} \mathbb{Z}_2$ is defined as:

$$(\sigma, a) \cdot (\tau, b) = (\sigma \circ \varphi_a(\tau), a + b)$$

where $\circ$ denotes permutation multiplication and $+$ denotes addition modulo 2.

Note:

- If φ is the trivial homomorphism (maps everything to the identity automorphism), the group becomes the **Direct Product** $S_n \times \mathbb{Z}_2$.
- To obtain a non-trivial semidirect product, θ is typically chosen to be an inner automorphism (conjugation by an element of order 2) or, for $n = 6$, an outer automorphism.

Proof. 1. **Proof for K_n :**

The complete graph K_n contains every possible edge between n distinct vertices. By definition, an automorphism is a permutation of vertices that preserves adjacency. Since every pair of distinct vertices in K_n is adjacent, any bijection (permutation) from the vertex set V to itself preserves adjacency. There are $n!$ such permutations on n elements. The group of all permutations on n elements is the symmetric group S_n . Thus, $\text{Aut}(K_n) \cong S_n$.

2. **Proof for $K_n - e$:**

Let $G = K_n - e$. Let the deleted edge be $e = \{u, v\}$. In G , the vertices u and v have degree $n - 2$ (connected to everyone except each other). All other $n - 2$ vertices (let's call this set W) have degree $n - 1$. Any automorphism ϕ must map vertices to vertices of the same degree. Therefore, ϕ must map the set $\{u, v\}$ to itself and the set W to itself.

- On the set W , the induced subgraph is a complete graph K_{n-2} . Any permutation of these vertices is valid. This gives the group S_{n-2} .
- On the set $\{u, v\}$, we can either fix them ($u \rightarrow u, v \rightarrow v$) or swap them ($u \rightarrow v, v \rightarrow u$). This gives the group S_2 .

Since these actions are independent, $\text{Aut}(G) \cong S_{n-2} \times S_2$.

3. Proof for K_n with two adjacent edges removed:

Let the removed edges be $\{u, v\}$ and $\{v, w\}$. These edges share the common vertex v . We analyze the degrees of the vertices in G :

- Vertex v is not connected to u or w . Its degree is $n - 1 - 2 = n - 3$.
- Vertices u and w are not connected to v . Their degree is $n - 1 - 1 = n - 2$.
- The remaining $n - 3$ vertices (set Z) are connected to everyone. Their degree is $n - 1$.

Since vertex degrees must be preserved: 1. Vertex v is unique (degree $n - 3$) and must be fixed ($\phi(v) = v$). 2. The set $\{u, w\}$ (degree $n - 2$) must map to itself. We can swap them or fix them ($\cong S_2$). 3. The set Z (degree $n - 1$) forms a K_{n-3} . Any permutation is allowed ($\cong S_{n-3}$). Thus, $\text{Aut}(G) \cong S_{n-3} \times S_2$.

4. Proof for K_n with two non-adjacent edges removed:

Let the removed edges be $e_1 = \{u, v\}$ and $e_2 = \{x, y\}$, where $\{u, v\} \cap \{x, y\} = \emptyset$. The vertices u, v, x, y have degree $n - 2$. The remaining $n - 4$ vertices (set Z) have degree $n - 1$. Any automorphism must map Z to Z and $\{u, v, x, y\}$ to $\{u, v, x, y\}$.

- Action on Z : The subgraph induced by Z is K_{n-4} , allowing any permutation ($\cong S_{n-4}$).
- Action on $\{u, v, x, y\}$: The induced subgraph on these four vertices is K_4 minus two disjoint edges (which is equivalent to a cycle C_4 : $u - x - v - y - u$), which gives rise to a group of rotations and reflections of a square. This group of symmetries is isomorphic to the Dihedral group of order 8, D_4 .

Thus, $\text{Aut}(G) \cong S_{n-4} \times D_4$.

5. Proof for C_n :

Let the vertices of the cycle be $1, 2, \dots, n$ in clockwise order. The cycle graph C_n allows for two types of symmetries: 1. Rotations: We can shift every vertex by k positions (n possibilities). 2. Reflections: We can reflect the graph across an axis of symmetry (n possibilities). A group generated by rotations and reflections of a regular n -gon is the Dihedral group D_n , which has order $2n$. Thus, $\text{Aut}(C_n) \cong D_n$.

6. Proof for $K_{m,n}$:

Let V_1 and V_2 be the partite sets with $|V_1| = m$ and $|V_2| = n$. In a complete bipartite graph, every vertex in V_1 is connected to every vertex in V_2 . First assume $m \neq n$;

- Since $m \neq n$, the degrees or the sizes of the partitions distinguish the sets V_1 and V_2 . An automorphism cannot map a vertex from V_1 to V_2 .
- Therefore, any automorphism ϕ acts as a permutation σ on V_1 and a permutation τ on V_2 .
- Since all vertices in V_1 are connected to all in V_2 , any choice of $\sigma \in S_m$ and $\tau \in S_n$ preserves adjacency.

Consequently, $\text{Aut}(K_{m,n}) \cong S_m \times S_n$.

Next, assume that $m = n$; Let $G = K_{n,n}$ with partite sets V_1 and V_2 . Since $\text{Aut}(G) = \text{Aut}(\overline{G})$, consider the complement $\overline{G}$. $\overline{G}$ consists of edges only within V_1 and within V_2 , making it the disjoint union of two complete graphs $K_n \cup K_n$.

Any automorphism of $\overline{G}$ must map connected components to connected components. Since the two components are isomorphic:

- (a) We can permute vertices within V_1 arbitrarily (S_n) and within V_2 arbitrarily (S_n). This forms the subgroup $S_n \times S_n$.
- (b) We can swap the sets V_1 and V_2 entirely. This action corresponds to $\mathbb{Z}_2$.

The group generated by these actions is the semi-direct product $(S_n \times S_n) \rtimes \mathbb{Z}_2$, also known as the wreath product $S_n \wr S_2$.

7. Proof of P_n : Let the vertices of P_n be labeled $1, 2, \dots, n$ in sequence. The vertices 1 and n are the only vertices with degree 1. Since automorphisms preserve degrees, any $\phi \in \text{Aut}(P_n)$ must map the set $\{1, n\}$ to itself. This allows only two possibilities for $\phi(1)$:

- (a) $\phi(1) = 1$: Since automorphisms preserve adjacency and the path is connected, fixing the first vertex fixes the entire path. Thus, $\phi(i) = i$ for all i (the identity map, id).
- (b) $\phi(1) = n$: To preserve the path structure, the neighbor of 1 must map to the neighbor of n , and so on. This forces $\phi(i) = n - i + 1$ (the reflection map, σ).

Thus, $\text{Aut}(P_n) = \{id, \sigma\}$. Since $\sigma^2 = id$ and $\sigma \neq id$, this group is cyclic of order 2, which is isomorphic to the symmetric group S_2 . ■

4.7 Edge Isomorphisms and Automorphisms

Definition. Two graphs $G(V, E)$ and $H(U, F)$ are said to be **edge-isomorphic** if there exists a bijection $\eta : E \rightarrow F$ that preserves edge-adjacency and edge-non-adjacency. Specifically, two edges $e, f \in E$ share a common vertex in G if and only if their images $\eta(e), \eta(f) \in F$ share a common vertex in H . Such a map η is called an *edge-isomorphism*.

An edge-isomorphism from a graph to itself is called an **edge-automorphism**. The set of all edge-automorphisms of a graph G forms a group under composition, known as the *edge-automorphism group* of G , denoted by $\text{Aut}_E(G)$.

It is important to observe that for a given graph G , the vertex automorphism group $\text{Aut}(G)$ and the edge-automorphism group $\text{Aut}_E(G)$ need not be isomorphic. This discrepancy arises because there may exist edge-automorphisms that are not induced by any vertex automorphism (i.e., they are not in the induced edge group $\text{Aut}_I(G)$).

Figure 1 illustrates three specific cases— $K_{1,3} + e$, $K_4 - e$, and K_4 —where each graph admits an edge-automorphism η that cannot be induced by any permutation of the vertices.

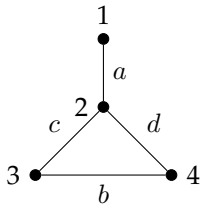


Figure 4.9: *
 $K_{1,3} + e$
 $\eta = (a\ b)(c\ d)$

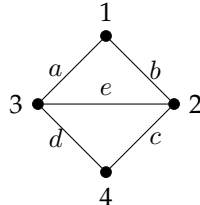


Figure 4.10: *
 $K_4 - e$
 $\eta = (a\ b\ c\ d)(e)$

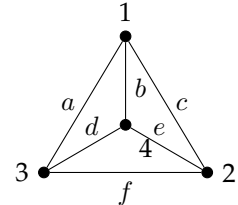


Figure 4.11: *
 K_4
 $\eta = (b\ f)(d)(e)(a)(c)$

Figure 4.12: Edge-automorphisms not induced by vertex automorphisms.

Theorem 4.7.1. Let η be an edge-isomorphism from a connected graph G to a connected graph H . If G is not isomorphic to any of the graphs K_3 , $K_{1,3}$, $K_{1,3} + e$, $K_4 - e$, or K_4 , then η is induced by a vertex isomorphism from G to H .

Theorem 4.7.2. For a non-empty graph G , $\Gamma_I(G) \cong \Gamma_E(G)$ if and only if the following two conditions hold.

- (a) Both K_3 and $K_{1,3}$ are not components of G .
- (b) None of the graphs $K_{1,3} + e$, $K_4 - e$ and K_4 is a component of G .

Proof. The necessity of these conditions follows from the existence of edge-automorphisms for the excluded graphs that are not induced by vertex automorphisms. The sufficiency is immediate for connected graphs by the preceding theorem.

We now establish sufficiency for a disconnected graph G . Let α be an arbitrary edge-automorphism of G . Our goal is to demonstrate that a vertex-automorphism of G induces α .

Consider any connected component H of G . The image of its edge set under α , denoted $\alpha(E(H))$, induces a subgraph $\langle \alpha(E(H)) \rangle$ which must also be a connected component of G .

If a component H is isomorphic to K_3 or $K_{1,3}$, condition (a) ensures that any such component maps to another isomorphic component in a way that preserves the structure. Specifically, we must have $H \cong \langle \alpha(E(H)) \rangle$. Although these specific graphs admit exceptional edge-automorphisms, within the context of the whole graph and the constraints on components, the restriction of α to H aligns with a vertex-automorphism of H .

If a component H is not isomorphic to K_3 or $K_{1,3}$, and by condition (b) is also not isomorphic to $K_{1,3} + e$, $K_4 - e$, or K_4 , then H falls under the purview of Theorem 1. Consequently, the restriction of α to the edges of H is induced by a vertex-isomorphism from H to its image component.

Since α decomposes into independent mappings on the disjoint components of G , and each restriction is induced by a vertex-isomorphism, we can construct a global vertex-automorphism for G by combining these individual maps. Thus, α is induced by a vertex-automorphism of G . ■

Corollary 4.7.3. *If G is a connected graph of order $n \geq 3$, then $\text{Aut}(G)$, $\text{Aut}_I(G)$ and $\text{Aut}_E(G)$ are all isomorphic if and only if G is none of the graphs $K_{1,3} + e$, $K_4 - e$, K_4 .*

4.8 Cayley Graphs of Groups

Let G be a group and let $S \subseteq G$ be a generating set of G . The **Cayley graph** $\Gamma = \text{Cay}(G, S)$ is a graph constructed as follows:

1. The set of vertices $V(\Gamma)$ corresponds to the elements of the group G . That is, $V(\Gamma) = G$.
2. The set of edges $E(\Gamma)$ is defined such that there is a directed edge from g to h if and only if $h = gs$ for some $s \in S$.

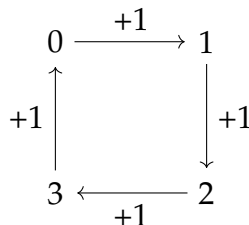
Notation: An edge is typically denoted as (g, gs) and acts as a directed edge colored or labeled by the generator s .

- Usually, the generating set S is chosen such that the identity element $1 \notin S$.
- If S is symmetric (i.e., $s \in S \implies s^{-1} \in S$), the graph is viewed as undirected, because an edge (g, gs) implies the existence of an edge $(gs, gss^{-1}) = (gs, g)$.

Example 4.8.1 (The Cyclic Group $\mathbb{Z}_4$). Let $G = \mathbb{Z}_4 = \{0, 1, 2, 3\}$ under addition modulo 4. Let the generating set be $S = \{1\}$.

- **Vertices:** 0, 1, 2, 3.
- **Edges:** Since the operation is $+$, edges are $(g, g + 1)$.

$$0 \rightarrow 1, \quad 1 \rightarrow 2, \quad 2 \rightarrow 3, \quad 3 \rightarrow 0$$

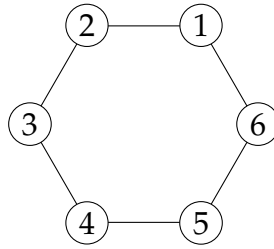


This forms a directed cycle C_4 .

Example 4.8.2 (Cyclic Group $\mathbb{Z}_6$). Let $G = \mathbb{Z}_6 = \{0, 1, 2, 3, 4, 5\}$ (under addition modulo 6). Let the generating set be $S = \{1\}$.

- Vertices: $\{0, 1, 2, 3, 4, 5\}$
- Edges: $(i, i + 1 \pmod{6})$.

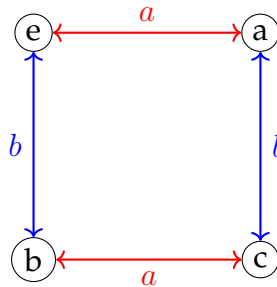
This forms a simple directed cycle C_6 .



Example 4.8.3 (The Klein 4-Group V_4). Let $G = V_4 = \{e, a, b, c\}$ where $a^2 = b^2 = c^2 = e$ and $ab = c$. Let the generating set be $S = \{a, b\}$.

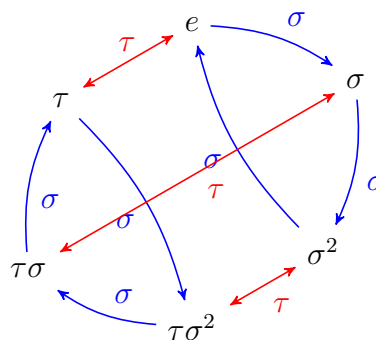
- Edges generated by a (Red): $(e, a), (a, e), (b, c), (c, b)$.
- Edges generated by b (Blue): $(e, b), (b, e), (a, c), (c, a)$.

Since $a = a^{-1}$ and $b = b^{-1}$, for every arrow $x \rightarrow y$, there is an arrow $y \rightarrow x$. We often draw these as undirected edges.



Example 4.8.4 (Symmetric Group S_3). Let $G = S_3$, the permutations of $\{1, 2, 3\}$. Let $S = \{\tau, \sigma\}$ where $\tau = (12)$ and $\sigma = (123)$.

- σ has order 3 (forms hexagons).
- τ has order 2 (forms bi-directional swaps).



Note: The red edges represent τ . Since $\tau^2 = e$, these are technically bidirectional arrows $x \leftrightarrow y$.

4.9 Undirected Cayley Graphs

While the definition implies a digraph (directed graph), we often treat them as undirected graphs under specific conditions.

Definition 4.9.1. $\text{Cay}(G, S)$ is considered an undirected graph if S is inverse-closed. That is, for every $s \in S$, we have $s^{-1} \in S$.

In this case, if there is an edge $g \rightarrow gs$, there is also an edge $gs \rightarrow (gs)s^{-1} = g$. These pairs of directed edges are treated as a single undirected edge.

Theorem 4.9.2. If S generates the group G , then the Cayley graph $\text{Cay}(G, S)$ is connected.

Proof. Let $u, v \in G$ be any two vertices in the graph. We wish to show there is a path from u to v .

Consider the element $g = u^{-1}v$. Since S generates G , the element g can be written as a product of generators and their inverses (if we consider the undirected case or if S is closed under inverses).

$$g = s_1 s_2 \dots s_k \quad \text{where } s_i \in S \cup S^{-1}$$

Therefore, $v = u(s_1 s_2 \dots s_k)$. We can construct a walk starting at u :

$$\begin{aligned} w_0 &= u \\ w_1 &= u s_1 \\ w_2 &= u s_1 s_2 \\ &\vdots \\ w_k &= u s_1 s_2 \dots s_k = v \end{aligned}$$

Since there is a sequence of edges connecting u to v , the graph is connected. ■

Theorem 4.9.3. The Cayley graph $\Gamma = \text{Cay}(G, S)$ is $|S|$ -regular (in terms of out-degree).

Proof. Let $g \in G$ be any vertex. The edges originating from g are determined by the definition:

$$E_g = \{(g, gs) \mid s \in S\}$$

Since group cancellation holds ($gs_1 = gs_2 \implies s_1 = s_2$), the number of distinct edges starting at g is exactly equal to the number of distinct elements in S . Thus, the out-degree of every vertex is $|S|$. ■

This is the most fundamental property of Cayley graphs. It implies that the graph "looks the same" from every vertex.

Theorem 4.9.4. Every Cayley graph $\text{Cay}(G, S)$ is vertex-transitive.

Proof. A graph Γ is vertex-transitive if, for any two vertices $u, v \in V(\Gamma)$, there exists an automorphism $\phi \in \text{Aut}(\Gamma)$ such that $\phi(u) = v$.

Let $u, v \in G$. Let $h = vu^{-1}$. We define the map $\lambda_h : G \rightarrow G$ by left multiplication:

$$\lambda_h(x) = hx$$

1. λ_h is a bijection: This follows from group theory axioms. The inverse is $\lambda_{h^{-1}}$.

2. λ_h preserves adjacency (is a graph homomorphism): Suppose there is an edge from x to y . By definition, this means $y = xs$ for some $s \in S$. We check the relationship between $\lambda_h(x)$ and $\lambda_h(y)$:

$$\lambda_h(y) = hy = h(xs) = (hx)s = \lambda_h(x)s$$

Since $\lambda_h(y) = \lambda_h(x)s$, there exists an edge from $\lambda_h(x)$ to $\lambda_h(y)$ with the same label s . Thus, λ_h is a graph automorphism.

3. Mapping u to v :

$$\lambda_h(u) = hu = (vu^{-1})u = v$$

Since we found an automorphism mapping u to v for arbitrary u, v , the graph is vertex-transitive. ■

Definition 4.9.5. In the Cayley digraph $D(\Gamma, \Lambda)$, if (g, k) is an arc, $k = gh$ for some $h \in \Lambda$, that is $g^{-1}k \in \Lambda$ and $g^{-1}k$ is called the color of (g, k) .

An automorphism of D is said to be color preserving if it preserves the colors of the arcs.

Theorem 4.9.6 (Automorphism Theorem for Colored Cayley Graphs). *The group of color-preserving automorphisms of the Cayley digraph $D(\Gamma, \Lambda)$ is isomorphic to Γ .*

Proof. Let $C(D)$ denote the group of color-preserving automorphisms of the digraph $D(\Gamma, \Lambda)$. We first establish that the group of left multiplications is a subgroup of $C(D)$. For each element $g \in \Gamma$, define the mapping $\alpha_g : \Gamma \rightarrow \Gamma$ by $\alpha_g(x) = gx$. Since Γ is a group, α_g is clearly a bijection. To see that α_g preserves colors, consider an arbitrary arc (u, v) in D with color $h = u^{-1}v \in \Lambda$. The image of this arc under α_g is $(\alpha_g(u), \alpha_g(v)) = (gu, gv)$. The color of the image arc is determined by

$$(gu)^{-1}(gv) = u^{-1}g^{-1}gv = u^{-1}v = h.$$

Since the image arc has the same color h as the original arc, α_g is a color-preserving automorphism. Thus, the set of left multiplications $L(\Gamma) = \{\alpha_g \mid g \in \Gamma\}$ is contained in $C(D)$.

Conversely, we show that every colour-preserving automorphism belongs to $L(\Gamma)$. Let $\phi \in C(D)$ be an arbitrary automorphism. Let e denote the identity of Γ , and let $\phi(e) = g$. We claim that $\phi(x) = gx$ for all $x \in \Gamma$. We proceed by induction on the length of the shortest path from e to x in terms of the generating set Λ . The base case $x = e$ holds by definition, as $\phi(e) = g = ge$.

For the inductive step, assume that $\phi(u) = gu$ for some vertex u , and let v be a vertex adjacent to u such that (u, v) is an arc with color $h \in \Lambda$. By the definition of color, $u^{-1}v = h$, or $v = uh$. Since ϕ is color-preserving, the arc $(\phi(u), \phi(v))$ must also have color h . Therefore,

$$\phi(u)^{-1}\phi(v) = h.$$

Substituting the inductive hypothesis $\phi(u) = gu$, we have

$$(gu)^{-1}\phi(v) = h \implies u^{-1}g^{-1}\phi(v) = h \implies \phi(v) = guh.$$

Since $v = uh$, it follows that $\phi(v) = gv$. Because the Cayley graph is connected (assuming Λ generates Γ), this property extends to all vertices $x \in \Gamma$. Thus, $\phi = \alpha_g$, implying $C(D) \subseteq L(\Gamma)$.

We have established that $C(D) = L(\Gamma)$. Finally, consider the mapping $\Psi : \Gamma \rightarrow C(D)$ defined by $\Psi(g) = \alpha_g$. This map is surjective by the argument above and injective because $\alpha_g(e) = g$ implies distinct group elements define distinct automorphisms. Furthermore, Ψ is a homomorphism because the composition of left multiplications corresponds to multiplication in the group:

$$(\alpha_g \circ \alpha_k)(x) = \alpha_g(kx) = g(kx) = (gk)x = \alpha_{gk}(x).$$

Therefore, Ψ is an isomorphism, and $\Gamma \cong C(D)$. ■

Theorem 4.9.7 (Frucht, 1939). *Let Γ be a finite group. There exists a finite, undirected graph G such that the automorphism group of G is isomorphic to Γ :*

$$\text{Aut}(G) \cong \Gamma$$

Proof. Let $\Lambda = \{h_1, h_2, \dots, h_n\}$ be a set of generators for the group Γ , and let $D = D(\Gamma, \Lambda)$ be the Cayley digraph of Γ . We have previously established that the group of colour-preserving automorphisms of this digraph, denoted $C(D)$, is isomorphic to Γ . To prove the theorem, we construct an undirected, uncoloured graph G whose full automorphism group coincides with $C(D)$.

The graph G is obtained from D by a specific topological substitution that encodes the direction and colour of the original arcs. For every directed arc (g_i, g_j) in D with color h_k (where $g_i^{-1}g_j = h_k$), we replace the arc with a subgraph connecting g_i and g_j . This subgraph consists of a path of length three, $g_i - u - v - g_j$, along with two identifying paths rooted at the internal vertices: a path P_{2k} of length $2k$ attached to u , and a path P_{2k+1} of length $2k+1$ attached to v . This construction is illustrated in Figure 3.1. Since this procedure is performed for every arc, the resulting graph G is undirected.

We now define the isomorphism between the automorphism groups. Any colour-preserving automorphism $\alpha \in C(D)$ naturally induces a unique automorphism on G . Since α maps an arc (g_i, g_j) of color h_k to another arc (g_x, g_y) of the same color h_k , it maps the substituting subgraph of the graph G to the identical subgraph of the isomorphic copy of G under α , preserving adjacency throughout. Thus, $C(D)$ is isomorphic to a subgroup of $\text{Aut}(G)$.

Conversely, let $\beta \in \text{Aut}(G)$ be an arbitrary automorphism of the constructed graph. We observe that the vertices of G can be partitioned based on their degrees and structural properties. The original vertices from Γ have degree at least $2|\Lambda|$, whereas the internal vertices u and v have degree 3, and the vertices on the attached paths have degree 1 or 2. Consequently, β must map the set of original vertices Γ to itself. Furthermore, β must map the internal structures connecting these vertices to isomorphic structures.

Consider the subgraph connecting two adjacent original vertices. Because the vertex u has an attached path of even length $2k$, and v has an attached path of odd length $2k + 1$, β cannot map u to v . This asymmetry forces β to preserve the orientation of the original arc (mapping u to a u -type vertex and v to a v -type vertex). Similarly, since the lengths of the attached paths depend strictly on the index k , β can only map a subgraph corresponding to generator h_k to another subgraph corresponding to h_k . This implies that β effectively acts on the original vertices Γ in a way that preserves both the adjacency direction and the edge colors. Thus, β corresponds to a unique element in $C(D)$.

We conclude that $\text{Aut}(G) \cong C(D)$. Since $C(D) \cong \Gamma$, it follows that $\text{Aut}(G) \cong \Gamma$. ■

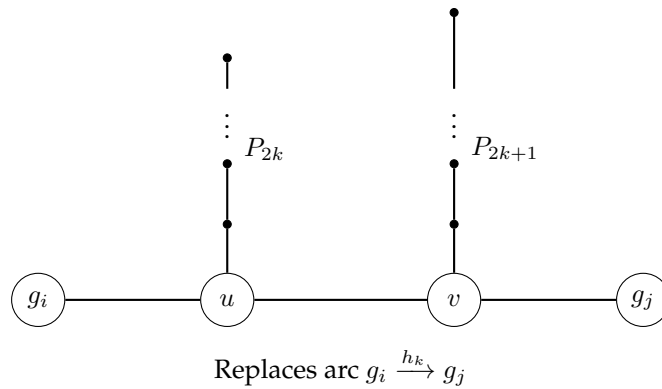


Figure 4.13: Frucht Graph: The subgraph replacing the directed edge from g_i to g_j with color h_k . The asymmetry of the paths P_{2k} and P_{2k+1} enforces the direction and color preservation.

Corollary 4.9.8. *Let Γ be a finite group of order n , and let Λ be a generating set of size $s = |\Lambda|$. The Frucht graph $G(\Gamma)$ (Fig 4.13) has exactly $n(s + 1)(2s + 1)$ vertices.*

Proof. To determine the order of the Frucht graph $G(\Gamma)$, we consider the vertices contributed by the original group elements and those introduced by the topological construction. The underlying Cayley digraph $D(\Gamma, \Lambda)$ contains n vertices, and each vertex possesses exactly s outgoing arcs corresponding to the k generators indexed $k = 1$ to s . In the transition to $G(\Gamma)$, each arc associated with the k -th generator is replaced by a subgraph that introduces $4k + 1$ new internal vertices. Consequently, for every element in the group, the set of outgoing arcs contributes a total of $\sum_{k=1}^s (4k + 1)$

additional vertices. Evaluating this sum yields $4\frac{s(s+1)}{2} + s = 2s^2 + 3s$ new vertices per group element. Across the entire group of n elements, the total number of added vertices is $n(2s^2 + 3s)$. Adding the n original vertices to this count, we find that the total order of the graph is $n(2s^2 + 3s + 1)$. Factoring the quadratic expression, we conclude that $|V(G(\Gamma))| = n(s + 1)(2s + 1)$. ■

Example 4.9.9. Frucht Graph Order for $n = 3$, and $s = 2$.

Solution. First, calculating directly using the formula:

$$|V(G)| = 3(2 + 1)(2(2) + 1) = 3(3)(5) = 45.$$

We now construct the count step-by-step based on the topological definition of the graph.

Every vertex in Γ has $s = 2$ outgoing arcs, one for each generator. The construction replaces these arcs with "gadgets" of different sizes depending on the index k . The number of internal vertices added for the k -th generator is $4k + 1$.

- For generator h_1 ($k = 1$): The gadget adds $4(1) + 1 = 5$ vertices.
- For generator h_2 ($k = 2$): The gadget adds $4(2) + 1 = 9$ vertices.

Since every group element (e, x, y) has exactly one outgoing arc for h_1 and one outgoing arc for h_2 , the total number of vertices added **per group element** is:

$$\text{Added per vertex} = 5 + 9 = 14.$$

Therefore,

$$\begin{aligned} \text{Total Vertices} &= n + n \sum_{k=1}^2 (4k + 1) \\ &= 3 + 3[(4(1) + 1) + (4(2) + 1)] \\ &= 3 + 3[5 + 9] \\ &= 3 + 42 \\ &= 45. \end{aligned}$$

■